

VDF PROJECT REPORT
HOME AUTOMATION SYSTEM
Group No. 13

Submitted by:

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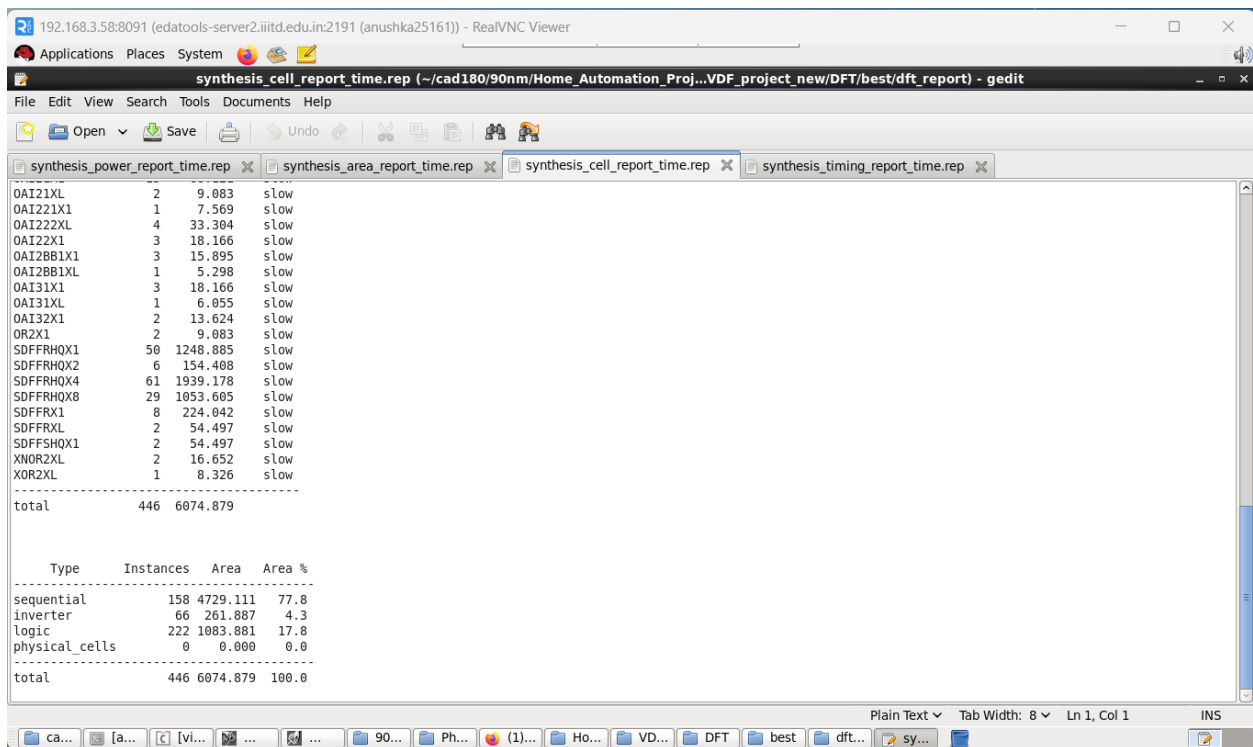
Introduction

Project Part –2

The physical design flow, which involves a number of critical elements like floor layout, placement, clock tree synthesis, and detailed routing, is applied to the design during this project phase. Physical design is the process of converting the netlist into the finished layout. The netlist used for this physical design stage was the result of the Design for Testability (DFT) scan insertion phase, which was finished in the project's preceding phase. The physical design flow entails converting the logical netlist representation of the design into a physical layout that details the exact placement and connections of the different integrated circuit components. This procedure is essential to guarantee that the finished design satisfies the required requirements and limitations for area utilization, power consumption, and performance.

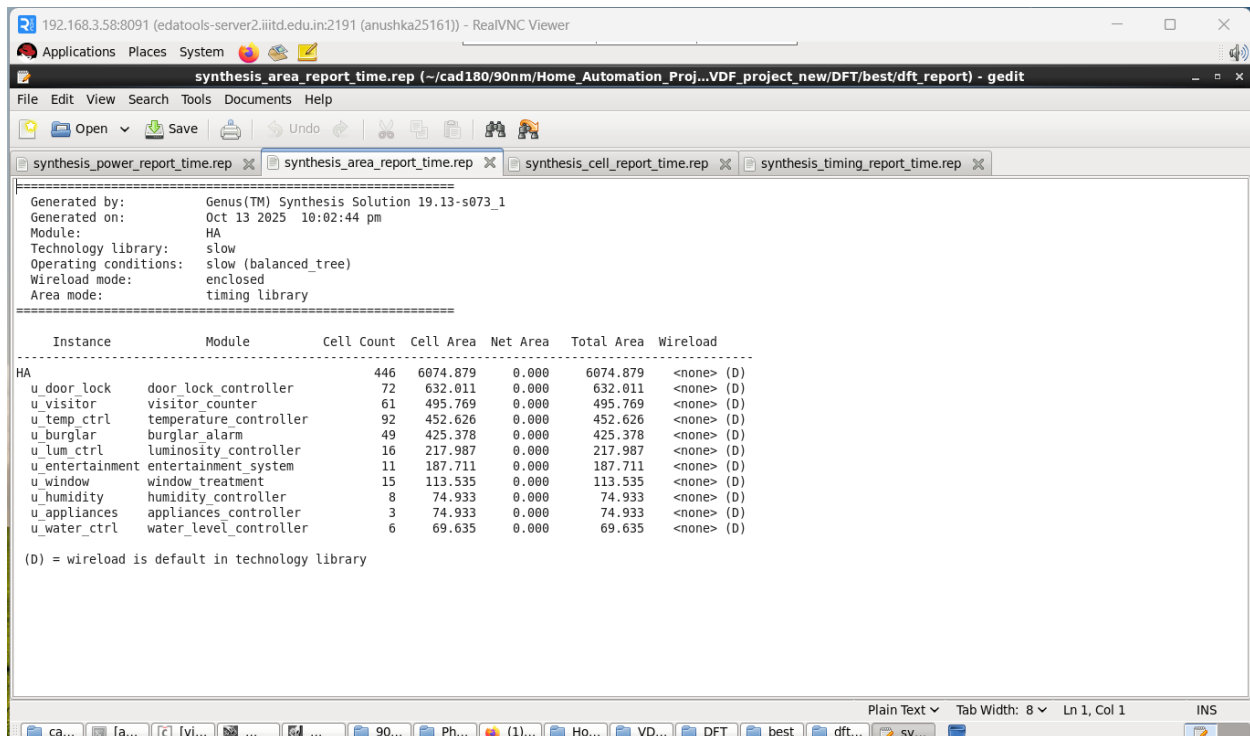
Before Physical Design: Scan insertion

1. Area for best timing constraint



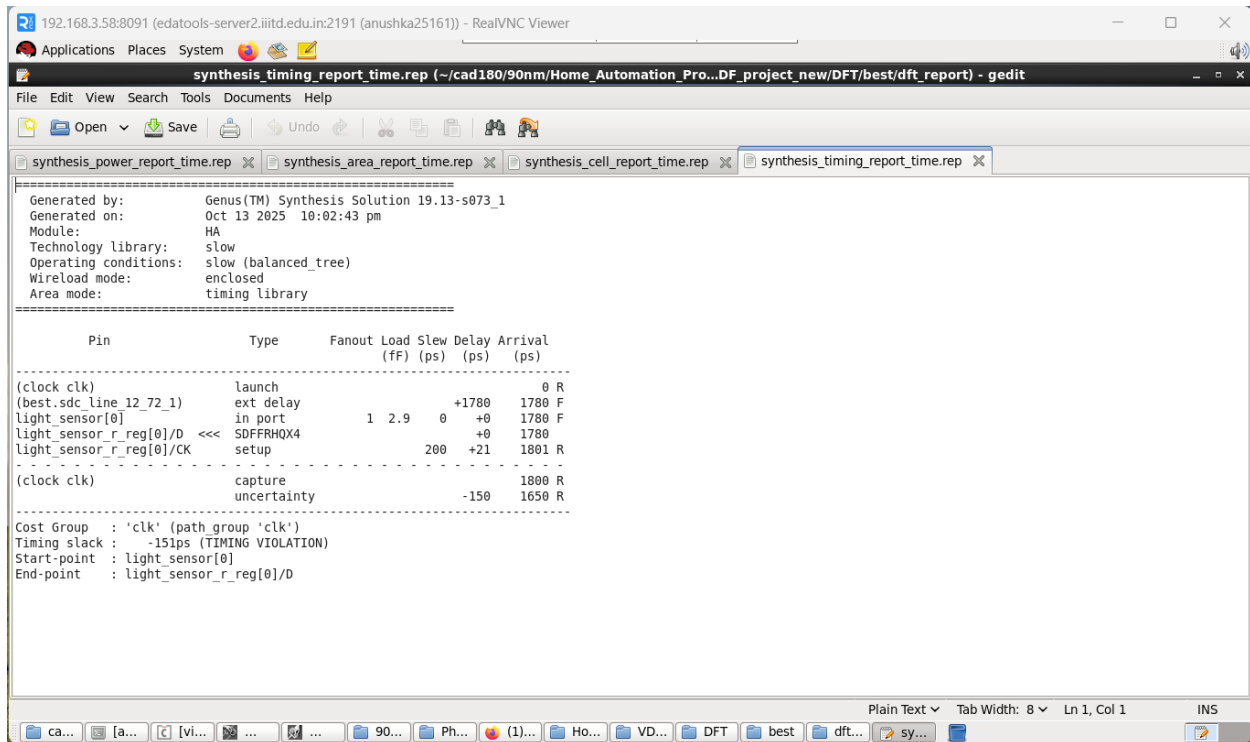
The screenshot shows a RealVNC Viewer window displaying a synthesis report. The report is titled "synthesis_cell_report_time.rep" and is located at the path "~/cad180/90nm/Home_Automation_Proj...VDF_project_new/DFT/best/dft_report". The report lists various components and their area contributions.

Type	Instances	Area	Area %
total	446	6074.879	
sequential	158	4729.111	77.8
inverter	66	261.887	4.3
logic	222	1083.881	17.8
physical_cells	0	0.000	0.0
total	446	6074.879	100.0



- Sequential Cells (e.g., flip-flops, latches): Highest area usage, accounting for nearly 78% of the design, crucial for pipelining and data storage.
- Inverters: Cover 4.3% area, used for signal inversion and buffering.
- Logic Cells (AND, OR, AOI/OAI/NAND/NOR/XOR etc.): About 18% of the total, representing the majority of combinational logic operations.
- Physical Cells: None reported in this summary.
- Total: 446 standard cell instances, occupying a total area of 6074.879

2. Timing for best timing constraint



```
Generated by:      Genus(TM) Synthesis Solution 19.13-s073_1
Generated on:      Oct 13 2025  10:02:43 pm
Module:           HA
Technology library: slow
Operating conditions: slow (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library

=====
Pin              Type      Fanout Load Slew Delay Arrival
              (ff) (ps) (ps) (ps)
-----
(clock clk)      launch              0 R
(best.sdc_line_12_72_1) ext delay          +1780 1780 F
light_sensor[0]  in port              +0 1780 F
light_sensor_r_reg[0]/D <<< SDFFRHGX4          +0 1780
light_sensor_r_reg[0]/CK setup          200 +21 1801 R
-----
(clock clk)      capture              1800 R
uncertainty              -150 1650 R
-----
Cost Group   : 'clk' (path group 'clk')
Timing slack : -151ps (TIMING VIOLATION)
Start-point  : light_sensor[0]
End-point    : light_sensor_r_reg[0]/D
```

Critical Path: The timing path starts at light_sensor and ends at light_sensor_r_reg/D, capturing the launch and capture points for this path on the "clk" clock domain.

Setup Time Analysis:

The path is launched by (clock clk) and propagates through pin light_sensor (input port), SDFFRHGX4 (setup at flip-flop), arriving at light_sensor_r_reg/CK.

Fanout and load information is provided for each pin, with propagation delays detailed.

Slack Information:

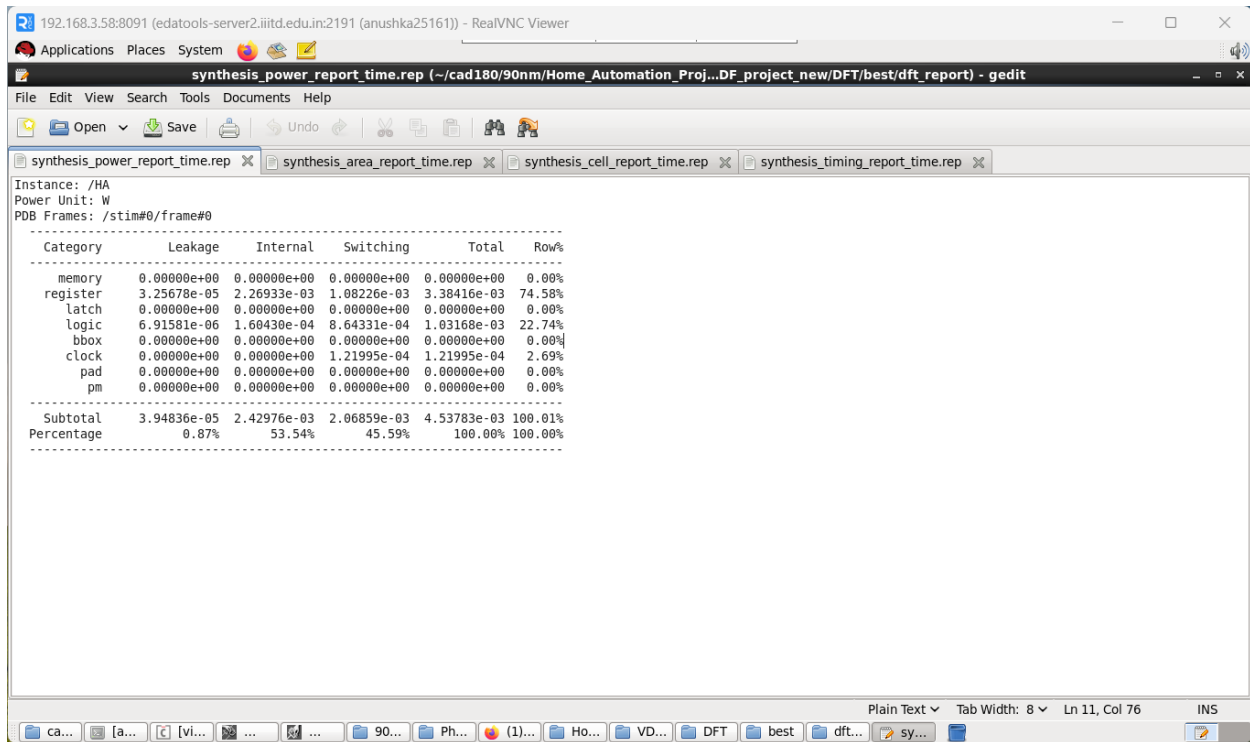
The cost group "clk" indicates a slack of -151ps, meaning the path fails the setup timing requirement by 151ps (timing violation).

Timing Elements Captured:

Launch edge and capture edge times are logged.

Arrival and delay times are shown for each relevant point of the path, including clock uncertainty (-150ps).

3. Power report for best timing constraint



Instance: /HA
Power Unit: W
PDB Frames: /stim#0/frame#0

Category	Leakage	Internal	Switching	Total	Row%
memory	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
register	3.25678e-05	2.26933e-03	1.08226e-03	3.38416e-03	74.58%
latch	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
logic	6.91581e-06	1.60430e-04	8.64331e-04	1.03168e-03	22.74%
bbox	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
clock	0.00000e+00	0.00000e+00	1.21995e-04	1.21995e-04	2.69%
pad	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
pm	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	0.00%
Subtotal	3.94836e-05	2.42976e-03	2.06859e-03	4.53783e-03	100.01%
Percentage	0.87%	53.54%	45.59%	100.00%	100.00%

Total Power Consumption:

Sum: 4.53783e-03 W (4.54 mW)

STA after Scan Insertion

The process of obtaining timing information from the design, including delay details of logic gates, connecting wires, and other circuit components, is known as static timing analysis. In order to identify important routes and calculate their time characteristics, it then simulates the circuit's behavior under various operating situations. The circuit's ability to operate at the clock frequency the designer had predicted is then confirmed using these characteristics.

Setup Analysis: Before the subsequent clock edge reaches the flip-flop, the input pin data for the capture flip-flop must be present during "Setup Time."

Hold Analysis: After the clock edge arrives at the flip-flop, no additional data should be received at the input pin of the capture flip-flop during the "Hold Time."

Best timing constraints:

a) Timing Report

Setup Time

This is the minimum time required before the clock edge that the data must be stable at the register's D input. Here, setup time is 0.299 ns, pulled from cell library characterization.

Phase Shift

Phase shift represents any intentional shift of clock edges between launch and capture registers, typically present in multi-clock domain paths. In this report, phase shift is 0 ns, usually for simple synchronous single-clock paths.

Uncertainty

Uncertainty combines clock jitter, skew, and modeling errors. It is a conservative adjustment that ensures robustness against variations. Reported as 0.100 ns in this example.

Required Time

This is the calculated time at which data must arrive at the capture register to safely meet setup. For synchronous paths, it generally equals the clock period minus setup, phase shift, and uncertainty:

$$\text{Required Time} = \text{Clock Period} - \text{Setup} - \text{Phase Shift} - \text{Uncertainty}$$

Here, required time is 4.601 ns.

Arrival Time

Arrival time is when the data actually arrives at the register's input, after propagating from the launching flip-flop through combinational logic. In this report, arrival time is 1.377 ns.

Slack

Slack is the difference:

$$\text{Slack} = \text{Required Time} - \text{Arrival Time}$$

Positive slack (3.224 ns) means the design meets setup requirements with margin

b) Path based analysis for hold slack

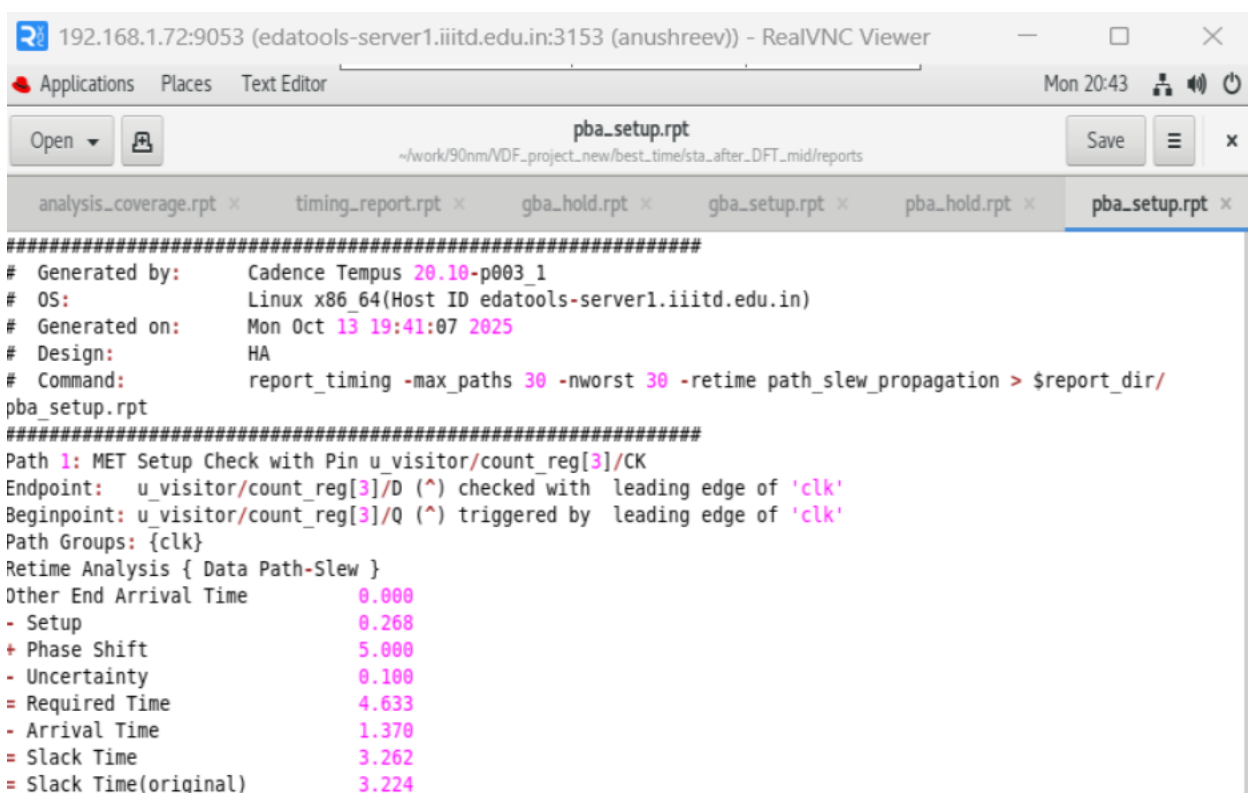
```
#####
# Generated by:      Cadence Tempus 20.10-p003_1
# OS:               Linux x86_64(Host ID edatools-server1.iiitd.edu.in)
# Generated on:      Mon Oct 13 19:41:07 2025
# Design:           HA
# Command:          report_timing -early -max_paths 30 -nworst 30 -retime path_slew_propagation > $report_dir/
pba_hold.rpt
#####
Path 1: VIOLATED Hold Check with Pin door_unlock_cmd_r_reg/CK
Endpoint:  door_unlock_cmd_r_reg/D (v) checked with  leading edge of 'clk'
Beginpoint: door_unlock_cmd      (v) triggered by  leading edge of '@'
Path Groups: {clk}
Retime Analysis { Data Path-Slew }
Other End Arrival Time      0.000
+ Hold                      0.206
+ Phase Shift               0.000
+ Uncertainty               0.100
= Required Time             0.306
Arrival Time                0.000
Slack Time                  -0.306
= Slack Time(original)     -0.319
#####
```

PBA traces the full path from launch (Q) to capture (D) considering the real slew and input arrival at each stage.

For hold, PBA calculates the minimum delay from launch to capture, with the least transition times and real input slews per cell across the path.

It therefore catches violated hold paths more precisely, sometimes reporting new violations missed by GBA due to more realistic timing aggregation

c) Path based analysis for setup slack



```
#####
# Generated by:      Cadence Tempus 20.10-p003_1
# OS:               Linux x86_64(Host ID edatools-server1.iiitd.edu.in)
# Generated on:      Mon Oct 13 19:41:07 2025
# Design:           HA
# Command:          report_timing -max_paths 30 -nworst 30 -retime path_slew_propagation > $report_dir/
pba_setup.rpt
#####
Path 1: MET Setup Check with Pin u_visitor/count_reg[3]/CK
Endpoint:  u_visitor/count_reg[3]/D (^) checked with  leading edge of 'clk'
Beginpoint: u_visitor/count_reg[3]/Q (^) triggered by  leading edge of 'clk'
Path Groups: {clk}
Retime Analysis { Data Path-Slew }
Other End Arrival Time          0.000
- Setup                        0.268
+ Phase Shift                   5.000
- Uncertainty                   0.100
= Required Time                 4.633
- Arrival Time                  1.370
= Slack Time                    3.262
= Slack Time(original)         3.224
```

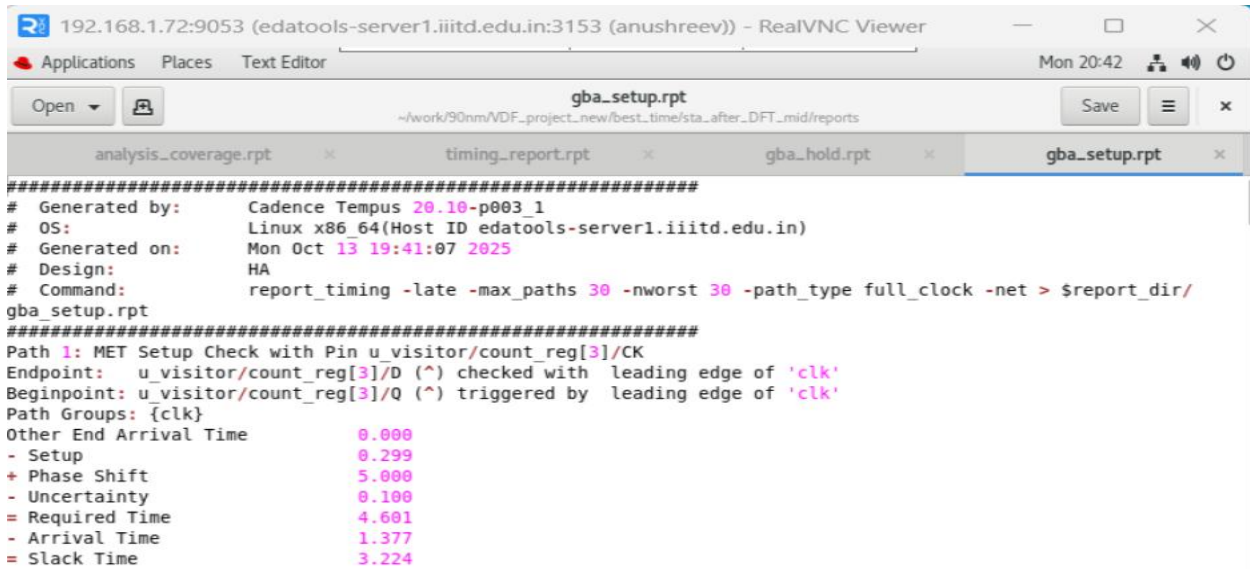
PBA traces the full path from launch (Q) to capture (D) considering the real slew and input arrival at each stage.

Arrival time at each gate/input is computed with the actual transition (slew) from the previous stage, so it reflects realistic worst-case conditions for the complete path, not just maximum individual cell delays.

Required time is not affected by PBA, it depends on constraints (such as setup/hold, clock, uncertainty). Arrival time becomes more accurate, reflecting real worst-case propagation.

Slack computation (Slack = Required Time – Arrival Time)

d) Graph based analysis for setup slack



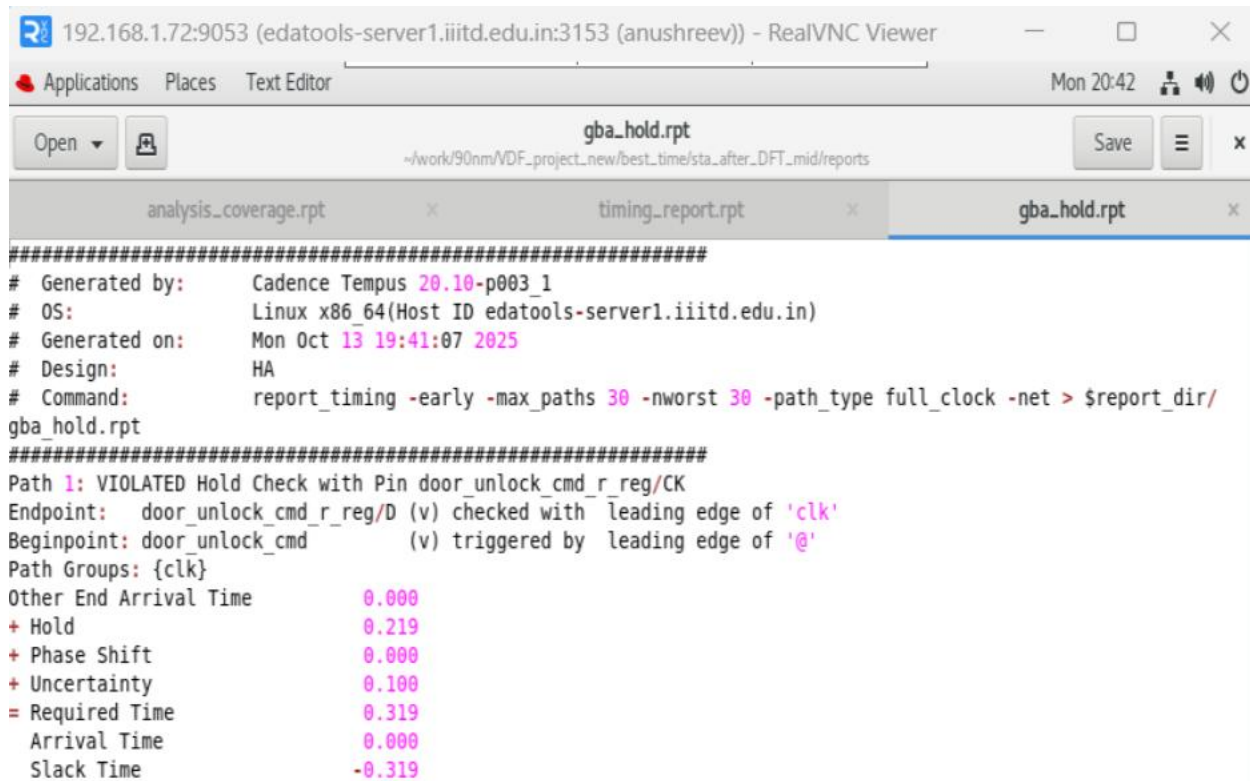
```
#####
# Generated by: Cadence Tempus 20.10-p003_1
# OS: Linux x86_64(Host ID edatools-server1.iiitd.edu.in)
# Generated on: Mon Oct 13 19:41:07 2025
# Design: HA
# Command: report_timing -late -max_paths 30 -nworst 30 -path_type full_clock -net > $report_dir/
gba_setup.rpt
#####
Path 1: MET Setup Check with Pin u_visitor/count_reg[3]/CK
Endpoint: u_visitor/count_reg[3]/D (^) checked with leading edge of 'clk'
Beginpoint: u_visitor/count_reg[3]/Q (^) triggered by leading edge of 'clk'
Path Groups: {clk}
Other End Arrival Time 0.000
- Setup 0.299
+ Phase Shift 5.000
- Uncertainty 0.100
= Required Time 4.601
- Arrival Time 1.377
= Slack Time 3.224
```

GBA computes each stage's delay using the cell's maximum delay for typical or worst slews, even if actual input slews could be less severe

Arrival time at each stage is computed assuming worst-case input for every cell, compounding pessimism along the path

This often underestimates timing margin and can produce negative slacks that don't reflect actual silicon delays.

e) Graph based analysis for Hold slack



```
#####
# Generated by:      Cadence Tempus 20.10-p003_1
# OS:                Linux x86_64(Host ID edatools-server1.iiitd.edu.in)
# Generated on:      Mon Oct 13 19:41:07 2025
# Design:            HA
# Command:           report_timing -early -max_paths 30 -nworst 30 -path_type full_clock -net > $report_dir/
gba_hold.rpt
#####
Path 1: VIOLATED Hold Check with Pin door_unlock_cmd_r_reg/CK
Endpoint:  door_unlock_cmd_r_reg/D (v) checked with leading edge of 'clk'
Beginpoint: door_unlock_cmd (v) triggered by leading edge of '@'
Path Groups: {clk}
Other End Arrival Time          0.000
+ Hold                          0.219
+ Phase Shift                   0.000
+ Uncertainty                   0.100
= Required Time                 0.319
  Arrival Time                 0.000
  Slack Time                    -0.319
```

For hold checks, GBA uses the minimum delay characterization, applied independently from stage to stage, considering fastest transitions and lowest possible delays

It can overestimate risk of hold violations, flagging failures that aren't physically realizable after layout and real parasitic extraction.

Effect of slew:

Slow slew (gentle transitions) increases gate delay, as gates respond slower when inputs change less sharply. Fast slew (sharp transitions) reduces gate delay for that stage.

Effect of load:

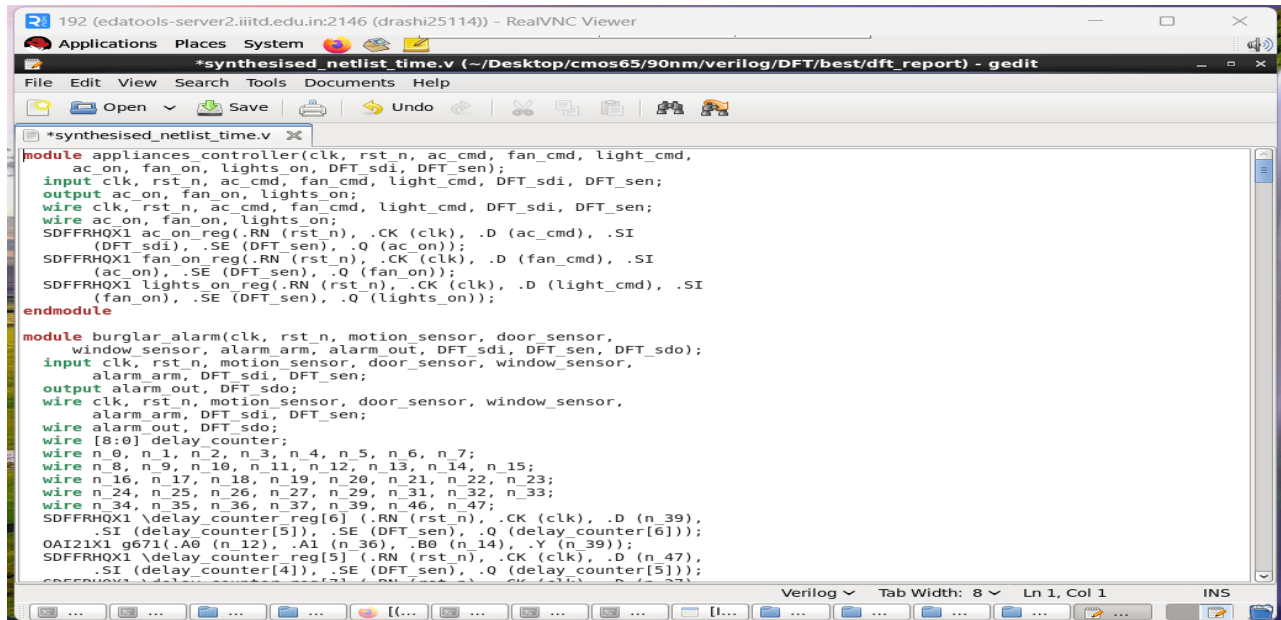
Higher output load increases propagation delay, as the gate needs to charge/discharge more capacitance. Delay tables in libraries specify delay as a function of output load and input slew

Effect of positive/negative unate-ness:

Positive Unate: Output transition is in the same direction as input transition (e.g., buffer, non-inverting gate).

Negative Unate: Output transition is opposite to the input transition (e.g., inverter, NAND, NOR).

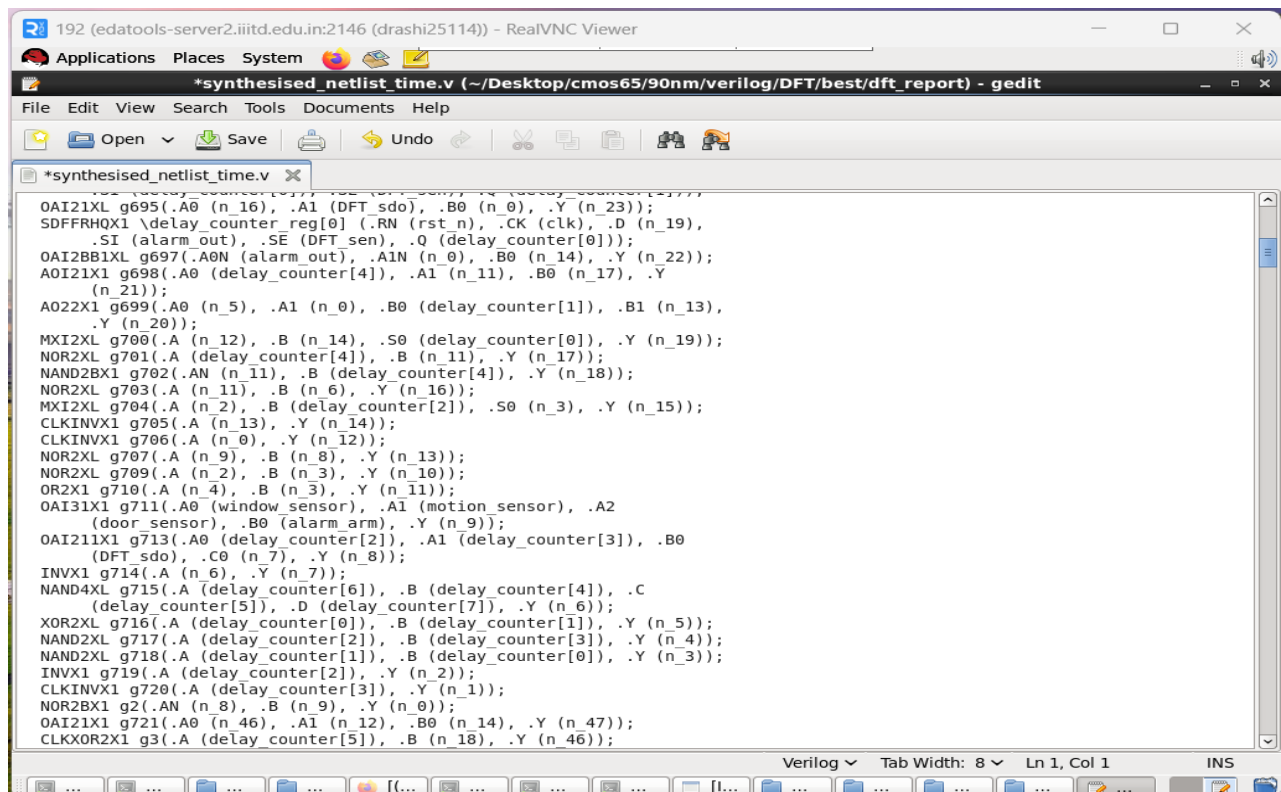
Synthesised Netlist:



The screenshot shows a Verilog synthesised netlist in a gedit editor. The netlist defines two modules: `appliances_controller` and `burglar_alarm`. The `appliances_controller` module has inputs `clk`, `rst_n`, `ac_cmd`, `fan_cmd`, `light_cmd`, `DFT_sdi`, and `DFT_sen`. It contains three registers: `ac_on_reg`, `DFT_sdi`, and `lights_on_reg`. The `burglar_alarm` module has inputs `clk`, `rst_n`, `motion_sensor`, `door_sensor`, `window_sensor`, `alarm arm`, `alarm out`, `DFT_sdi`, and `DFT_sen`. It contains a register `alarm out` and a delay counter `delay_counter`. The netlist is synthesized using Synopsys Design Compiler (SDC) and is in Verilog format.

```
*synthesised_netlist_time.v
module appliances_controller(clk, rst_n, ac_cmd, fan_cmd, light_cmd,
    ac_on, fan_on, lights_on, DFT_sdi, DFT_sen);
input clk, rst_n, ac_cmd, fan_cmd, light_cmd, DFT_sdi, DFT_sen;
output ac_on, fan_on, lights_on;
wire clk, rst_n, ac_cmd, fan_cmd, light_cmd, DFT_sdi, DFT_sen;
wire ac_on, fan_on, lights_on;
SDFFRHX1 ac_on_reg(.RN(rst_n), .CK(clk), .D(ac_cmd), .SI
    (DFT_sdi), .SE(DFT_sen), .Q(ac_on));
SDFFRHX1 fan_on_reg(.RN(rst_n), .CK(clk), .D(fan_cmd), .SI
    (ac_on), .SE(DFT_sen), .Q(fan_on));
SDFFRHX1 lights_on_reg(.RN(rst_n), .CK(clk), .D(light_cmd), .SI
    (fan_on), .SE(DFT_sen), .Q(lights_on));
endmodule

module burglar_alarm(clk, rst_n, motion_sensor, door_sensor,
    window_sensor, alarm arm, alarm out, DFT_sdi, DFT_sen);
input clk, rst_n, motion_sensor, door_sensor, window_sensor,
    alarm arm, DFT_sdi, DFT_sen;
output alarm out, DFT_sdi;
wire clk, rst_n, motion_sensor, door_sensor, window_sensor,
    alarm arm, DFT_sdi, DFT_sen;
wire alarm out, DFT_sdi;
wire [8:0] delay_counter;
wire n_0, n_1, n_2, n_3, n_4, n_5, n_6, n_7;
wire n_8, n_9, n_10, n_11, n_12, n_13, n_14, n_15;
wire n_16, n_17, n_18, n_19, n_20, n_21, n_22, n_23;
wire n_24, n_25, n_26, n_27, n_29, n_31, n_32, n_33;
wire n_34, n_35, n_36, n_37, n_39, n_46, n_47;
SDFFRHX1 delay_counter_reg[6] (.RN(rst_n), .CK(clk), .D(n_39),
    .SI(delay_counter[5]), .SE(DFT_sen), .Q(delay_counter[6]));
OAI21X1 g697(.A0(n_12), .A1(n_36), .B0(n_14), .Y(n_39));
SDFFRHX1 delay_counter_reg[5] (.RN(rst_n), .CK(clk), .D(n_47),
    .SI(delay_counter[4]), .SE(DFT_sen), .Q(delay_counter[5]));
```



The screenshot shows the continuation of the Verilog synthesised netlist in a gedit editor. The netlist defines the logic for the `burglar_alarm` module, including the `delay_counter` and the `alarm out` register. The netlist is synthesized using Synopsys Design Compiler (SDC) and is in Verilog format.

```
OAI21X1 g695(.A0(n_16), .A1(DFT_sdi), .B0(n_0), .Y(n_23));
SDFFRHX1 delay_counter_reg[0] (.RN(rst_n), .CK(clk), .D(n_19),
    .SI(alarm out), .SE(DFT_sen), .Q(delay_counter[0]));
OAI2BB1X1 g697(.A0N(alarm out), .A1N(n_0), .B0(n_14), .Y(n_22));
AOI21X1 g698(.A0(delay_counter[4]), .A1(n_11), .B0(n_17), .Y
    (n_21));
AO22X1 g699(.A0(n_5), .A1(n_0), .B0(delay_counter[1]), .B1(n_13),
    .Y(n_20));
MXI2XL g700(.A(n_12), .B(n_14), .S0(delay_counter[0]), .Y(n_19));
NOR2XL g701(.A(delay_counter[4]), .B(n_11), .Y(n_17));
NAND2BX1 g702(.AN(n_11), .B(delay_counter[4]), .Y(n_18));
NOR2XL g703(.A(n_11), .B(n_6), .Y(n_16));
MXI2XL g704(.A(n_2), .B(delay_counter[2]), .S0(n_3), .Y(n_15));
CLKINX1 g705(.A(n_13), .Y(n_14));
CLKINX1 g706(.A(n_0), .Y(n_12));
NOR2XL g707(.A(n_9), .B(n_8), .Y(n_13));
NOR2XL g709(.A(n_2), .B(n_3), .Y(n_10));
OR2X1 g710(.A(n_4), .B(n_3), .Y(n_11));
OAI31X1 g711(.A0(window_sensor), .A1(motion_sensor), .A2
    (door_sensor), .B0(alarm arm), .Y(n_9));
OAI211X1 g713(.A0(delay_counter[2]), .A1(delay_counter[3]), .B0
    (DFT_sdi), .C0(n_7), .Y(n_8));
INVX1 g714(.A(n_6), .Y(n_7));
NAND4XL g715(.A(delay_counter[6]), .B(delay_counter[4]), .C
    (delay_counter[5]), .D(delay_counter[7]), .Y(n_6));
XOR2XL g716(.A(delay_counter[0]), .B(delay_counter[1]), .Y(n_5));
NAND2XL g717(.A(delay_counter[2]), .B(delay_counter[3]), .Y(n_4));
NAND2XL g718(.A(delay_counter[1]), .B(delay_counter[0]), .Y(n_3));
INVX1 g719(.A(delay_counter[2]), .Y(n_2));
CLKINX1 g720(.A(delay_counter[3]), .Y(n_1));
NOR2BX1 g2(.AN(n_8), .B(n_9), .Y(n_0));
OAI21X1 g721(.A0(n_46), .A1(n_12), .B0(n_14), .Y(n_47));
CLKXOR2X1 g3(.A(delay_counter[5]), .B(n_18), .Y(n_46));
```

192 (edatools-server2.iitd.edu.in:2146 (drashi25114)) - RealVNC Viewer

Applications Places System

*synthesised_netlist_time.v (~/Desktop/cmos65/90nm/verilog/DFT/best/dft_report) - gedit

File Edit View Search Tools Documents Help

Open Save Undo

*synthesised_netlist_time.v

```
CLKX0R2X1 g3(.A (delay_counter[5]), .B (n_18), .Y (n_46));
endmodule

module door_lock_controller(clk, rst_n, lock_cmd, unlock_cmd,
    pin_input, door_locked, lock_status, DFT_sdi, DFT_sen, DFT_sdo);
    input clk, rst_n, lock_cmd, unlock_cmd, DFT_sdi, DFT_sen;
    input [3:0] pin_input;
    output door_locked, DFT_sdo;
    output [1:0] lock_status;
    wire clk, rst_n, lock_cmd, unlock_cmd, DFT_sdi, DFT_sen;
    wire [3:0] pin_input;
    wire door_locked, DFT_sdo;
    wire [1:0] lock_status;
    wire [2:0] attempt_counter;
    wire [1:0] state;
    wire [7:0] timeout_counter;
    wire n_0, n_1, n_2, n_3, n_4, n_5, n_6, n_7;
    wire n_8, n_9, n_10, n_11, n_12, n_13, n_14, n_15;
    wire n_16, n_17, n_18, n_19, n_20, n_21, n_22, n_23;
    wire n_24, n_25, n_26, n_27, n_28, n_29, n_30, n_31;
    wire n_32, n_33, n_34, n_35, n_36, n_37, n_38, n_39;
    wire n_40, n_41, n_42, n_43, n_44, n_45, n_46, n_47;
    wire n_48, n_49, n_50, n_51, n_52, n_53, n_54, n_55;
    SDFFRHQX1 door_locked_reg(.RN (rst_n), .CK (clk), .D (n_55), .SI
        (attempt_counter[2]), .SE (DFT_sen), .Q (door_locked));
    SDFFRHQX1 \lock_status_reg[1] (.RN (rst_n), .CK (clk), .D (state[1]),
        .SI (lock_status[0]), .SE (DFT_sen), .Q (lock_status[1]));
    SDFFRHQX1 \lock_status_reg[0] (.RN (rst_n), .CK (clk), .D (state[0]),
        .SI (door_locked), .SE (DFT_sen), .Q (lock_status[0]));
    OR2X1 g754(.A (state[0]), .B (state[1]), .Y (n_55));
    SDFFRHQX1 \timeout_counter_reg[7] (.RN (rst_n), .CK (clk), .D (n_54),
        .SI (timeout_counter[6]), .SE (DFT_sen), .Q (DFT_sdo));
    SDFFRHQX1 \timeout_counter_reg[6] (.RN (rst_n), .CK (clk), .D (n_52),
```

Verilog Tab Width: 8 Ln 1, Col 1 INS

192 (edatools-server2.iitd.edu.in:2146 (drashi25114)) - RealVNC Viewer

Applications Places System

*synthesised_netlist_time.v (~/Desktop/cmos65/90nm/verilog/DFT/best/dft_report) - gedit

File Edit View Search Tools Documents Help

Open Save Undo

*synthesised_netlist_time.v

```
SDFFRHQX1 \timeout_counter_reg[6] (.RN (rst_n), .CK (clk), .D (n_52),
    .SI (timeout_counter[5]), .SE (DFT_sen), .Q
        (timeout_counter[6]));
NOR2XL g2229(.A (n_30), .B (n_53), .Y (n_54));
AOI21XL g2230(.A0 (timeout_counter[6]), .A1 (n_49), .B0 (DFT_sdo), .Y
    (n_53));
NOR2XL g2231(.A (n_30), .B (n_51), .Y (n_52));
SDFFRHQX1 \timeout_counter_reg[5] (.RN (rst_n), .CK (clk), .D (n_50),
    .SI (timeout_counter[4]), .SE (DFT_sen), .Q
        (timeout_counter[5]));
MXI2XL g2233(.A (n_49), .B (n_46), .S0 (timeout_counter[6]), .Y
    (n_51));
NOR2XL g2234(.A (n_48), .B (n_30), .Y (n_50));
CLKINVX1 g2235(.A (n_46), .Y (n_49));
SDFFRHQX1 \timeout_counter_reg[4] (.RN (rst_n), .CK (clk), .D (n_45),
    .SI (timeout_counter[3]), .SE (DFT_sen), .Q
        (timeout_counter[4]));
AOI21XL g2237(.A0 (timeout_counter[5]), .A1 (n_42), .B0 (n_47), .Y
    (n_48));
NOR2XL g2238(.A (timeout_counter[5]), .B (n_42), .Y (n_47));
NOR2XL g2239(.A (n_44), .B (n_30), .Y (n_45));
NAND2BX1 g2240(.AN (n_42), .B (timeout_counter[5]), .Y (n_46));
SDFFRHQX1 \timeout_counter_reg[3] (.RN (rst_n), .CK (clk), .D (n_41),
    .SI (timeout_counter[2]), .SE (DFT_sen), .Q
        (timeout_counter[3]));
AOI21XL g2242(.A0 (timeout_counter[4]), .A1 (n_37), .B0 (n_43), .Y
    (n_44));
NOR2XL g2243(.A (timeout_counter[4]), .B (n_37), .Y (n_43));
NOR2XL g2244(.A (n_40), .B (n_30), .Y (n_41));
NAND2BX1 g2245(.AN (n_37), .B (timeout_counter[4]), .Y (n_42));
SDFFRHQX1 \attempt_counter_reg[0] (.RN (rst_n), .CK (clk), .D (n_39),
    .SI (DFT_sdi), .SE (DFT_sen), .Q (attempt_counter[0]));
SDFFRHQX1 \attempt_counter_reg[1] (.RN (rst_n), .CK (clk), .D (n_38),
    .SI (attempt_counter[0]), .SE (DFT_sen), .Q
```

Verilog Tab Width: 8 Ln 1, Col 1 INS

192 (edatools-server2.iitd.edu.in:2146 (drashi25114)) - RealVNC Viewer

Applications Places System

synthesised_netlist.v (~/Desktop/cmos65/90nm/pd) - gedit

File Edit View Search Tools Documents Help

Open Save Undo

*synthesised_netlist_time.v synthesised_netlist.v

```
DFFSRHQX4 \volume_cmd_r_reg[2] (.RN (rst_n), .SN (1'b1), .CK (clk),
.D (volume_cmd[2]), .Q (volume_cmd_r[2]));
DFFSRHQX4 \temp_sensor_r_reg[2] (.RN (rst_n), .SN (1'b1), .CK (clk),
.D (temp_sensor[2]), .Q (temp_sensor_r[2]));
DFFRHQX8 \volume_level_r_reg[3] (.RN (rst_n), .CK (clk), .D
(volume_level_int[3]), .Q (volume_level[3]));
DFFSRHQX4 \humidity_sensor_r_reg[0] (.RN (rst_n), .SN (1'b1), .CK
(clk), .D (humidity_sensor[0]), .Q (humidity_sensor_r[0]));
DFFRHQX8 \volume_level_r_reg[2] (.RN (rst_n), .CK (clk), .D
(volume_level_int[2]), .Q (volume_level[2]));
DFFRHQX8 \volume_level_r_reg[1] (.RN (rst_n), .CK (clk), .D
(volume_level_int[1]), .Q (volume_level[1]));
DFFRHQX8 \volume_level_r_reg[0] (.RN (rst_n), .CK (clk), .D
(volume_level_int[0]), .Q (volume_level[0]));
DFFSRHQX4 \window_manual_open_r_reg (.RN (rst_n), .SN (1'b1), .CK
(clk), .D (window_manual_open), .Q (window_manual_open_r));
DFFSRHQX4 \temp_sensor_r_reg[0] (.RN (rst_n), .SN (1'b1), .CK (clk),
.D (temp_sensor[0]), .Q (temp_sensor_r[0]));
DFFSRHQX4 \temp_sensor_r_reg[1] (.RN (rst_n), .SN (1'b1), .CK (clk),
.D (temp_sensor[1]), .Q (temp_sensor_r[1]));
DFFSRHQX4 \light_sensor_r_reg[3] (.RN (rst_n), .SN (1'b1), .CK (clk),
.D (light_sensor[3]), .Q (light_sensor_r[3]));
DFFRX1 \water_pump_on_r_reg (.RN (rst_n), .CK (clk), .D
(water_pump_on_int), .Q (UNCONNECTED3), .QN
(water_pump_on_r_370));
DFFSRHQX4 \light_sensor_r_reg[5] (.RN (rst_n), .SN (1'b1), .CK (clk),
.D (light_sensor[5]), .Q (light_sensor_r[5]));
DFFSRHQX4 \light_sensor_r_reg[7] (.RN (rst_n), .SN (1'b1), .CK (clk),
.D (light_sensor[7]), .Q (light_sensor_r[7]));
SDFFRHQX4 \temp_setpoint_r_reg[1] (.RN (rst_n), .CK (clk), .D
(temp_setpoint[1]), .SI (1'b0), .SE (1'b0), .Q
(temp_setpoint_r[1]));
SDFFRHQX4 \temp_setpoint_r_reg[3] (.RN (rst_n), .CK (clk), .D
(temp_setpoint[3]), .SI (1'b0), .SE (1'b0), .Q
```

Verilog Tab Width: 8 Ln 879, Col 30 INS

192 (edatools-server2.iitd.edu.in:2146 (drashi25114)) - RealVNC Viewer

Applications Places System

synthesised_netlist.v (~/Desktop/cmos65/90nm/pd) - gedit

File Edit View Search Tools Documents Help

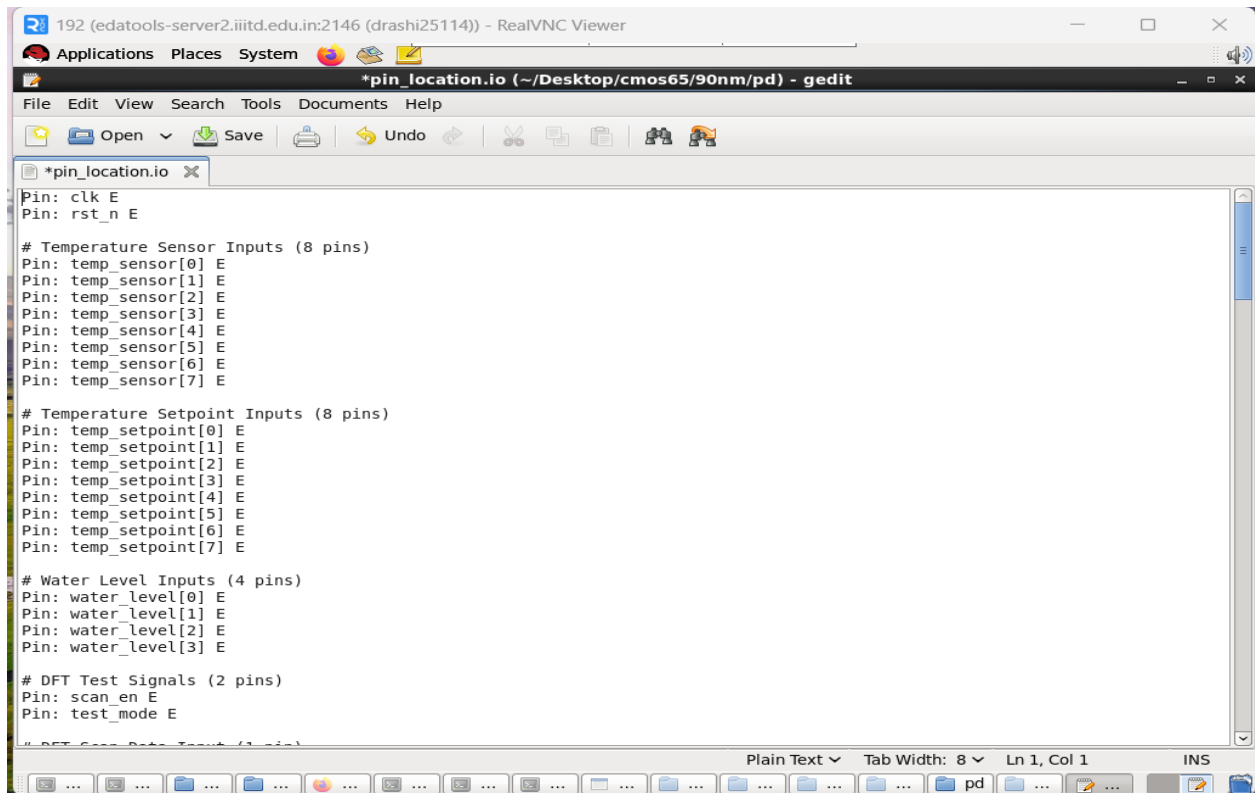
Open Save Undo

*synthesised_netlist_time.v synthesised_netlist.v

```
(temp_setpoint[3]), .SI (1'b0), .SE (1'b0), .Q
(temp_setpoint_r[3]));
DFFRHQX4 \temp_setpoint_r_reg[2] (.RN (rst_n), .CK (clk), .D
(temp_setpoint[2]), .Q (temp_setpoint_r[2]));
DFFRHQX8 \visitor_count_r_reg[1] (.RN (rst_n), .CK (clk), .D
(visitor_count_int[1]), .Q (visitor_count[1]));
DFFRHQX8 \visitor_count_r_reg[5] (.RN (rst_n), .CK (clk), .D
(visitor_count_int[5]), .Q (visitor_count[5]));
DFFRX1 \burglar_alarm_out_r_reg (.RN (rst_n), .CK (clk), .D
(burglar_alarm_out_int), .Q (UNCONNECTED4), .QN
(burglar_alarm_out_r_372));
DFFRX1 \heater_on_r_reg (.RN (rst_n), .CK (clk), .D (heater_on_int), .Q
(UNCONNECTED5), .QN (heater_on_r_368));
DFFRX1 \cooler_on_r_reg (.RN (rst_n), .CK (clk), .D (cooler_on_int), .Q
(UNCONNECTED6), .QN (cooler_on_r_369));
DFFRHQX8 \window_state_r_reg[1] (.RN (rst_n), .CK (clk), .D
(window_state_int[1]), .Q (window_state[1]));
DFFRHQX8 \visitor_count_r_reg[2] (.RN (rst_n), .CK (clk), .D
(visitor_count_int[2]), .Q (visitor_count[2]));
DFFRHQX8 \visitor_count_r_reg[3] (.RN (rst_n), .CK (clk), .D
(visitor_count_int[3]), .Q (visitor_count[3]));
DFFRHQX8 \visitor_count_r_reg[0] (.RN (rst_n), .CK (clk), .D
(visitor_count_int[0]), .Q (visitor_count[0]));
DFFRHQX8 \visitor_count_r_reg[4] (.RN (rst_n), .CK (clk), .D
(visitor_count_int[4]), .Q (visitor_count[4]));
DFFRHQX8 \window_state_r_reg[0] (.RN (rst_n), .CK (clk), .D
(window_state_int[0]), .Q (window_state[0]));
DFFRHQX8 \visitor_count_r_reg[7] (.RN (rst_n), .CK (clk), .D
(visitor_count_int[7]), .Q (visitor_count[7]));
DFFRHQX8 \visitor_count_r_reg[6] (.RN (rst_n), .CK (clk), .D
(visitor_count_int[6]), .Q (visitor_count[6]));
endmodule
```

Verilog Tab Width: 8 Ln 1, Col 1 INS

PIN_LOCATION.IO



```
192 (edatools-server2.iitd.edu.in:2146 (drashi25114)) - RealVNC Viewer
Applications Places System
*pin_location.io (~/.Desktop/cmos65/90nm/pd) - gedit
File Edit View Search Tools Documents Help
Open Save Undo
*pin_location.io
Pin: clk E
Pin: rst_n E

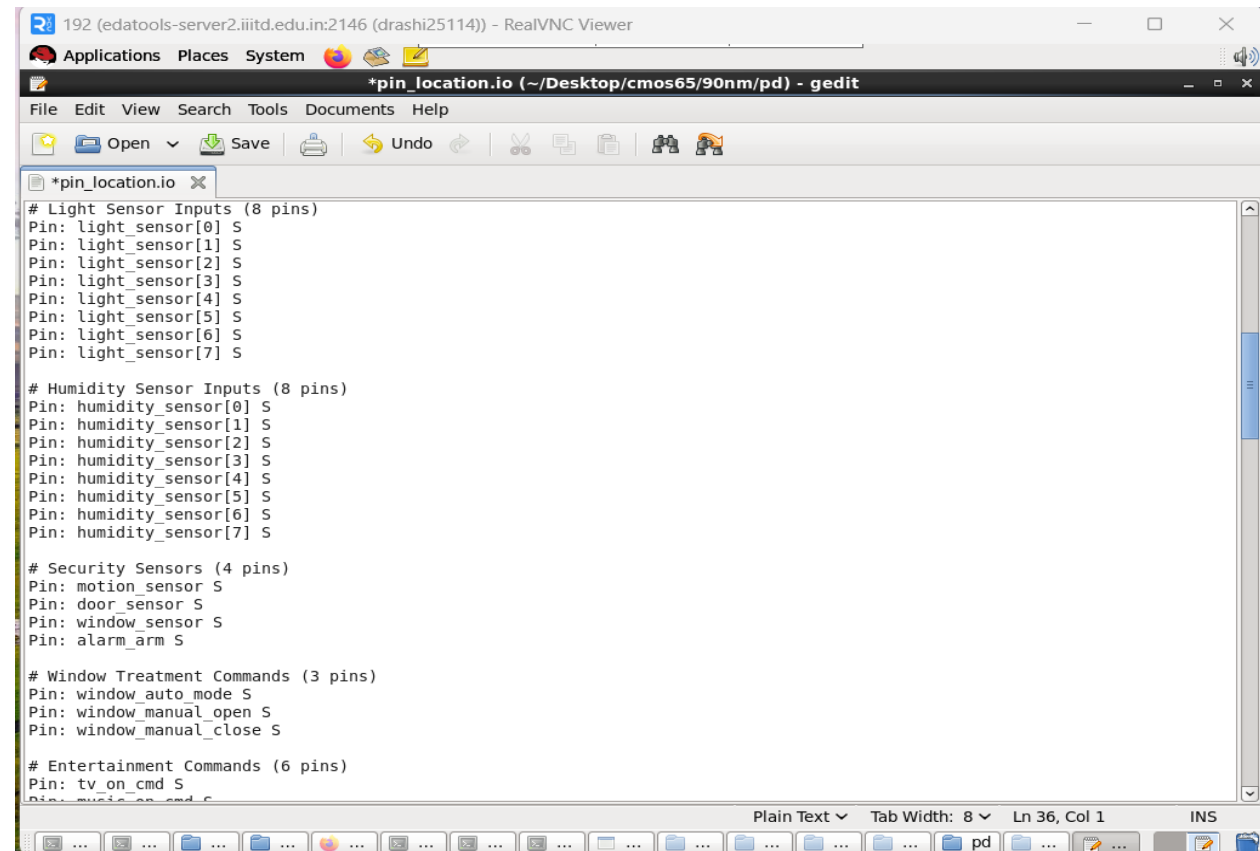
# Temperature Sensor Inputs (8 pins)
Pin: temp_sensor[0] E
Pin: temp_sensor[1] E
Pin: temp_sensor[2] E
Pin: temp_sensor[3] E
Pin: temp_sensor[4] E
Pin: temp_sensor[5] E
Pin: temp_sensor[6] E
Pin: temp_sensor[7] E

# Temperature Setpoint Inputs (8 pins)
Pin: temp_setpoint[0] E
Pin: temp_setpoint[1] E
Pin: temp_setpoint[2] E
Pin: temp_setpoint[3] E
Pin: temp_setpoint[4] E
Pin: temp_setpoint[5] E
Pin: temp_setpoint[6] E
Pin: temp_setpoint[7] E

# Water Level Inputs (4 pins)
Pin: water_level[0] E
Pin: water_level[1] E
Pin: water_level[2] E
Pin: water_level[3] E

# DFT Test Signals (2 pins)
Pin: scan_en E
Pin: test_mode E

# DFT Scan Data Inputs (1 pin)
```



```
192 (edatools-server2.iitd.edu.in:2146 (drashi25114)) - RealVNC Viewer
Applications Places System
*pin_location.io (~/.Desktop/cmos65/90nm/pd) - gedit
File Edit View Search Tools Documents Help
Open Save Undo
*pin_location.io
# Light Sensor Inputs (8 pins)
Pin: light_sensor[0] S
Pin: light_sensor[1] S
Pin: light_sensor[2] S
Pin: light_sensor[3] S
Pin: light_sensor[4] S
Pin: light_sensor[5] S
Pin: light_sensor[6] S
Pin: light_sensor[7] S

# Humidity Sensor Inputs (8 pins)
Pin: humidity_sensor[0] S
Pin: humidity_sensor[1] S
Pin: humidity_sensor[2] S
Pin: humidity_sensor[3] S
Pin: humidity_sensor[4] S
Pin: humidity_sensor[5] S
Pin: humidity_sensor[6] S
Pin: humidity_sensor[7] S

# Security Sensors (4 pins)
Pin: motion_sensor S
Pin: door_sensor S
Pin: window_sensor S
Pin: alarm_arm S

# Window Treatment Commands (3 pins)
Pin: window_auto_mode S
Pin: window_manual_open S
Pin: window_manual_close S

# Entertainment Commands (6 pins)
Pin: tv_on_cmd S
Pin: music_on_cmd S
```

192 (edatools-server2.iitd.edu.in:2146 (drashi25114)) - RealVNC Viewer

Applications Places System

*pin_location.io (~/Desktop/cmos65/90nm/pd) - gedit

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Open Save Undo

*pin_location.io X

```
Pin: window_manual_close S

# Entertainment Commands (6 pins)
Pin: tv_on_cmd S
Pin: music_on_cmd S
Pin: volume_cmd[0] S
Pin: volume_cmd[1] S
Pin: volume_cmd[2] S
Pin: volume_cmd[3] S
|

# Visitor Counter Inputs (2 pins)
Pin: entry_sensor W
Pin: exit_sensor W

# Appliance Control Commands (3 pins)
Pin: ac_cmd W
Pin: fan_cmd W
Pin: light_cmd W

# Door Lock Commands (2 pins)
Pin: door_lock_cmd W
Pin: door_unlock_cmd W

# Door Lock PIN Input (4 pins)
Pin: pin_input[0] W
Pin: pin_input[1] W
Pin: pin_input[2] W
Pin: pin_input[3] W

# DFT Scan Input (1 pin)
Pin: DFT_sdi_2 W

# System Status Outputs (8 pins)
Pin: system_status[0] W
```

Plain Text Tab Width: 8 Ln 75, Col 1 INS

192 (edatools-server2.iitd.edu.in:2146 (drashi25114)) - RealVNC Viewer

Applications Places System

*pin_location.io (~/Desktop/cmos65/90nm/pd) - gedit

File Edit View Search Tools Documents Help

Open Save Undo

*pin_location.io X

```
Pin: DFT_sdi_2 W

# System Status Outputs (8 pins)
Pin: system_status[0] W
Pin: system_status[1] W
Pin: system_status[2] W
Pin: system_status[3] W
Pin: system_status[4] W
Pin: system_status[5] W
Pin: system_status[6] W
Pin: system_status[7] W

# LED Brightness Outputs (8 pins)
Pin: led_brightness[0] W
Pin: led_brightness[1] W
Pin: led_brightness[2] W
Pin: led_brightness[3] W
Pin: led_brightness[4] W
Pin: led_brightness[5] W
Pin: led_brightness[6] W
Pin: led_brightness[7] W

# DFT Scan Output (1 pin)
Pin: DFT_sdo_1 W
|

# Temperature Controller Outputs (2 pins)
Pin: heater_on N
Pin: cooler_on N

# Water Controller Outputs (2 pins)
Pin: water_pump_on N
Pin: water_valve_close N

# Security Alarm Output (1 pin)
```

Plain Text Tab Width: 8 Ln 120, Col 1 INS

192 (edatools-server2.iiitd.edu.in:2146 (drashi25114)) - RealVNC Viewer

Applications Places System

*pin_location.io (~/Desktop/cmos65/90nm/pd) - gedit

File Edit View Search Tools Documents Help

Open Save Undo

*pin_location.io

```
# Security Alarm Output (1 pin)
Pin: burglar_alarm_out N

# Visitor Counter Outputs (8 pins)
Pin: visitor_count[0] N
Pin: visitor_count[1] N
Pin: visitor_count[2] N
Pin: visitor_count[3] N
Pin: visitor_count[4] N
Pin: visitor_count[5] N
Pin: visitor_count[6] N
Pin: visitor_count[7] N

# Window Treatment Outputs (2 pins)
Pin: window_state[0] N
Pin: window_state[1] N

# Entertainment System Outputs (6 pins)
Pin: tv_on N
Pin: music_on N
Pin: volume_level[0] N
Pin: volume_level[1] N
Pin: volume_level[2] N
Pin: volume_level[3] N

# Appliance Outputs (3 pins)
Pin: ac_on N
Pin: fan_on N
Pin: lights_on N

# Door Lock Outputs (3 pins)
Pin: door_locked N
Pin: lock_status[0] N
```

Plain Text Tab Width: 8 Ln 120, Col 1 INS

pd

192 (edatools-server2.iiitd.edu.in:2146 (drashi25114)) - RealVNC Viewer

Applications Places System

*pin_location.io (~/Desktop/cmos65/90nm/pd) - gedit

File Edit View Search Tools Documents Help

Open Save Undo

*pin_location.io

```
Pin: visitor_count[4] N
Pin: visitor_count[5] N
Pin: visitor_count[6] N
Pin: visitor_count[7] N

# Window Treatment Outputs (2 pins)
Pin: window_state[0] N
Pin: window_state[1] N

# Entertainment System Outputs (6 pins)
Pin: tv_on N
Pin: music_on N
Pin: volume_level[0] N
Pin: volume_level[1] N
Pin: volume_level[2] N
Pin: volume_level[3] N

# Appliance Outputs (3 pins)
Pin: ac_on N
Pin: fan_on N
Pin: lights_on N

# Door Lock Outputs (3 pins)
Pin: door_locked N
Pin: lock_status[0] N
Pin: lock_status[1] N

# HVAC Humidity Outputs (2 pins)
Pin: humidifier_on N
Pin: dehumidifier_on N

# DFT Scan Output (1 pin)
Pin: DFT_sdo_2 N
```

Plain Text Tab Width: 8 Ln 120, Col 1 INS

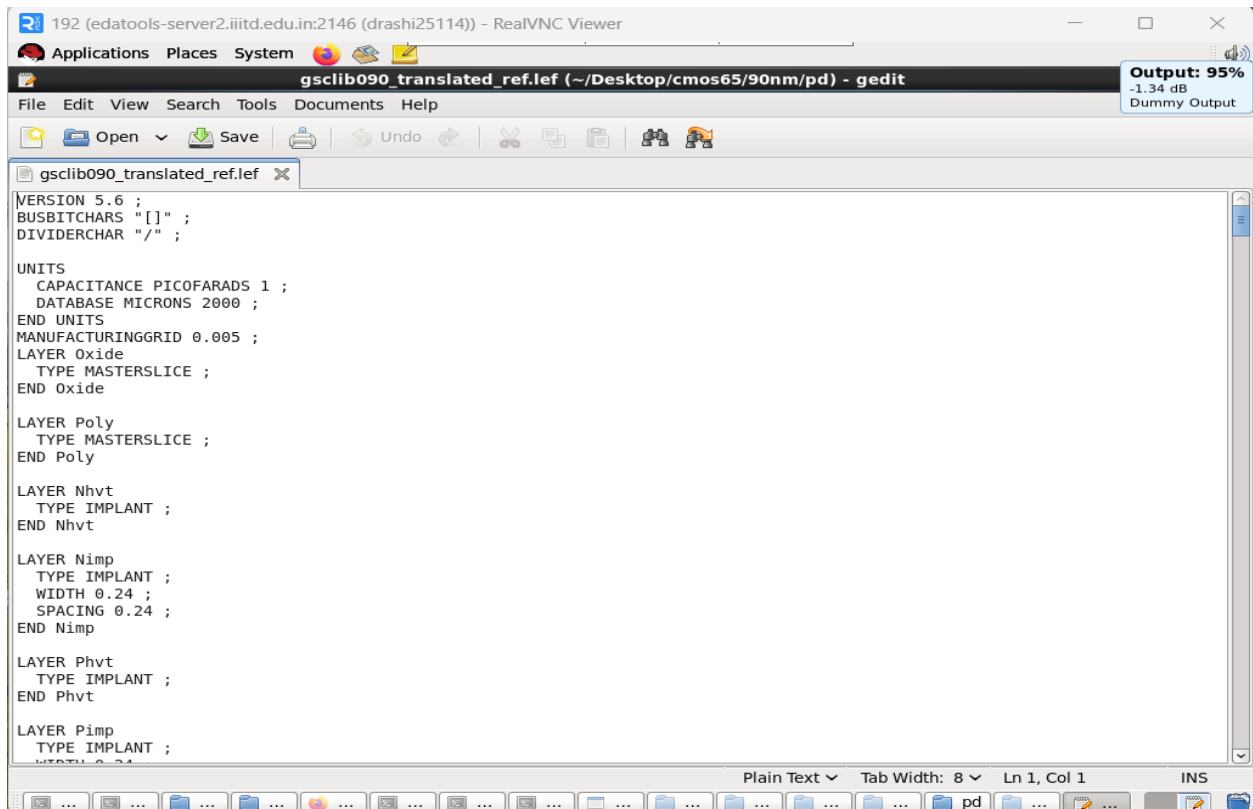
pd

The pin_position.io file defines a total of 116 pins for the Home Automation Top Module.

These pins are symmetrically distributed across four sides of the chip/package — North, South, East, and West, with 29 pins per side, ensuring layout symmetry, balanced routing, and minimal signal skew during physical design.

Side	Functional Group Summary	Total Pins
East	Clock, Reset, Temperature & Water Sensors, DFT Inputs	29
South	Light, Humidity, Security, and Entertainment Commands	29
West	Command Inputs, Status Outputs, DFT I/O	29
North	Control Outputs (HVAC, Water, Security, Entertainment)	29
Total		116 Pins

LEF FILE



```
VERSION 5.6 ;
BUSBITCHARS "[" ;
DIVIDERCHAR "/" ;

UNITS
  CAPACITANCE PICO FARADS 1 ;
  DATABASE MICRONS 2000 ;
END UNITS
MANUFACTURINGGRID 0.005 ;
LAYER Oxide
  TYPE MASTERSLICE ;
END Oxide

LAYER Poly
  TYPE MASTERSLICE ;
END Poly

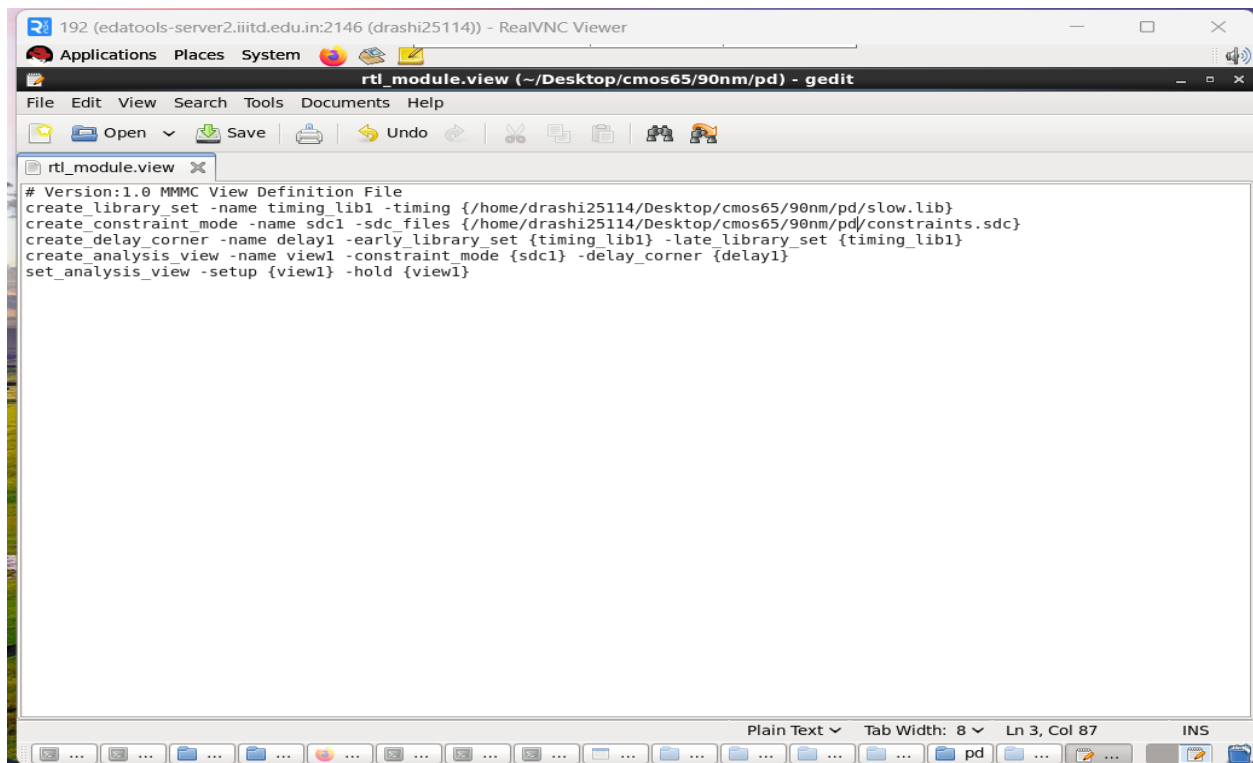
LAYER Nhvt
  TYPE IMPLANT ;
END Nhvt

LAYER Nimp
  TYPE IMPLANT ;
  WIDTH 0.24 ;
  SPACING 0.24 ;
END Nimp

LAYER Phvt
  TYPE IMPLANT ;
END Phvt

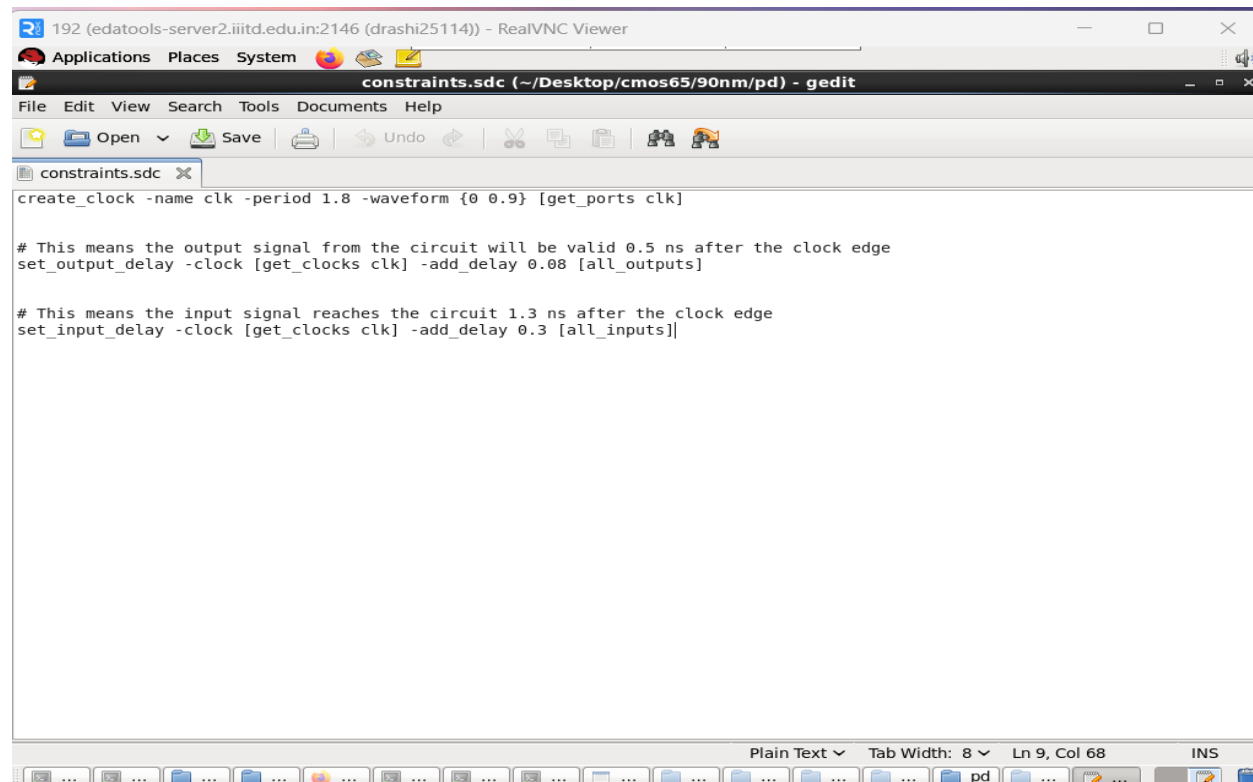
LAYER Pimp
  TYPE IMPLANT ;
  WIDTH 0.24 ;
END Pimp
```

RTL module.view



```
# Version:1.0 MMMC View Definition File
create_library_set -name timing_lib1 -timing {/home/drashi25114/Desktop/cmos65/90nm/pd/slow.lib}
create_constraint_mode -name sdc1 -sdc_files {/home/drashi25114/Desktop/cmos65/90nm/pd/constraints.sdc}
create_delay_corner -name delay1 -early_library_set {timing_lib1} -late_library_set {timing_lib1}
create_analysis_view -name view1 -constraint_mode {sdc1} -delay_corner {delay1}
set_analysis_view -setup {view1} -hold {view1}
```

Constraints.sdc



```
create_clock -name clk -period 1.8 -waveform {0 0.9} [get_ports clk]

# This means the output signal from the circuit will be valid 0.5 ns after the clock edge
set_output_delay -clock [get_clocks clk] -add_delay 0.08 [all_outputs]

# This means the input signal reaches the circuit 1.3 ns after the clock edge
set_input_delay -clock [get_clocks clk] -add_delay 0.3 [all_inputs]
```

How Metal Layers Affect Timing

- Higher metal layers (e.g., M6, M7) are thicker and have lower resistance. Assigning long, timing-critical nets to these layers results in reduced RC delay, improving timing and reducing the number of required buffers along the path.
- Lower metal layers (e.g., M1, M2) have increased resistance and capacitance, leading to higher signal delay if used for long or critical nets.
- Increased spacing in upper metal layers can further minimize sidewall capacitance, lessening performance degradation for the fastest paths.
- Too much congestion in upper layers may force critical nets onto lower layers, worsening timing.
- The choice of metal layer directly determines the resistance (R) and capacitance (C) for a given interconnect, impacting the RC time constant and thus the arrival time shown in timing reports.

Utilization Factor-0.5

After Placement reports:

1. Area

The area report provides module-wise details about cell counts, flop counts, combinational area, and total area for your physical design after placement and optimization. When referencing 0.5 utilization, this indicates that half (50%) of the available placement rows are occupied by standard cells, and the remaining 50% are left as whitespace to accommodate routing, buffering, and congestion avoidance

Meaning of 0.5 Utilization

- Utilization Ratio: In physical design, 0.5 utilization means that only 50% of the core area is filled with standard cells; the rest is whitespace. This is intentional to:
 - Prevent routing congestion.
 - Allow space for buffering and clock tree synthesis.

PD.tcl				area.rpt		
Hinst Name	Module Name	Inst Count		Total Area		Buffer
Inverter	Combinational	Flop		Latch	Clock Gate	
Macro	Physical					
HA		446		6074.879		0.000
261.887	1083.881	4729.111		0.000	0.000	
0.000	0.000					
u_appliances	appliances_controller	3		74.933		0.000
0.000	0.000	74.933		0.000	0.000	
0.000	0.000					
u_burglar	burglar_alarm	49		425.378		0.000
11.354	164.247	249.777		0.000	0.000	
0.000	0.000					
u_door_lock	door_lock_controller	72		632.012		0.000
20.436	211.932	399.643		0.000	0.000	
0.000	0.000					
u_entertainment	entertainment_system	11		187.711		0.000
0.000	33.304	154.408		0.000	0.000	
0.000	0.000					
u_humidity	humidity_controller	8		74.933		0.000
2.271	22.707	49.955		0.000	0.000	
0.000	0.000					
u_lum_ctrl	luminosity_controller	16		217.987		0.000
18.166	0.000	199.822		0.000	0.000	
0.000	0.000					
u_temp_ctrl	temperature_controller	92		452.626		0.000
46.171	351.959	54.497		0.000	0.000	
0.000	0.000					
u_visitor	visitor_counter	61		495.769		0.000
15.895	218.744	261.130		0.000	0.000	
0.000	0.000					
u_water_ctrl	water_level_controller	6		69.635		0.000
0.000	19.679	49.955		0.000	0.000	

PD.tcl				area.rpt		
Hinst Name	Module Name	Inst Count		Total Area		Buffer
Inverter	Combinational	Flop		Latch	Clock Gate	
Macro	Physical					
HA		446		6074.879		0.000
261.887	1083.881	4729.111		0.000	0.000	
0.000	0.000					
u_appliances	appliances_controller	3		74.933		0.000
0.000	0.000	74.933		0.000	0.000	
0.000	0.000					
u_burglar	burglar_alarm	49		425.378		0.000
11.354	164.247	249.777		0.000	0.000	
0.000	0.000					
u_door_lock	door_lock_controller	72		632.012		0.000
20.436	211.932	399.643		0.000	0.000	
0.000	0.000					
u_entertainment	entertainment_system	11		187.711		0.000
0.000	33.304	154.408		0.000	0.000	
0.000	0.000					
u_humidity	humidity_controller	8		74.933		0.000
2.271	22.707	49.955		0.000	0.000	
0.000	0.000					
u_lum_ctrl	luminosity_controller	16		217.987		0.000
18.166	0.000	199.822		0.000	0.000	
0.000	0.000					
u_temp_ctrl	temperature_controller	92		452.626		0.000
46.171	351.959	54.497		0.000	0.000	
0.000	0.000					
u_visitor	visitor_counter	61		495.769		0.000
15.895	218.744	261.130		0.000	0.000	
0.000	0.000					
u_water_ctrl	water_level_controller	6		69.635		0.000
0.000	19.679	49.955		0.000	0.000	
0.000	0.000					
u_window	window_treatment	15		113.535		0.000
2.271	61.309	49.955		0.000	0.000	
0.000	0.000					

- Inverters: The total inverter area is 261.887, serving as distributed signal inverters across various modules.
- Buffers: Total area for buffers is 0.000, so no explicit buffer cells are present or counted separately in this report.
- Flip-Flops: Flip-flops make up 4729.111 of the area, dominant in storage and sequential logic paths.
- Combinational Cells (Other): Combinational (non-buffer/inverter) logic cells consume 1083.881, supporting all module logic functions.
- Latches: Area for latches is 446, used for level-sensitive storage and pipeline timing.
- Macros: Macro area is 0.000, indicating no memory or hard macros in this section.
- Total Standard Cell Area: Cumulative standard cell area is 6074.879, showing the total silicon used for all placed logic and storage

2. Power

power.rpt					
~/work/90nm/VDF_project_new/Physical_design/placement_reports/power					
light_sensor_r_reg[0]	0.01973	0.00117	0.02112	0.0002171	SDFFRHQX4
light_sensor_r_reg[1]	0.01952	0.001328	0.02106	0.0002171	SDFFRHQX4
humidity_sensor_r_reg[3]	0.01966	0.001057	0.02093	0.0002171	SDFFRHQX4
water_level_r_reg[2]	0.01967	0.0009878	0.02087	0.0002171	SDFFRHQX4
alarm_arm_r_reg	0.01983	0.0007831	0.02083	0.0002171	SDFFRHQX4
water_level_r_reg[3]	0.01967	0.0009359	0.02082	0.0002171	SDFFRHQX4
humidity_sensor_r_reg[5]	0.01967	0.0009218	0.02081	0.0002171	SDFFRHQX4
door_sensor_r_reg	0.01975	0.0007639	0.02073	0.0002171	SDFFRHQX4
temp_sensor_r_reg[0]	0.01967	0.0008298	0.02072	0.0002171	SDFFRHQX4
light_sensor_r_reg[0]	0.01941	0.001076	0.0207	0.0002171	SDFFRHQX4
u_burglar/delay_counter_reg[1]	0.01854	0.002096	0.0207	0.0001526	SDFFRHQX1
water_level_r_reg[0]	0.01968	0.000767	0.02066	0.0002171	SDFFRHQX4
temp_sensor_r_reg[1]	0.01968	0.0007628	0.02066	0.0002171	SDFFRHQX4
window_sensor_r_reg	0.01968	0.0007491	0.02064	0.0002171	SDFFRHQX4
pin_input_r_reg[2]	0.0197	0.0007207	0.02064	0.0002171	SDFFRHQX4
humidity_sensor_r_reg[2]	0.01969	0.0006526	0.02056	0.0002171	SDFFRHQX4
u_entertainment/music_on_reg	0.01903	0.001369	0.02055	0.0001526	SDFFRHQX1
motion_sensor_r_reg	0.0197	0.0006106	0.02053	0.0002171	SDFFRHQX4
humidity_sensor_r_reg[6]	0.0197	0.0006015	0.02052	0.0002171	SDFFRHQX4
water_level_r_reg[1]	0.0197	0.0005926	0.02051	0.0002171	SDFFRHQX4
humidity_sensor_r_reg[7]	0.01971	0.0005466	0.02048	0.0002171	SDFFRHQX4
humidity_sensor_r_reg[4]	0.01972	0.0004765	0.02042	0.0002171	SDFFRHQX4
u_lum_ctrl/led_brightness_reg[6]	0.01843	0.001765	0.02034	0.0001526	SDFFRHQX1
light_cmd_r_reg	0.01917	0.0008769	0.02026	0.0002171	SDFFRHQX4
u_burglar/delay_counter_reg[0]	0.01816	0.001721	0.02003	0.0001526	SDFFRHQX1
u_appliances/fan_on_reg	0.01843	0.001361	0.01994	0.0001526	SDFFRHQX1
u_humidity/humidifier_on_reg	0.01855	0.001232	0.01994	0.0001526	SDFFRHQX1
u_burglar/alarm_out_reg	0.01814	0.001641	0.01994	0.0001526	SDFFRHQX1
u_door_lock/attempt_counter_reg[2]	0.01794	0.001592	0.01969	0.0001526	SDFFRHQX1
pin_input_r_reg[0]	0.01868	0.0006449	0.01955	0.0002171	SDFFRHQX4
u_water_ctrl/valve_close_reg	0.0182	0.001144	0.01949	0.0001526	SDFFRHQX1
u_lum_ctrl/led_brightness_reg[2]	0.01802	0.001311	0.01948	0.0001526	SDFFRHQX1
u_appliances/ac_on_reg	0.01827	0.0007802	0.01936	0.0001526	SDFFRHQX1
u_lum_ctrl/led_brightness_reg[0]	0.01796	0.001107	0.01922	0.0001526	SDFFRHQX1
u_lum_ctrl/led_brightness_reg[4]	0.01768	0.001355	0.01919	0.0001526	SDFFRHQX1
u_lum_ctrl/led_brightness_reg[1]	0.0178	0.001073	0.01903	0.0001526	SDFFRHQX1
u_burglar/delay_counter_reg[3]	0.01693	0.001938	0.01902	0.0001526	SDFFRHQX1
u_burglar/delay_counter_reg[7]	0.01704	0.001362	0.01855	0.0001526	SDFFRHQX1
pin_input_r_reg[1]	0.01783	0.0005091	0.01855	0.0002171	SDFFRHQX4
u_burglar/delay_counter_reg[2]	0.01657	0.001817	0.01854	0.0001526	SDFFRHQX1
g50	0.01794	0.0003395	0.01838	0.0001012	CLKINVX8
u_door_lock/timeout_counter_reg[7]	0.01678	0.001387	0.01832	0.0001526	SDFFRHQX1
g51	0.01799	0.0002197	0.01832	0.0001012	CLKINVX8
u_burglar/delay_counter_reg[4]	0.01644	0.001687	0.01828	0.0001526	SDFFRHQX1
u_visitor/exit_prev_reg	0.01692	0.0008649	0.01794	0.0001526	SDFFRHQX1
u_door_lock/timeout_counter_reg[3]	0.01634	0.001405	0.0179	0.0001526	SDFFRHQX1
u_burglar/delay_counter_reg[8]	0.01653	0.00121	0.0179	0.0001526	SDFFRHQX1
u_water_ctrl/pump_on_reg	0.01639	0.001262	0.0178	0.0001526	SDFFRHQX1
pin_input_r_reg[3]	0.0168	0.0003517	0.01737	0.0002171	SDFFRHQX4
u_door_lock/timeout_counter_reg[2]	0.01583	0.001299	0.01728	0.0001526	SDFFRHQX1
g48	0.01671	0.0004605	0.01727	0.0001012	CLKINVX8
g49	0.01683	0.0001779	0.01711	0.0001012	CLKINVX8
u_visitor/entry_prev_reg	0.01506	0.0005635	0.01577	0.0001526	SDFFRHQX1
u_door_lock/timeout_counter_reg[6]	0.01422	0.001173	0.01555	0.0001526	SDFFRHQX1
u_door_lock/timeout_counter_reg[5]	0.01348	0.000996	0.01463	0.0001526	SDFFRHQX1
u_door_lock/timeout_counter_reg[4]	0.01304	0.0007354	0.01393	0.0001526	SDFFRHQX1
g73	0.01318	0.0006113	0.0139	0.0001012	CLKINVX8
g72	0.01333	0.000237	0.01367	0.0001012	CLKINVX8
g70	0.01251	0.000291	0.0129	0.0001012	CLKINVX8
g46	0.01208	0.0006601	0.01284	0.0001012	CLKINVX8
g71	0.01257	0.0001554	0.01282	0.0001012	CLKINVX8
g47	0.01215	0.0004997	0.01275	0.0001012	CLKINVX8
g74	0.01091	0.001206	0.01221	0.0001012	CLKINVX8
g76	0.01088	0.001146	0.01213	0.0001012	CLKINVX8
g44	0.01038	0.001176	0.01165	0.0001012	CLKINVX8
g75	0.01135	0.0001403	0.01159	0.0001012	CLKINVX8
g77	0.01132	0.000115	0.01153	0.0001012	CLKINVX8
g45	0.01081	0.0001377	0.01105	0.0001012	CLKINVX8
u_visitor/g1101	0.003267	0.0003012	0.006296	1.816e-05	NOR2X1
u_visitor/g1092	0.00577	0.0000048	0.006196	2.093e-05	NOR2X1
lock_status_r_reg[0]	0.00729	0.001153	0.00374	0.0001009	SDFFRHQX8
u_door_lock/g2	0.0001843	0.0001415	0.0003608	3.499e-05	NAND2BX1
u_temp_ctrl/g1694	0.0001758	0.0001507	0.0003358	9.289e-06	NAND2XL
u_temp_ctrl/g1739	0.0002293	3.252e-05	0.0002837	2.186e-05	NOR3BX1
u_visitor/g1068	0.0001568	0.0001036	0.0002727	1.236e-05	NOR2XL
u_temp_ctrl/g1685	0.0001166	9.264e-05	0.0002191	9.93e-06	NAND3XL
u_door_lock/g2257	0.0001479	4.882e-05	0.0002091	1.236e-05	NOR2XL
u_burglar/g710	8.388e-05	7.983e-05	0.0002063	4.256e-05	OR2X1
u_temp_ctrl/g1697	6.454e-05	0.0001218	0.0001956	9.289e-06	NAND2XL
u_door_lock/g2258	6.335e-05	4.702e-05	0.0001454	3.499e-05	NAND2BX1
u_visitor/g1057	3.632e-05	9.832e-05	0.0001439	9.289e-06	NAND2XL
u_temp_ctrl/g1665	7.587e-05	5.744e-05	0.0001438	1.046e-05	NAND4XL
u_door_lock/g2243	9.217e-05	2.479e-05	0.0001293	1.236e-05	NOR2XL
u_burglar/g714	5.963e-05	2.93e-05	0.0001019	1.301e-05	INVX1
u_burglar/g715	3.842e-05	5.198e-05	0.0001009	1.046e-05	NAND4XL
u_burglar/g701	6.061e-05	2.464e-05	9.761e-05	1.236e-05	NOR2XL
u_door_lock/g2238	4.734e-05	1.232e-05	7.201e-05	1.236e-05	NOR2XL
u_door_lock/g2245	1.419e-05	1.313e-05	6.23e-05	3.499e-05	NAND2BX1
u_burglar/g702	1.199e-05	1.379e-05	6.077e-05	3.499e-05	NAND2BX1
u_burglar/g689	1.077e-05	7.985e-06	5.374e-05	3.499e-05	NAND2BX1
u_door_lock/g2240	9.502e-06	6.93e-06	5.142e-05	3.499e-05	NAND2BX1
u_burglar/g705	4.29e-06	2.24e-05	3.97e-05	1.301e-05	CLKINVX1
u_burglar/g692	3.327e-06	2.012e-06	3.953e-05	3.419e-05	NAND3BX1
u_burglar/g713	1.166e-05	7.607e-06	3.337e-05	1.41e-05	OAI211X1
u_visitor/g1082	6.989e-06	9.928e-06	2.621e-05	9.289e-06	NAND2XL
u_door_lock/g2235	6.434e-06	4.677e-06	2.413e-05	1.301e-05	CLKINVX1
u_burglar/g678	7.926e-06	3.299e-06	2.358e-05	1.236e-05	NOR2XL
u_burglar/g707	5.825e-06	4.276e-06	2.246e-05	1.236e-05	NOR2XL
u_burglar/g684	2.293e-06	1.056e-06	1.571e-05	1.236e-05	NOR2XL
u_burglar/g703	7.44e-07	1.878e-07	1.329e-05	1.236e-05	NOR2XL

Total (	446 of	446)	4.2	0.5166	
Total Capacitance			4.047e-12 F	4.756	0.03948
Power Density			*** No Die Area ***		

Total Power: 4.756mW

Internal Power: 4.2mW (88.3%)

Switching Power: 0.5166mW (10.9%)

Leakage Power: 0.03948mW (0.8%)

3. Timing Reports:

a. Setup (Late time)

Instance	Arc	Cell	Delay	Arrival Time	Required Time
u_visitor/count_reg[5]	CK ^			0.000	-0.325
u_visitor/count_reg[5]	CK ^ -> Q ^	SDFFRHQX2	0.456	0.456	0.131
u_visitor/g1096	B ^ -> Y v	NAND2XL	0.123	0.579	0.254
u_visitor/g1085	C v -> Y ^	NOR3X1	0.144	0.723	0.398
u_visitor/g1082	B ^ -> Y v	NAND2XL	0.152	0.875	0.550
u_visitor/g1080	C v -> Y ^	NAND3BX1	0.216	1.092	0.767
u_visitor/g1078	A ^ -> Y v	CLKINVX1	0.120	1.212	0.887
u_visitor/g1075	B v -> Y ^	NOR2X1	0.220	1.432	1.107

Arrival Time: Actual time when data reaches the destination D pin.

Required Time: Latest permissible arrival time for correct capture.

Slack: Difference between required and arrival times.

Formula (for setup): $\text{Slack} = \text{Required Time} - \text{Arrival Time}$

Slack = -0.325

b. Hold (Early time)

Instance	Arc	Cell	Delay	Arrival Time	Required Time
light_sensor_r_reg[7]	light_sensor[7] v	SDFFRHQX4	0.000	0.300	0.222
light_sensor_r_reg[7]	D v			0.300	0.222

Arrival Time: Actual time when new data reaches the destination pin.

Required Time: Hold requirement calculated from clock and library values.

Slack: Difference between arrival and required time.

Formula (for hold): $\text{Slack} = \text{Arrival Time} - \text{Required Time}$

Required = 0.222 ns, Arrival = 0.300 ns → Slack = +0.078 ns → Hold met

CTS reports:

1. Area:

area_post_pnr2.rpt					
~/work/90nm/VDF_project_new/Physical_design/cts_reports/area					
Inst Name	Module Name	Inst Count	Total Area	Buffer	
Inverter	Combinational	Flop	Latch	Clock Gate	
Macro	Physical				
HA		469	6264.104	110.507	
168.032	1243.587	4741.979	0.000	0.000	
0.000	0.000				
u_appliances	appliances_controller	3	74.933	0.000	0.000
0.000	0.000	74.933	0.000	0.000	
0.000	0.000				
u_burglar	burglar_alarm	49	436.731	0.000	0.000
11.354	175.601	249.777	0.000	0.000	
0.000	0.000				
u_door_lock	door_lock_controller	73	648.663	0.000	0.000
20.436	228.584	399.643	0.000	0.000	
0.000	0.000				
u_entertainment	entertainment_system	11	187.711	0.000	0.000
0.000	33.304	154.408	0.000	0.000	
0.000	0.000				
u_humidity	humidity_controller	8	74.933	0.000	0.000
2.271	22.707	49.955	0.000	0.000	
0.000	0.000				
u_lum_ctrl	luminescence_controller	16	217.987	0.000	0.000
18.166	0.000	199.822	0.000	0.000	
0.000	0.000				
u_temp_ctrl	temperature_controller	100	567.675	0.000	0.000
56.011	457.168	54.497	0.000	0.000	
0.000	0.000				
u_visitor	visitor_counter	63	544.968	0.000	4.541
21.193	245.236	273.998	0.000	0.000	
0.000	0.000				
u_water_ctrl	water_level_controller	6	69.635	0.000	0.000
0.000	19.679	49.955	0.000	0.000	
0.000	0.000				
u_window	window_treatment	15	113.535	0.000	

- Inverters: The inverter area is 168.032, used extensively for logic inversion and signal restoration throughout modules.
- Buffers: Buffers occupy a total area of 110.507, highlighting significant buffer insertion for drive-strengthening, likely after clock tree synthesis or optimization.
- Flip-Flops: The reported area for flip-flops is 4741.979, forming the backbone of sequential logic in the design.
- Combinational Cells (Other): Combinational cells excluding buffers/inverters occupy 1243.587 of the area, implementing the majority of the main logic.
- Latches: Latch area is 469, used for level-sensitive timing elements and pipeline holding stages.
- Macros: The macro area is 0.000, so no large black-box or embedded memory blocks are included here.
- Total Standard Cell Area: Summed, the total reported standard cell area is 6264.104, representing full silicon utilization of all placed cell

2. Timing Reports:

Setup:

```
#####
# Generated by: Cadence Innovus 20.10-p004.1
# OS: Linux x86_64 (Host ID edatools-server1.iiitd.edu.in)
# Generated on: Sun Nov 9 19:40:22 2025
# Design: HA
# Command: report_timing -late -max_paths 100 > cts_reports/timing/timing_post_PnR_late.txt
#####
Path 1: MET Setup Check with Pin u_temp_ctrl/cooler_on_reg/CK
Endpoint: u_temp_ctrl/cooler_on_reg/D (^) checked with leading edge of 'clk'
Beginpoint: temp_setpoint_r_reg[3]/Q (^) triggered by leading edge of 'clk'
Path Groups: {clk}
Analysis View: view1
Other End Arrival Time -0.001
- Setup 0.217
+ Phase Shift 1.800
= Required Time 1.581
- Arrival Time 1.553
= Slack Time 0.028
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) -0.001
= Beginpoint Arrival Time -0.001
#####
```

Instance	Arc	Cell	Delay	Arrival Time	Required Time
temp_setpoint_r_reg[3]	CK ^			-0.001	0.027
temp_setpoint_r_reg[3]	CK ^ -> Q ^	SDFFRHQX4	0.412	0.411	0.439
FE_OPC1 temp_setpoint_r_3	A ^ -> Y ^	BUFEX6	0.120	0.531	0.559
u_temp_ctrl/g1709	B ^ -> Y v	NAND2X1	0.086	0.617	0.645
u_temp_ctrl/g1692	B v -> Y ^	NOR3X1	0.184	0.801	0.829
u_temp_ctrl/g1691	B0 ^ -> Y v	A0I21X2	0.079	0.879	0.907
u_temp_ctrl/g1681	B v -> Y ^	NAND2X2	0.067	0.947	0.975
u_temp_ctrl/g1659	B0 ^ -> Y v	A0T2BR1X1	0.073	1.020	1.048

Arrival Time: Actual time when data reaches the D input.

Required Time: Latest allowable time for data to arrive.

Slack:

Formula (for setup): $\text{Slack} = \text{Required Time} - \text{Arrival Time}$

Required = 1.553 ns, Arrival = 1.528 ns → Slack = +0.025 ns → Setup MET

```
#####
# Generated by: Cadence Innovus 20.10-p004.1
# OS: Linux x86_64 (Host ID edatools-server1.iiitd.edu.in)
# Generated on: Sun Nov 9 19:40:22 2025
# Design: HA
# Command: report_timing -early -view {view1} -max_paths 100 > cts_reports/timing/timing_post_PnR_early.txt
#####
Path 1: MET Removal Check with Pin u_burglar/delay_counter_reg[1]/CK
Endpoint: u_burglar/delay_counter_reg[1]/RN (^) checked with leading edge of 'clk'
Beginpoint: rst_n (^) triggered by leading edge of 'clk'
Path Groups: {async_default}
Analysis View: view1
Other End Arrival Time -0.004
+ Removal 0.221
+ Phase Shift 0.000
= Required Time 0.218
Arrival Time 0.317
Slack Time 0.099
Clock Rise Edge 0.000
+ Input Delay 0.300
= Beginpoint Arrival Time 0.300
#####
```

Instance	Arc	Cell	Delay	Arrival Time	Required Time
u_burglar/delay_counter_reg[1]	rst_n ^			0.300	0.201
	RN ^	SDFFRHQX1	0.017	0.317	0.218

Path 2: MET Removal Check with Pin humidity_sensor_r_reg[1]/CK
Endpoint: humidity_sensor_r_reg[1]/RN (^) checked with leading edge of 'clk'

1.

2.

Arrival Time: Time when the reset release reaches the flip-flop.

Required Time: Time at which reset must remain stable to avoid violation.

Slack:

Formula (for removal): $\text{Slack} = \text{Arrival Time} - \text{Required Time}$

Required = 0.317 ns, Arrival = 0.300 ns → Slack = +0.017 ns → Removal MET

3. Power Report

```
Open [icon] ctsafteranalysis.rpt ~/work/90nm/VDF_project_new/Physical_design/cts_reports Save [icon] x
ctsafteranalysis.rpt x timing_post_PnR_late.txt x area_post_pnr4.rpt x timing_post_PnR_early.txt x
*
* Innovus 20.10-p004_1 (64bit) 05/07/2020 20:02 (Linux 2.6.32-431.11.2.el6.x86_64)
*
* Date & Time: 2025-Nov-09 19:40:22 (2025-Nov-09 14:10:22 GMT)
*
* -----
*
* Design: HA
*
* Liberty Libraries used:
*   view1: /home/anushreev/work/90nm/VDF_project_new/Physical_design/lib/slow.lib
*
* Power Domain used:
*   Rail: VDD Voltage: 0.9
*
* Power View : view1
*
* User-Defined Activity : N.A.
*
* Activity File: N.A.
*
* Hierarchical Global Activity: N.A.
*
* Global Activity: N.A.
*
* Sequential Element Activity: 0.200000
*
* Primary Input Activity: 0.200000
*
* Default icg ratio: N.A.
*
* -----
* Global Comb ClockGate Ratio: N.A.
```

```
Open [icon] ctsafteranalysis.rpt ~/work/90nm/VDF_project_new/Physical_design/cts_reports Save [icon] x
ctsafteranalysis.rpt x timing_post_PnR_late.txt x area_post_pnr4.rpt x timing_post_PnR_early.txt x
*
* report_power -outfile cts_reports/ctsafteranalysis.rpt -rail_analysis_format VS
*
* -----
```

Cell	Internal Power	Switching Power	Total Power	Leakage Power	Cell Name
CTS_ccl_a_buf_00004	0.07978	0.09222	0.1722	0.0001864	CLKBUF16
CTS_ccl_a_buf_00006	0.07977	0.08992	0.1699	0.0001864	CLKBUF16
led_brightness_r_reg[4]	0.01974	0.0003435	0.02038	0.0003009	SDFFRHQX8
led_brightness_r_reg[7]	0.01974	0.0003296	0.02037	0.0003009	SDFFRHQX8
volume_level_r_reg[3]	0.01927	0.0007793	0.02035	0.0003009	SDFFRHQX8
led_brightness_r_reg[5]	0.01975	0.0002986	0.02035	0.0003009	SDFFRHQX8
led_brightness_r_reg[2]	0.01975	0.0002954	0.02034	0.0003009	SDFFRHQX8
led_brightness_r_reg[1]	0.01975	0.0002906	0.02034	0.0003009	SDFFRHQX8
visitor_count_r_reg[1]	0.01975	0.0002803	0.02033	0.0003009	SDFFRHQX8
visitor_count_r_reg[4]	0.01969	0.0003257	0.02031	0.0003009	SDFFRHQX8
led_brightness_r_reg[6]	0.01976	0.0002385	0.0203	0.0003009	SDFFRHQX8
led_brightness_r_reg[3]	0.01976	0.0002341	0.0203	0.0003009	SDFFRHQX8
visitor_count_r_reg[0]	0.01976	0.0002273	0.02029	0.0003009	SDFFRHQX8
visitor_count_r_reg[3]	0.0197	0.000272	0.02027	0.0003009	SDFFRHQX8
visitor_count_r_reg[2]	0.0197	0.0002566	0.02026	0.0003009	SDFFRHQX8
window_state_r_reg[1]	0.01929	0.0006634	0.02025	0.0003009	SDFFRHQX8
visitor_count_r_reg[7]	0.0197	0.0002504	0.02025	0.0003009	SDFFRHQX8
visitor_count_r_reg[6]	0.0197	0.0002467	0.02025	0.0003009	SDFFRHQX8
window_state_r_reg[0]	0.01937	0.0005295	0.0202	0.0003009	SDFFRHQX8
dehumidifier_on_r_reg	0.01932	0.0005205	0.02014	0.0003009	SDFFRHQX8
water_valve_close_r_reg	0.01932	0.0005095	0.02013	0.0003009	SDFFRHQX8
music_on_r_reg	0.01932	0.0004994	0.02012	0.0003009	SDFFRHQX8
humidifier_on_r_reg	0.01935	0.000397	0.02005	0.0003009	SDFFRHQX8
volume_level_r_reg[2]	0.01935	0.0003864	0.02004	0.0003009	SDFFRHQX8
tv_on_r_reg	0.01935	0.0003478	0.02	0.0003009	SDFFRHQX8
volume_level_r_reg[0]	0.01936	0.0003442	0.02	0.0003009	SDFFRHQX8

ctsafteranalysis.rpt	timing_post_PnR_late.txt	area_post_pnr4.rpt	timing_post_PnR_early.txt
u_visitor/g1091	0.0001352	0.0001373	0.0002818
u_visitor/g1086	0.0001798	7.504e-05	0.000273
u_temp_ctrl/g1660	0.0001376	0.0001253	0.0002722
u_burglar/g705	3.974e-05	0.0002063	0.0002591
u_door_lock/g2257	0.0001847	5.651e-05	0.0002536
u_burglar/FE_0FC0_n_6	0.0001351	9.41e-05	0.0002422
u_visitor/g1086	8.834e-05	0.0001391	0.0002415
u_visitor/g1077	0.0001077	9.533e-05	0.0002212
u_burglar/g701	0.0001158	4.654e-05	0.0001747
u_burglar/g689	7.738e-05	5.872e-05	0.0001711
u_burglar/g713	9.356e-05	6.012e-05	0.0001678
u_door_lock/g2243	0.0001155	3.058e-05	0.0001585
u_temp_ctrl/g1739	0.0001145	1.63e-05	0.0001527
u_visitor/g1057	3.395e-05	9.314e-05	0.0001364
u_temp_ctrl/g1685	5.834e-05	4.646e-05	0.0001147
u_burglar/g692	4.783e-05	3.089e-05	0.0001129
u_temp_ctrl/g1665	6.244e-05	2.838e-05	0.0001069
u_burglar/g707	5.437e-05	4.018e-05	0.0001069
u_door_lock/g2238	6.924e-05	1.784e-05	9.945e-05
u_door_lock/g2235	4.855e-05	3.662e-05	9.819e-05
u_temp_ctrl/g1697_dup	3.207e-05	3.611e-05	7.747e-05
u_temp_ctrl/g1697	3.202e-05	3.195e-05	7.326e-05
u_burglar/g678	4.038e-05	1.669e-05	6.943e-05
u_visitor/g1082	1.729e-05	2.226e-05	5.357e-05
u_visitor/g1064	2.024e-05	1.85e-05	4.803e-05
u_burglar/g684	2.315e-05	1.055e-05	4.606e-05
u_burglar/g703	2.353e-05	6.093e-06	4.198e-05

Total (469 of 469)	2.809	0.4242	3.274
Total Capacitance	5.724e-12 F		0.04041
Power Density	*** No Die Area ***		

Total Power: 3.274mW

Internal Power: 2.809mW (85.8%)

Switching Power: 0.4242mW (13.0%)

Leakage Power: 0.04041mW (1.2%)

Optimized CTS:

1. Area:

Hinst Name	Module Name	Inst Count	Total Area	Buffer
Macro	Combinational	Flop	Latch	Clock Gate
HA		469	5925.013	101.425
167.275	1243.587	4412.727	0.000	0.000
0.000	0.000			
u_appliances	appliances_controller	3	74.933	0.000
0.000	0.000	74.933	0.000	0.000
0.000	0.000			
u_burglar	burglar_alarm	49	436.731	0.000
11.354	175.601	249.777	0.000	0.000
0.000	0.000			
u_door_lock	door_lock_controller	73	648.663	0.000
20.436	228.584	399.643	0.000	0.000
0.000	0.000			
u_entertainment	entertainment_system	11	187.711	0.000
0.000	33.304	154.408	0.000	0.000
0.000	0.000			
u_humidity	humidity_controller	8	74.933	0.000
2.271	22.707	49.955	0.000	0.000
0.000	0.000			
u_lum_ctrl	luminosity_controller	16	217.987	0.000
18.166	0.000	199.822	0.000	0.000
0.000	0.000			
u_temp_ctrl	temperature_controller	100	567.675	0.000
56.011	457.168	54.497	0.000	0.000
0.000	0.000			
u_visitor	visitor_counter	63	544.211	4.541
20.436	245.236	273.998	0.000	0.000
0.000	0.000			
u_water_ctrl	water_level_controller	6	69.635	0.000
0.000	19.679	49.955	0.000	0.000
0.000	0.000			
u_window	window_treatment	15	113.535	0.000

- Inverters: The inverter area is 167.275, distributed throughout the design for logic inversion.
- Buffers: Buffers sum up to 101.425, reflecting post-route or clock tree-driven placement for signal strength/stability.

- Flip-Flops: The total area for flip-flops is 4412.727, serving as the primary sequential storage cells in the design.
- Combinational Cells (Other): Combinational cells (apart from buffers/inverters) together use 1243.587 area, implementing main logic functionality.
- Latches: Latch area is 469, used in key timing paths and for level-sensitive state storage.
- Macros: No macro area is reported (0.000), so no embedded memories or custom blocks are included here.
- Total Standard Cell Area: The sum for all standard cells is 5925.013, representing total silicon consumed by all placed logic/building blocks.

2. Timing reports

Hold:

```

#####
# Generated by: Cadence Innovus 20.10-p004.1
# OS: Linux x86_64 (Host ID edatools-server1.iiitd.edu.in)
# Generated on: Sun Nov 9 19:40:42 2025
# Design: HA
# Command: report_timing -early -view {view1} -max_paths 100 > optimised_cts_reports/timing/timing_post_PnR_early.txt
#####
Path 1: MET Removal Check with Pin humidity_sensor_r_reg[5]/CK
Endpoint: humidity_sensor_r_reg[5]/RN (^) checked with leading edge of 'clk'
Beginpoint: rst_n
Path Groups: {async default}
Analysis View: view1
Other End Arrival Time -0.004
+ Removal 0.219
+ Phase Shift 0.000
= Required Time 0.215
Arrival Time 0.315
Slack Time 0.100
Clock Rise Edge 0.000
+ Input Delay 0.300
= Beginpoint Arrival Time 0.300
+-----+-----+-----+-----+-----+-----+
| Instance | Arc | Cell | Delay | Arrival Time | Required Time |
+-----+-----+-----+-----+-----+-----+
| humidity_sensor_r_reg[5] | RN ^ | SDFRHOX4 | 0.015 | 0.300 | 0.200 |
| | | | | 0.315 | 0.215 |
+-----+-----+-----+-----+-----+-----+
Path 2: MET Removal Check with Pin humidity_sensor_r_reg[0]/CK
Endpoint: humidity_sensor_r_reg[0]/RN (^) checked with leading edge of 'clk'
Beginpoint: rst_n
Path Groups: {async default}

```

Arrival Time: Time when reset is released at the flip-flop.

Required Time: The time reset must not change before.

Slack:

Formula: $\text{Slack} = \text{Arrival Time} - \text{Required Time}$

Positive Slack → Removal met (safe)

Negative Slack → Removal violation

Required = 0.315 ns, Arrival = 0.300 ns → Slack = +0.015 ns → Removal MET

Setup:

area_post_pnr3.rpt

timing_post_PnR_early.txt

ctsafteranalysis.rpt

timing_post_PnR_late.txt

#####

Generated by: Cadence Innovus 20.10-p004_1

OS: Linux x86_64(Host ID edatools-server1.iiitd.edu.in)

Generated on: Sun Nov 9 19:40:43 2025

Design: HA

Command: report_timing -late -max_paths 100 > optimised_cts_reports/timing/timing_post_PnR_late.txt

#####

Path 1: MET Setup Check with Pin u_temp_ctrl/cooler_on_reg/CK

Endpoint: u_temp_ctrl/cooler_on_reg/D (^) checked with leading edge of 'clk'

Beginpoint: temp_setpoint_r_reg[3]/Q (^) triggered by leading edge of 'clk'

Path Groups: {clk}

Analysis View: view1

Other End Arrival Time -0.001

- Setup 0.219

+ Phase Shift 1.800

= Required Time 1.580

= Arrival Time 1.555

= Slack Time 0.025

Clock Rise Edge 0.000

+ Clock Network Latency (Prop) -0.001

= Beginpoint Arrival Time -0.001

#####

Instance	Arc	Cell	Delay	Arrival Time	Required Time
temp_setpoint_r_reg[3]	CK ^			-0.001	0.024
temp_setpoint_r_reg[3]	CK ^ -> Q ^	SDFFRHQX4	0.412	0.411	0.436
FE_OPC1 temp_setpoint_r_3	A ^ -> Y ^	BUFX6	0.120	0.531	0.556
u_temp_ctrl/g1709	B ^ -> Y v	NAND2X1	0.086	0.617	0.642
u_temp_ctrl/g1692	B v -> Y ^	NOR3X1	0.184	0.801	0.826
u_temp_ctrl/g1691	B0 ^ -> Y v	AOI21X2	0.078	0.880	0.904
u_temp_ctrl/g1681	B v -> Y ^	NAND2X2	0.067	0.947	0.972
u_temp_ctrl/g1659	B0 ^ -> Y v	OAI2BR1X1	0.073	1.020	1.045

Plain Text

Tab Width: 8

Ln 1, Col 1

INS

The arrival time is the actual time when the new data reaches the D input of the destination flip-flop.

The required time indicates the latest time by which the data must arrive to be correctly captured.

The slack is the difference between the required time and the arrival time, calculated as:

Slack = Required Time – Arrival Time

A positive slack means the setup requirement is met and the path is safe.

A negative slack indicates a setup violation, meaning the data arrived too late.

In this report, the required time is 1.553 ns and the arrival time is 1.528 ns, giving a slack of +0.025 ns.

The cumulative delay from all these components results in the total arrival time of 1.528 ns, which is compared against the required time of 1.553 ns to calculate the final slack.

Optimizing cell placement

3. Power Reports:

ctsafteranalysispostoptimization.rpt					
~/work/90nm/VDF_project_new/Phy...cal.design/optimised_cts_reports					
timing_post_PnR_early.txt					
Cell	Internal Power	Switching Power	Total Power	Leakage Power	Cell Name
CTS_ccl_a_buf_00004	0.07978	0.09237	0.1723	0.0001864	CLKBUF16
CTS_ccl_a_buf_00006	0.07977	0.09008	0.17	0.0001864	CLKBUF16
u_visitor/count_reg[5]	0.01684	0.001562	0.01862	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[1]	0.01663	0.001665	0.01851	0.0002171	SDFFRHQX4
u_visitor/count_reg[0]	0.01724	0.0009879	0.01845	0.0002171	SDFFRHQX4
u_visitor/count_reg[1]	0.01722	0.0009439	0.01838	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[5]	0.0163	0.001849	0.01837	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[2]	0.01631	0.001578	0.01811	0.0002171	SDFFRHQX4
temp_sensor_r_reg[2]	0.01664	0.001203	0.01806	0.0002171	SDFFRHQX4
temp_sensor_r_reg[6]	0.01664	0.001144	0.018	0.0002171	SDFFRHQX4
humidity_sensor_r_reg[1]	0.01665	0.001055	0.01792	0.0002171	SDFFRHQX4
temp_sensor_r_reg[3]	0.01664	0.0009998	0.01786	0.0002171	SDFFRHQX4
window_auto_mode_reg	0.01664	0.0009664	0.01782	0.0002171	SDFFRHQX4
temp_sensor_r_reg[5]	0.01664	0.0009114	0.01777	0.0002171	SDFFRHQX4
volume_cmd_r_reg[2]	0.01664	0.0008841	0.01774	0.0002171	SDFFRHQX4
volume_cmd_r_reg[3]	0.01664	0.0008556	0.01771	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[6]	0.01632	0.001164	0.0177	0.0002171	SDFFRHQX4
temp_sensor_r_reg[4]	0.01664	0.0008385	0.0177	0.0002171	SDFFRHQX4
music_on_cmd_r_reg	0.01664	0.0008239	0.01768	0.0002171	SDFFRHQX4
volume_cmd_r_reg[1]	0.01664	0.0008166	0.01768	0.0002171	SDFFRHQX4
entry_sensor_r_reg	0.01662	0.0008181	0.01766	0.0002171	SDFFRHQX4
volume_cmd_r_reg[0]	0.01664	0.0007845	0.01765	0.0002171	SDFFRHQX4
temp_sensor_r_reg[7]	0.01665	0.0007352	0.0176	0.0002171	SDFFRHQX4
tv_on_cmd_r_reg	0.01665	0.0007277	0.0176	0.0002171	SDFFRHQX4
light_sensor_r_reg[5]	0.01663	0.0007347	0.01758	0.0002171	SDFFRHQX4
light_sensor_r_reg[7]	0.01663	0.0007292	0.01757	0.0002171	SDFFRHQX4
light_sensor_r_reg[4]	0.01663	0.0007217	0.01757	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[0]	0.01665	0.0006869	0.01755	0.0002171	SDFFRHQX4
window_manual_close_r_reg	0.01665	0.0006447	0.01751	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[3]	0.01632	0.0009761	0.01751	0.0002171	SDFFRHQX4

ctsafteranalysispostoptimization.rpt					
~/work/90nm/VDF_project_new/Phy...cal.design/optimised_cts_reports					
timing_post_PnR_early.txt					
u_lum_ctrl/led_brightness_reg[2]	0.01328	0.0004135	0.01385	0.0001526	SDFFRHQX1
u_burglar/alarm_out_reg	0.01315	0.0005366	0.01384	0.0001526	SDFFRHQX1
u_humidity/dehumidifier_on_reg	0.01327	0.0004139	0.01384	0.0001526	SDFFRHQX1
u_water_ctrl/valve_close_reg	0.01333	0.0003545	0.01384	0.0001526	SDFFRHQX1
u_entertainment/volume_level_reg[1]	0.01318	0.0004999	0.01383	0.0001526	SDFFRHQX1
u_door_lock/timeout_counter_reg[1]	0.01305	0.0006271	0.01383	0.0001526	SDFFRHQX1
u_lum_ctrl/led_brightness_reg[3]	0.01328	0.0003622	0.0138	0.0001526	SDFFRHQX1
u_door_lock/timeout_counter_reg[3]	0.01302	0.0006219	0.01379	0.0001526	SDFFRHQX1
window_state_r_reg[1]	0.01296	0.0006733	0.01378	0.0001526	SDFFRHQX1
u_lum_ctrl/led_brightness_reg[1]	0.01328	0.0003478	0.01378	0.0001526	SDFFRHQX1
u_lum_ctrl/led_brightness_reg[0]	0.01328	0.0003434	0.01378	0.0001526	SDFFRHQX1
led_brightness_r_reg[7]	0.01328	0.00033	0.01377	0.0001526	SDFFRHQX1
u_appliances/lights_on_reg	0.01328	0.0003251	0.01376	0.0001526	SDFFRHQX1
led_brightness_r_reg[4]	0.01328	0.0003205	0.01375	0.0001526	SDFFRHQX1
u_door_lock/timeout_counter_reg[0]	0.01298	0.0006156	0.01375	0.0001526	SDFFRHQX1
u_door_lock/timeout_counter_reg[7]	0.01302	0.0005722	0.01374	0.0001526	SDFFRHQX1
visitor_count_r_reg[4]	0.01328	0.0003061	0.01374	0.0001526	SDFFRHQX1
u_appliances/ac_on_reg	0.01327	0.0003055	0.01373	0.0001526	SDFFRHQX1
u_lum_ctrl/led_brightness_reg[7]	0.01296	0.0005968	0.01371	0.0001526	SDFFRHQX1
led_brightness_r_reg[5]	0.01328	0.0002718	0.01371	0.0001526	SDFFRHQX1
visitor_count_r_reg[1]	0.01328	0.0002707	0.0137	0.0001526	SDFFRHQX1
led_brightness_r_reg[2]	0.01328	0.0002668	0.0137	0.0001526	SDFFRHQX1
led_brightness_r_reg[1]	0.01328	0.0002658	0.0137	0.0001526	SDFFRHQX1
visitor_count_r_reg[3]	0.01328	0.0002642	0.0137	0.0001526	SDFFRHQX1
visitor_count_r_reg[7]	0.01328	0.0002558	0.01369	0.0001526	SDFFRHQX1
led_brightness_r_reg[6]	0.01328	0.0002436	0.01368	0.0001526	SDFFRHQX1
led_brightness_r_reg[3]	0.01328	0.0002411	0.01367	0.0001526	SDFFRHQX1
visitor_count_r_reg[6]	0.01328	0.0002369	0.01367	0.0001526	SDFFRHQX1
visitor_count_r_reg[2]	0.01328	0.0002333	0.01366	0.0001526	SDFFRHQX1
visitor_count_r_reg[0]	0.01328	0.0002258	0.01366	0.0001526	SDFFRHQX1
u_lum_ctrl/led_brightness_reg[6]	0.01296	0.0005408	0.01366	0.0001526	SDFFRHQX1
window_state_r_reg[0]	0.01296	0.0005354	0.01365	0.0001526	SDFFRHQX1
u_door_lock/lock_status_reg[1]	0.01298	0.0004995	0.01363	0.0001526	SDFFRHQX1

Open



ctsafteranalysispostoptimization.rpt

~/work/90nm/VDF-project_new/Phys_cal_design/optimised_cts_reports

Save





timing_post_PnR_early.txt

ctsafteranalysispostoptimization.rpt

humidity_sensor_r_reg[5]	0.01667	0.0004661	0.01735	0.0002171	SDFFRHX4
temp_sensor_r_reg[1]	0.01667	0.0004603	0.01734	0.0002171	SDFFRHX4
light_cmd_r_reg	0.01665	0.0004575	0.01732	0.0002171	SDFFRHX4
ac_cmd_r_reg	0.01665	0.0004509	0.01731	0.0002171	SDFFRHX4
fan_cmd_r_reg	0.01665	0.0004442	0.01731	0.0002171	SDFFRHX4
temp_setpoint_r_reg[7]	0.01633	0.0007651	0.01731	0.0002171	SDFFRHX4
temp_sensor_r_reg[0]	0.01667	0.0004131	0.0173	0.0002171	SDFFRHX4
alarm_arm_r_reg	0.01667	0.0003811	0.01727	0.0002171	SDFFRHX4
window_sensor_r_reg	0.01667	0.0003759	0.01726	0.0002171	SDFFRHX4
door_sensor_r_reg	0.01667	0.0003741	0.01726	0.0002171	SDFFRHX4
door_lock_cmd_r_reg	0.01665	0.0003813	0.01725	0.0002171	SDFFRHX4
door_unlock_cmd_r_reg	0.01665	0.0003734	0.01724	0.0002171	SDFFRHX4
pin_input_r_reg[0]	0.01665	0.0003555	0.01723	0.0002171	SDFFRHX4
pin_input_r_reg[2]	0.01665	0.0003494	0.01722	0.0002171	SDFFRHX4
pin_input_r_reg[1]	0.01666	0.0003334	0.01721	0.0002171	SDFFRHX4
motion_sensor_r_reg	0.01668	0.0003036	0.0172	0.0002171	SDFFRHX4
humidity_sensor_r_reg[2]	0.01666	0.0003253	0.0172	0.0002171	SDFFRHX4
humidity_sensor_r_reg[6]	0.01668	0.000298	0.0172	0.0002171	SDFFRHX4
humidity_sensor_r_reg[7]	0.01669	0.0002719	0.01718	0.0002171	SDFFRHX4
pin_input_r_reg[3]	0.01667	0.0002812	0.01717	0.0002171	SDFFRHX4
humidity_sensor_r_reg[4]	0.01669	0.0002349	0.01715	0.0002171	SDFFRHX4
water_level_r_reg[3]	0.01635	0.0004967	0.01706	0.0002171	SDFFRHX4
water_level_r_reg[2]	0.01635	0.0004906	0.01705	0.0002171	SDFFRHX4
water_level_r_reg[0]	0.01636	0.0003591	0.01693	0.0002171	SDFFRHX4
water_level_r_reg[1]	0.01637	0.0002951	0.01688	0.0002171	SDFFRHX4
u_visitor/count_reg[2]	0.01444	0.001023	0.01563	0.0001737	SDFFRHX2
u_visitor/count_reg[3]	0.01412	0.00127	0.01556	0.0001737	SDFFRHX2
u_visitor/count_reg[6]	0.01408	0.001089	0.01535	0.0001737	SDFFRHX2
u_door_lock/state_reg[1]	0.01365	0.001527	0.01533	0.0001526	SDFFRHX1
u_visitor/count_reg[4]	0.01411	0.0009327	0.01522	0.0001737	SDFFRHX2
u_door_lock/state_reg[0]	0.01353	0.001244	0.01492	0.0001526	SDFFRHX1
u_window/window_state_reg[0]	0.01388	0.0007112	0.01474	0.0001526	SDFFRHX1

Open		ctsafteranalysispostoptimization.rpt		~/work/90nm/VDF-project_new/Phys_cal_design/optimised_cts_reports		Save			
timing_post_PnR_early.txt		ctsafteranalysispostoptimization.rpt							
u_visitor/g1091	0.0001354	0.0001372	0.0002819	9.289e-06	NAND2XL				
u_visitor/g1068	0.0001798	7.508e-05	0.000273	1.816e-05	NOR2X1				
u_temp_ctrl/g1660	0.0001376	0.0001251	0.000272	9.289e-06	NAND2XL				
u_burglar/g705	3.974e-05	0.0002066	0.0002593	1.301e-05	CLKINXV1				
u_door_lock/g2257	0.0001847	5.658e-05	0.0002536	1.236e-05	NOR2XL				
u_visitor/g1086	8.836e-05	0.0001421	0.0002445	1.403e-05	NAND2X1				
u_burglar/FE_OF00_n_6	0.0001351	9.421e-05	0.0002423	1.301e-05	INVX1				
u_visitor/g1077	0.0001077	9.334e-05	0.0002192	1.816e-05	NOR2X1				
u_burglar/g701	0.0001158	4.655e-05	0.0001747	1.236e-05	NOR2XL				
u_burglar/g689	7.738e-05	5.878e-05	0.0001712	3.499e-05	NAND2BX1				
u_burglar/g713	9.356e-05	6.004e-05	0.0001677	1.41e-05	OAI211X1				
u_door_lock/g2243	0.0001155	3.059e-05	0.0001585	1.236e-05	NOR2XL				
u_temp_ctrl/g1739	0.0001145	1.633e-05	0.0001527	2.186e-05	NOR3BX1				
u_visitor/g1057	3.395e-05	9.324e-05	0.0001365	9.289e-06	NAND2XL				
u_temp_ctrl/g1685	5.834e-05	4.646e-05	0.0001147	9.93e-06	NAND3XL				
u_burglar/g692	4.782e-05	3.05e-05	0.0001125	3.419e-05	NAND3BX1				
u_temp_ctrl/g1665	6.244e-05	2.838e-05	0.0001069	1.612e-05	NAND4X1				
u_burglar/g707	5.437e-05	4.013e-05	0.0001069	1.236e-05	NOR2XL				
u_door_lock/g2238	6.925e-05	1.785e-05	9.946e-05	1.236e-05	NOR2XL				
u_door_lock/g2235	4.856e-05	3.664e-05	9.822e-05	1.301e-05	CLKINXV1				
u_temp_ctrl/g1697_dup	3.207e-05	3.619e-05	7.756e-05	9.289e-06	NAND2XL				
u_temp_ctrl/g1697	3.202e-05	3.202e-05	7.333e-05	9.289e-06	NAND2XL				
u_burglar/g678	4.038e-05	1.669e-05	6.943e-05	1.236e-05	NOR2XL				
u_visitor/g1082	1.729e-05	2.229e-05	5.361e-05	1.403e-05	NAND2X1				
u_visitor/g1064	2.021e-05	1.852e-05	4.802e-05	9.289e-06	NAND2XL				
u_burglar/g684	2.315e-05	1.056e-05	4.606e-05	1.236e-05	NOR2XL				
u_burglar/g703	2.353e-05	6.099e-06	4.199e-05	1.236e-05	NOR2XL				

Total (	469 of	469)	2.618	0.4163	3.071	0.03594			
Total Capacitance			5.543e-12 F						
Power Density			*** No Die Area ***						

Total Power: 3.071mW

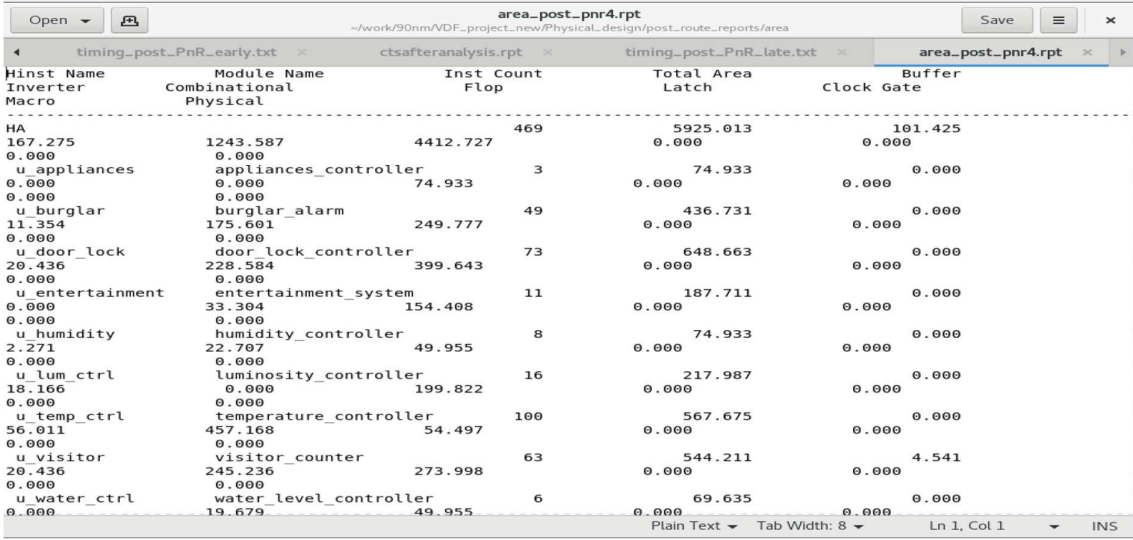
Internal Power: 2.618mW (85.2%)

Switching Power: 0.4163mW (13.6%)

Leakage Power: 0.03594mW (1.2%)

Post Routing:

1. Area:



Hinst Name	Module Name	Inst Count	Total Area	Buffer
Inverter	Combinational	Flop	Latch	Clock Gate
Macro	Physical			
HA		469	5925.013	101.425
167.275	1243.587	4412.727	0.000	0.000
0.000	0.000			
0.000	appliances	appliances_controller	74.933	0.000
0.000	0.000	74.933	0.000	0.000
0.000	0.000			
u_burglar	burglar_alarm	49	436.731	0.000
11.354	175.601	249.777	0.000	0.000
0.000	0.000			
u_door_lock	door_lock_controller	73	648.663	0.000
20.436	228.564	399.643	0.000	0.000
0.000	0.000			
u_entertainment	entertainment_system	11	187.711	0.000
0.000	33.304	154.408	0.000	0.000
0.000	0.000			
u_humidity	humidity_controller	8	74.933	0.000
2.271	22.707	49.955	0.000	0.000
0.000	0.000			
u_lum_ctrl	luminosity_controller	16	217.987	0.000
18.166	0.000	199.822	0.000	0.000
0.000	0.000			
u_temp_ctrl	temperature_controller	100	567.675	0.000
56.011	457.168	54.497	0.000	0.000
0.000	0.000			
u_visitor	visitor_counter	63	544.211	4.541
20.436	245.236	273.998	0.000	0.000
0.000	0.000			
u_water_ctrl	water_level_controller	6	69.635	0.000
0.000	19.679	49.955	0.000	0.000

- Inverters: The inverter area is 167.275, providing distributed logic inversion and signal stabilization.
- Buffers: Buffers take up 101.425 of area, used for strengthening signals or re-driving long/nets-heavy loads.
- Flip-Flops: The area dedicated to flip-flops is 4412.727, making up most of the sequential logic building blocks in the design.
- Combinational Cells (Other): Other combinational cells utilize 1243.587 of the total area, responsible for all core logic (AND/OR/NAND/complex combo implementations).
- Latches: Area for latches is 469, important for timing-path management and level-sensitive data storage.
- Macros: Macro area is 0.000, indicating no SRAM or other custom blocks in this particular report.
- Total Standard Cell Area: The cumulative standard cell area is 5925.013, summing up every implemented functional and sequential logic block.

2. Timing Reports

Hold:

```
#####
# Generated by: Cadence Innovus 20.10-p004.1
# OS: Linux x86_64 (Host ID edatools-server1.iiitd.edu.in)
# Generated on: Sun Nov 9 19:40:52 2025
# Design: HA
# Command: report_timing -early -view {view1} -max_paths 100 > post_route_reports/timing/
timing_post_PnR_early.txt
#####
Path 1: MET Removal Check with Pin humidity_sensor_r_reg[5]/CK
Endpoint: humidity_sensor_r_reg[5]/RN (^) checked with leading edge of 'clk'
Beginpoint: rst_n (^) triggered by leading edge of 'clk'
Path Groups: {async default}
Analysis View: view1
Other End Arrival Time -0.002
+ Removal 0.214
+ Phase Shift 0.000
= Required Time 0.212
Arrival Time 0.311
Slack Time 0.100
Clock Rise Edge 0.000
+ Input Delay 0.300
= Beginpoint Arrival Time 0.300
+-----+-----+-----+-----+-----+-----+
| Instance | Arc | Cell | Delay | Arrival | Required |
| Time | Time |
+-----+-----+-----+-----+-----+-----+
| humidity_sensor_r_reg[5] | RN ^ | SDFFRHX4 | 0.011 | 0.311 | 0.212 |
+-----+-----+-----+-----+-----+-----+
Path 2: MET Removal Check with Pin humidity_sensor_r_reg[2]/CK
Endpoint: humidity_sensor_r_reg[2]/RN (^) checked with leading edge of 'clk'
Beginpoint: rst_n (^) triggered by leading edge of 'clk'
Path Groups: {async default}
```

Arrival Time: 0.311 ns (when reset reaches the flip-flop)

Required Time: 0.212 ns (latest safe de-assertion time)

Slack Calculation:

Slack = Arrival Time - Required Time

0.311 - 0.212 = +0.100 ns

Result: Positive slack → Removal MET (safe)

Setup:

```
#####
# Generated by: Cadence Innovus 20.10-p004.1
# OS: Linux x86_64 (Host ID edatools-server1.iiitd.edu.in)
# Generated on: Sun Nov 9 19:40:52 2025
# Design: HA
# Command: report_timing -late -max_paths 100 > post_route_reports/timing/timing_post_PnR_late.txt
timing_post_PnR_late.txt
#####
Path 1: MET Setup Check with Pin u_temp_ctrl/cooler_on reg/CK
Endpoint: u_temp_ctrl/cooler_on reg/D (^) checked with leading edge of 'clk'
Beginpoint: temp_setpoint_r_reg[1]/Q (^) triggered by leading edge of 'clk'
Path Groups: {clk}
Analysis View: view1
Other End Arrival Time 0.000
- Setup 0.217
+ Phase Shift 1.800
= Required Time 1.583
Arrival Time 1.546
= Slack Time 0.037
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) -0.001
= Beginpoint Arrival Time -0.001
+-----+-----+-----+-----+-----+-----+
| Instance | Arc | Cell | Delay | Arrival | Required |
| Time | Time |
+-----+-----+-----+-----+-----+-----+
| temp_setpoint_r_reg[1] | CK ^ | SDFFRHX4 | 0.442 | 0.441 | 0.478 |
| temp_setpoint_r_reg[1] | CK ^ -> Q ^ | SDFFRHX4 | 0.442 | 0.441 | 0.478 |
| u_temp_ctrl/g1716 | A ^ -> Y v | NAND2X1 | 0.211 | 0.652 | 0.688 |
| u_temp_ctrl/g1680 | A2 v -> Y ^ | AOI31X1 | 0.136 | 0.788 | 0.824 |
| u_temp_ctrl/g1662 | C0 ^ -> Y v | AOI221X1 | 0.179 | 0.966 | 1.003 |
| u_temp_ctrl/g1643 | A0 v -> Y ^ | AOI211X2 | 0.160 | 1.126 | 1.163 |
| u_temp_ctrl/FE RC 29_0 | A ^ -> Y v | NAND2X1 | 0.109 | 1.235 | 1.271 |
| u_temp_ctrl/FE RC 28_0 | C v -> Y ^ | NAND3X2 | 0.097 | 1.331 | 1.368 |
+-----+-----+-----+-----+-----+-----+

```

Required Time: 1.583 ns (latest safe arrival time)

Arrival Time: 1.546 ns (actual data arrival time)

Slack Calculation:

Slack = Required Time - Arrival Time

1.583 - 1.546 = +0.037 ns

Result: Positive slack → Setup MET (safe)

4. Power Reports:

routeafteranalysis.rpt					
~/work/90nm/VDF_project_new/Physical_design/post_route_reports					
timing_post_PnR_early.txt	ctsafteranalysispostoptimization.rpt	timing_post_PnR_early.txt	routeafteranalysis.rpt		
*u_door_lock/timeout_counter_reg[7]	0.01302	0.0005644	0.01374	0.0001526	SDFFRHQX1
*u_lum_ctrl/led_brightness_reg[7]	0.01296	0.0006171	0.01373	0.0001526	SDFFRHQX1
*led_brightness_r_reg[2]	0.01328	0.000291	0.01373	0.0001526	SDFFRHQX1
*u_appliances/ac_on_reg	0.01327	0.0002979	0.01372	0.0001526	SDFFRHQX1
*window_state_r_reg[0]	0.01297	0.000603	0.01372	0.0001526	SDFFRHQX1
*led_brightness_r_reg[5]	0.01328	0.0002751	0.01371	0.0001526	SDFFRHQX1
*visitor_count_r_reg[1]	0.01328	0.0002667	0.0137	0.0001526	SDFFRHQX1
*led_brightness_r_reg[1]	0.01328	0.0002666	0.0137	0.0001526	SDFFRHQX1
*visitor_count_r_reg[3]	0.01328	0.000254	0.01369	0.0001526	SDFFRHQX1
*led_brightness_r_reg[3]	0.01328	0.000246	0.01368	0.0001526	SDFFRHQX1
*visitor_count_r_reg[7]	0.01328	0.0002487	0.01368	0.0001526	SDFFRHQX1
*led_brightness_r_reg[6]	0.01328	0.0002421	0.01368	0.0001526	SDFFRHQX1
*u_lum_ctrl/led_brightness_reg[6]	0.01296	0.0005532	0.01367	0.0001526	SDFFRHQX1
*visitor_count_r_reg[0]	0.01328	0.0002332	0.01367	0.0001526	SDFFRHQX1
*visitor_count_r_reg[2]	0.01328	0.0002342	0.01367	0.0001526	SDFFRHQX1
*u_door_lock/lock_status_reg[1]	0.01299	0.0005115	0.01365	0.0001526	SDFFRHQX1
*visitor_count_r_reg[5]	0.01328	0.0002147	0.01365	0.0001526	SDFFRHQX1
*music_on_r_reg	0.01296	0.0005379	0.01365	0.0001526	SDFFRHQX1
*dehumidifier_on_r_reg	0.01296	0.0005215	0.01363	0.0001526	SDFFRHQX1
*u_door_lock/lock_status_reg[0]	0.01298	0.0004707	0.0136	0.0001526	SDFFRHQX1
*water_valve_close_r_reg	0.01295	0.0004778	0.01358	0.0001526	SDFFRHQX1
*u_lum_ctrl/led_brightness_reg[5]	0.01296	0.0004551	0.01357	0.0001526	SDFFRHQX1
*humidifier_on_r_reg	0.01296	0.0004325	0.01355	0.0001526	SDFFRHQX1
*u_entertainment/tv_on_reg	0.01296	0.0004374	0.01355	0.0001526	SDFFRHQX1
*u_visitor/exit_prev_reg	0.01296	0.0004314	0.01355	0.0001526	SDFFRHQX1
*u_entertainment/music_on_reg	0.01296	0.0004258	0.01353	0.0001526	SDFFRHQX1
*volume_level_r_reg[2]	0.01296	0.0003589	0.01347	0.0001526	SDFFRHQX1
*u_humidity/humidifier_on_reg	0.01293	0.0003828	0.01347	0.0001526	SDFFRHQX1
*volume_level_r_reg[0]	0.01296	0.000331	0.01344	0.0001526	SDFFRHQX1
*volume_level_r_reg[1]	0.01296	0.0003277	0.01344	0.0001526	SDFFRHQX1
*lock_status_r_reg[0]	0.01296	0.0003217	0.01343	0.0001526	SDFFRHQX1
*tv_on_r_reg	0.01296	0.0003236	0.01343	0.0001526	SDFFRHQX1
*u_visitor/entry_prev_reg	0.01297	0.0003113	0.01343	0.0001526	SDFFRHQX1

routeafteranalysis.rpt					
~/work/90nm/VDF_project_new/Physical_design/post_route_reports					
timing_post_PnR_early.txt	ctsafteranalysispostoptimization.rpt	timing_post_PnR_early.txt	routeafteranalysis.rpt		
Cell	Internal Power	Switching Power	Total Power	Leakage Power	Cell Name
CT5_ccl_a_buf_00004	0.07979	0.09398	0.174	0.0001864	CLKBUF16
CT5_ccl_a_buf_00006	0.07978	0.09198	0.172	0.0001864	CLKBUF16
u_visitor/count_reg[5]	0.01684	0.00153	0.01859	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[1]	0.01664	0.001677	0.01853	0.0002171	SDFFRHQX4
u_visitor/count_reg[0]	0.01724	0.001008	0.01847	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[5]	0.01631	0.001853	0.01838	0.0002171	SDFFRHQX4
u_visitor/count_reg[1]	0.01722	0.0009288	0.01837	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[2]	0.01631	0.001592	0.01812	0.0002171	SDFFRHQX4
temp_sensor_r_reg[2]	0.01664	0.001212	0.01807	0.0002171	SDFFRHQX4
temp_sensor_r_reg[6]	0.01664	0.001166	0.01803	0.0002171	SDFFRHQX4
humidity_sensor_r_reg[1]	0.01665	0.001048	0.01792	0.0002171	SDFFRHQX4
temp_sensor_r_reg[3]	0.01665	0.00102	0.01788	0.0002171	SDFFRHQX4
window_auto_mode_r_reg	0.01665	0.0009958	0.01786	0.0002171	SDFFRHQX4
temp_sensor_r_reg[5]	0.01665	0.0009118	0.01778	0.0002171	SDFFRHQX4
temp_sensor_r_reg[4]	0.01665	0.0008705	0.01774	0.0002171	SDFFRHQX4
volume_cmd_r_reg[3]	0.01665	0.0008517	0.01772	0.0002171	SDFFRHQX4
volume_cmd_r_reg[2]	0.01665	0.0008467	0.01771	0.0002171	SDFFRHQX4
volume_cmd_r_reg[0]	0.01665	0.0008443	0.01771	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[6]	0.01632	0.001164	0.0177	0.0002171	SDFFRHQX4
volume_cmd_r_reg[1]	0.01665	0.0008312	0.0177	0.0002171	SDFFRHQX4
music_on_cmd_r_reg	0.01665	0.0008304	0.0177	0.0002171	SDFFRHQX4
entry_sensor_r_reg	0.01662	0.0008497	0.01768	0.0002171	SDFFRHQX4
temp_sensor_r_reg[7]	0.01666	0.0007463	0.01762	0.0002171	SDFFRHQX4
tv_on_cmd_r_reg	0.01666	0.0007273	0.01761	0.0002171	SDFFRHQX4
light_sensor_r_reg[5]	0.01663	0.0007534	0.0176	0.0002171	SDFFRHQX4
light_sensor_r_reg[7]	0.01663	0.0007297	0.01758	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[0]	0.01666	0.0006955	0.01757	0.0002171	SDFFRHQX4
light_sensor_r_reg[4]	0.01663	0.0007136	0.01756	0.0002171	SDFFRHQX4
window_manual_close_r_reg	0.01666	0.000658	0.01754	0.0002171	SDFFRHQX4
light_sensor_r_reg[3]	0.01663	0.000661	0.01753	0.0002171	SDFFRHQX4

Open

routeafteranalysis.rpt

Save

~/work/90nm/VDF_project_new/Physical_design/post_route_reports

timing_post_PnR_early.txt

ctsafteranalysispostoptimization.rpt

timing_post_PnR_early.txt

routeafteranalysis.rpt

u_burglar/delay_counter_reg[1]

0.01354

0.00079

0.01449

0.0001526

SDFFRHQX1

u_burglar/delay_counter_reg[3]

0.01355

0.0007469

0.01445

0.0001526

SDFFRHQX1

u_burglar/delay_counter_reg[2]

0.01354

0.0007153

0.01441

0.0001526

SDFFRHQX1

u_burglar/delay_counter_reg[4]

0.01322

0.001017

0.01439

0.0001526

SDFFRHQX1

u_burglar/delay_counter_reg[0]

0.01348

0.0007406

0.01437

0.0001526

SDFFRHQX1

u_door_lock/attempt_counter_reg[1]

0.01346

0.0007404

0.01435

0.0001526

SDFFRHQX1

u_door_lock/attempt_counter_reg[0]

0.01346

0.0007255

0.01434

0.0001526

SDFFRHQX1

u_door_lock/timeout_counter_reg[5]

0.01302

0.001149

0.01433

0.0001526

SDFFRHQX1

u_burglar/delay_counter_reg[5]

0.01323

0.0009221

0.0143

0.0001526

SDFFRHQX1

FE_OFC9_scan_en

0.001212

0.01302

0.0143

6.202e-05

CLKBUF3

u_burglar/delay_counter_reg[6]

0.01319

0.0008838

0.01422

0.0001526

SDFFRHQX1

u_entertainment/volume_level_reg[0]

0.01348

0.0005867

0.01421

0.0001436

SDFFSHQX1

u_entertainment/volume_level_reg[2]

0.01348

0.0005433

0.01417

0.0001436

SDFFSHQX1

u_burglar/delay_counter_reg[8]

0.0135

0.0004952

0.01415

0.0001526

SDFFRHQX1

u_water_ctrl/pump_on_reg

0.01347

0.0004997

0.01412

0.0001526

SDFFRHQX1

u_burglar/delay_counter_reg[7]

0.01316

0.000702

0.01402

0.0001526

SDFFRHQX1

u_door_lock/timeout_counter_reg[4]

0.01305

0.0007648

0.01397

0.0001526

SDFFRHQX1

u_door_lock/timeout_counter_reg[2]

0.01299

0.0007905

0.01394

0.0001526

SDFFRHQX1

u_door_lock/timeout_counter_reg[6]

0.01302

0.0007371

0.01391

0.0001526

SDFFRHQX1

u_lum_ctrl/led_brightness_reg[4]

0.01328

0.0004713

0.01391

0.0001526

SDFFRHQX1

volume_level_r_reg[3]

0.01296

0.0007811

0.01389

0.0001526

SDFFRHQX1

u_entertainment/volume_level_reg[3]

0.01319

0.0005537

0.01389

0.0001526

SDFFRHQX1

u_appliances/fan_on_reg

0.01327

0.0004467

0.01387

0.0001526

SDFFRHQX1

u_water_ctrl/valve_close_reg

0.01334

0.0003638

0.01385

0.0001526

SDFFRHQX1

u_burglar/alarm_out_reg

0.01315

0.00054

0.01385

0.0001526

SDFFRHQX1

u_lum_ctrl/led_brightness_reg[2]

0.01328

0.0004099

0.01384

0.0001526

SDFFRHQX1

u_door_lock/timeout_counter_reg[1]

0.01305

0.0006346

0.01384

0.0001526

SDFFRHQX1

u_humidity/dehumidifier_on_reg

0.01328

0.0004015

0.01383

0.0001526

SDFFRHQX1

u_entertainment/volume_level_reg[1]

0.01319

0.0004887

0.01383

0.0001526

SDFFRHQX1

u_lum_ctrl/led_brightness_reg[3]

0.01328

0.000382

0.01382

0.0001526

SDFFRHQX1

u_door_lock/timeout_counter_reg[3]

0.01302

0.0006313

0.0138

0.0001526

SDFFRHQX1

window_state_r_reg[1]

0.01296

0.0006843

0.0138

0.0001526

SDFFRHQX1

u

Open

routeafteranalysis.rpt

Save

~/work/90nm/VDF_project_new/Physical_design/post_route_reports

timing_post_PnR_early.txt

ctsafteranalysispostoptimization.rpt

timing_post_PnR_early.txt

routeafteranalysis.rpt

temp_setpoint_r_reg[3]

0.01632

0.0009585

0.0175

0.0002171

SDFFRHQX4

temp_setpoint_r_reg[4]

0.01632

0.0009314

0.01747

0.0002171

SDFFRHQX4

light_sensor_r_reg[6]

0.01664

0.0006014

0.01746

0.0002171

SDFFRHQX4

exit_sensor_r_reg

0.01664

0.0005941

0.01745

0.0002171

SDFFRHQX4

window_manual_open_r_reg

0.01667

0.0005536

0.01744

0.0002171

SDFFRHQX4

light_sensor_r_reg[0]

0.01664

0.0005626

0.01742

0.0002171

SDFFRHQX4

humidity_sensor_r_reg[3]

0.01667

0.000507

0.01739

0.0002171

SDFFRHQX4

humidity_sensor_r_reg[0]

0.01664

0.0005259

0.01739

0.0002171

SDFFRHQX4

humidity_sensor_r_reg[5]

0.01667

0.0004622

0.01735

0.0002171

SDFFRHQX4

temp_sensor_r_reg[1]

0.01667

0.0004604

0.01735

0.0002171

SDFFRHQX4

light_cmd_r_reg

0.01665

0.0004785

0.01734

0.0002171

SDFFRHQX4

ac_cmd_r_reg

0.01665

0.000466

0.01733

0.0002171

SDFFRHQX4

temp_sensor_r_reg[0]

0.01668

0.0004173

0.01731

0.0002171

SDFFRHQX4

temp_setpoint_r_reg[7]

0.01633

0.0007547

0.0173

0.0002171

SDFFRHQX4

fan_cmd_r_reg

0.01665

0.00043

0.0173

0.0002171

SDFFRHQX4

alarm_arm_r_reg

0.01668

0.0003922

0.01729

0.0002171

SDFFRHQX4

window_sensor_r_reg

0.01668

0.0003744

0.01727

0.0002171

SDFFRHQX4

door_sensor_r_reg

0.01668

0.0003732

0.01727

0.0002171

SDFFRHQX4

door_lock_cmd_r_reg

0.01665

0.000381

0.01725

0.0002171

SDFFRHQX4

door_unlock_cmd_r_reg

0.01666

0.0003683

0.01724

0.0002171

SDFFRHQX4

pin_input_r_reg[0]

0.01665

0.0003681

0.01724

0.0002171

SDFFRHQX4

motion_sensor_r_reg

0.01669

0.0003155

0.01722

0.0002171

SDFFRHQX4

humidity_sensor_r_reg[2]

0.01666

0.0003347

0.01721

0.0002171

SDFFRHQX4

pin_input_r_reg[2]

0.01666

0.0003342

0.01721

0.0002171

SDFFRHQX4

pin_input_r_reg[1]

0.01666

0.0003254

0.0172

0.0002171

SDFFRHQX4

humidity_sensor_r_reg[6]

0.01669

0.0002885

0.0172

0.0002171

SDFFRHQX4

humidity_sensor_r_reg[7]

0.0167

0.0002691

0.01718

0.0002171

SDFFRHQX4

pin_input_r_reg[3]

0.01668

0.0002745

0.01717

0.0002171

SDFFRHQX4

humidity_sensor_r_reg[4]

0.0167

0.0002353

0.01715

0.0002171

SDFFRHQX4

water_level_r_reg[2]

0.01635

0.0005051

0.01707

0.0002171

SDFFRHQX4

water_level_r_reg[3]

0.01635

0.0004912

0.01706

0.0002171

SDFFRHQX4

water_level_r_reg[0]

0.01636

0.0003411

0.01692

0.0002171

SDFFRHQX4

water_level_r_reg[1]

0.01637

0.0002956

0.01688

0.0002171

SDFFRHQX4

Total Capacitance

5.578e-12 F

Power Density

*** No Die Area ***

Total Power: 3.076mW

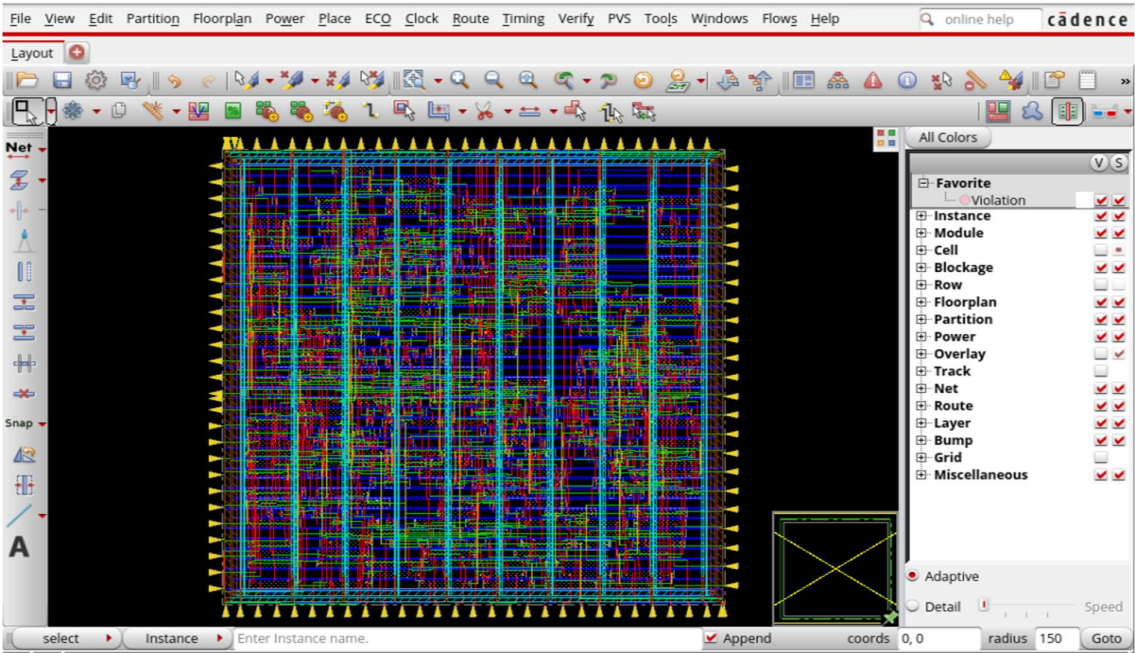
Internal Power: 2.619mW (85.1%)

Switching Power: 0.4211mW (13.7%)

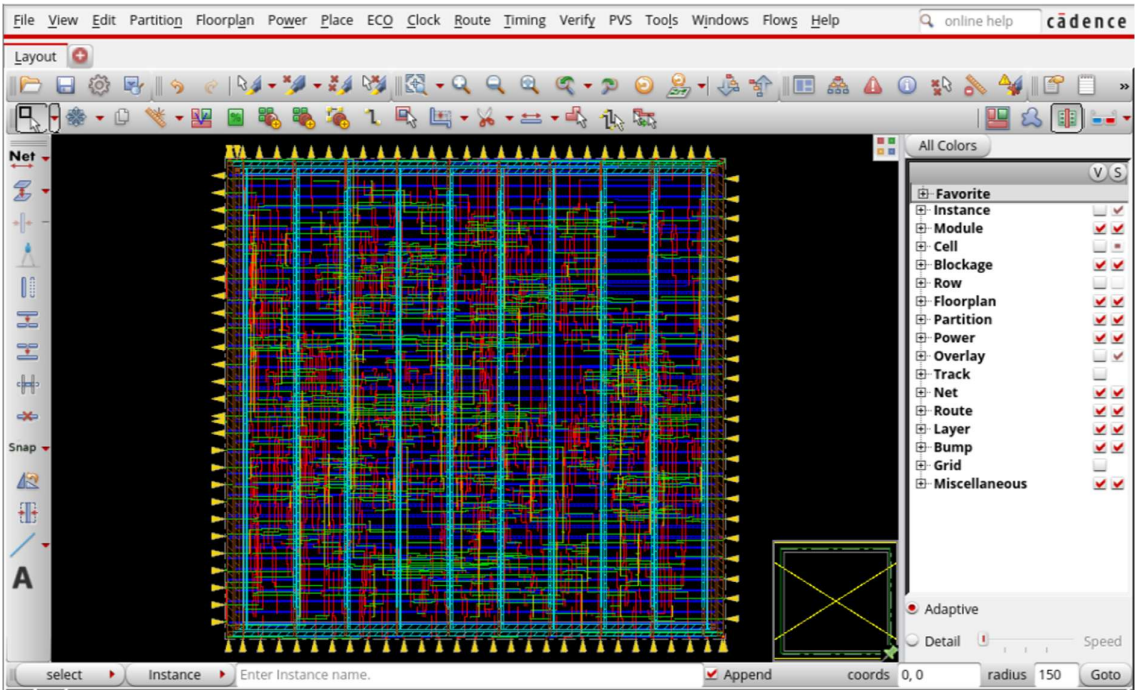
Leakage Power: 0.03594mW (1.2%)

Utilization Factor 0.5:

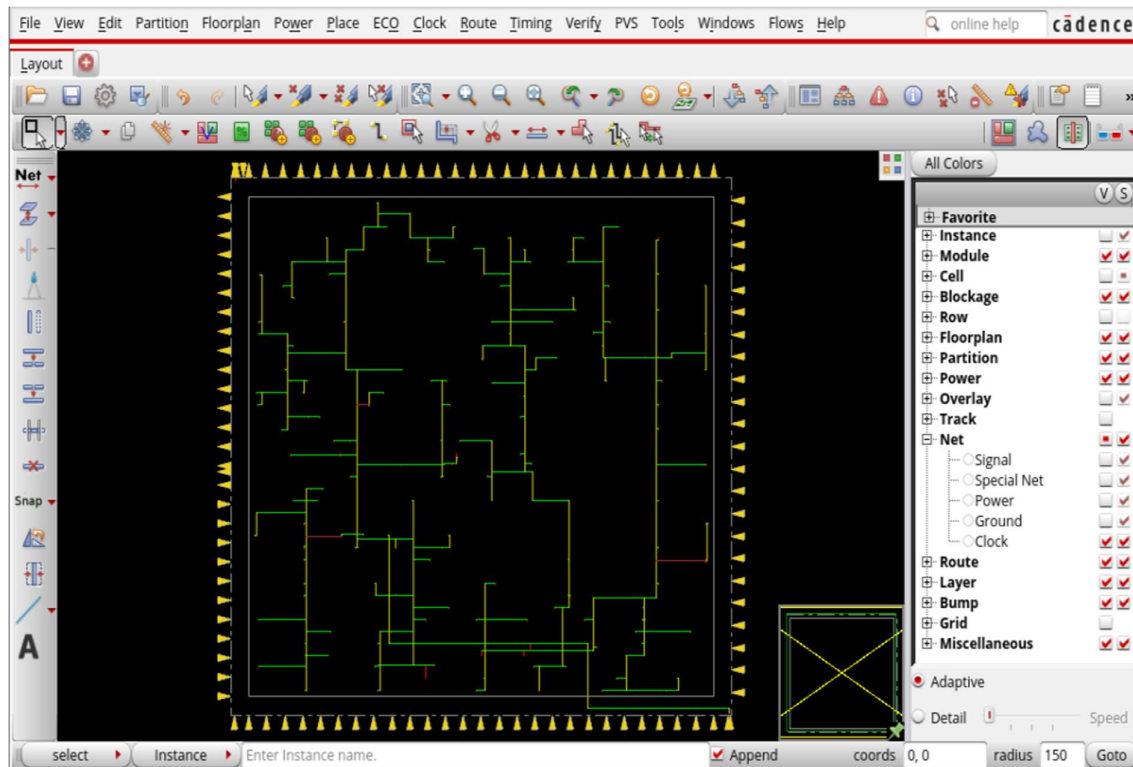
Core:



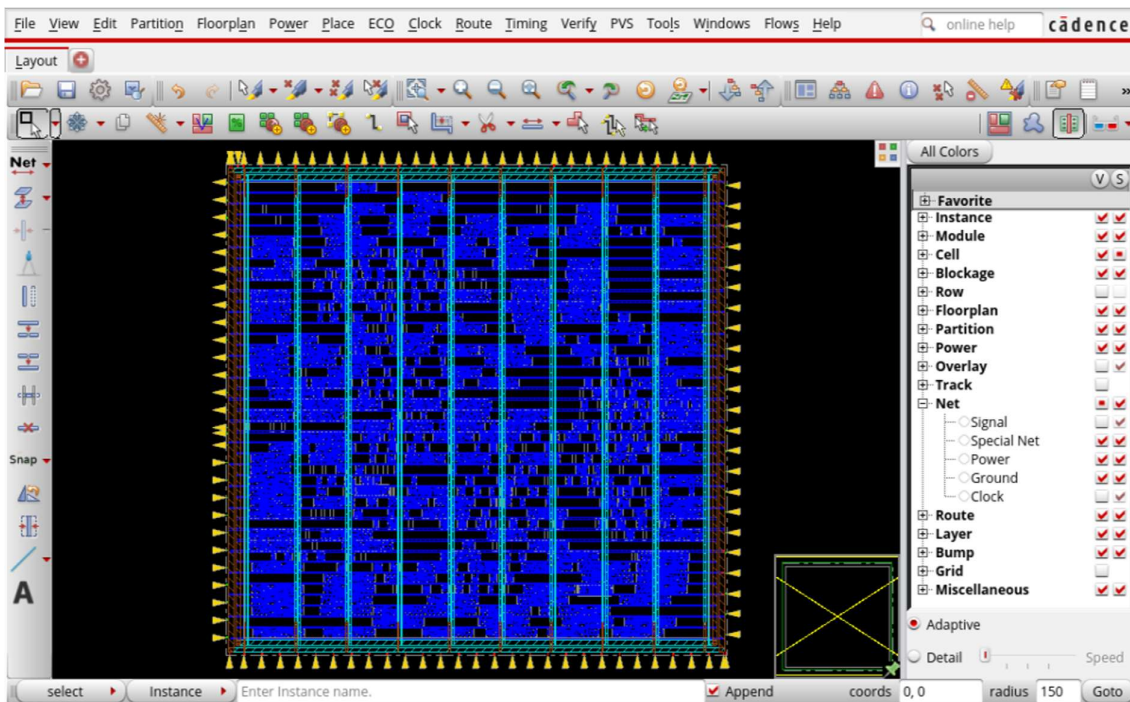
Routing of wires:



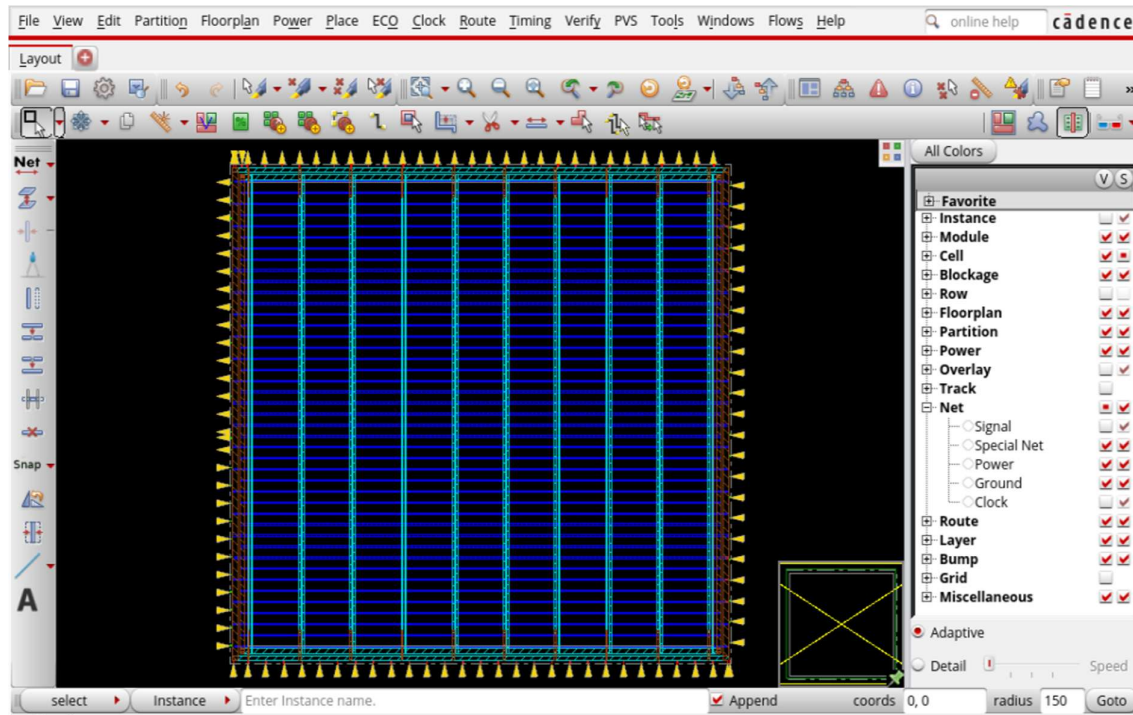
After clock tree synthesis



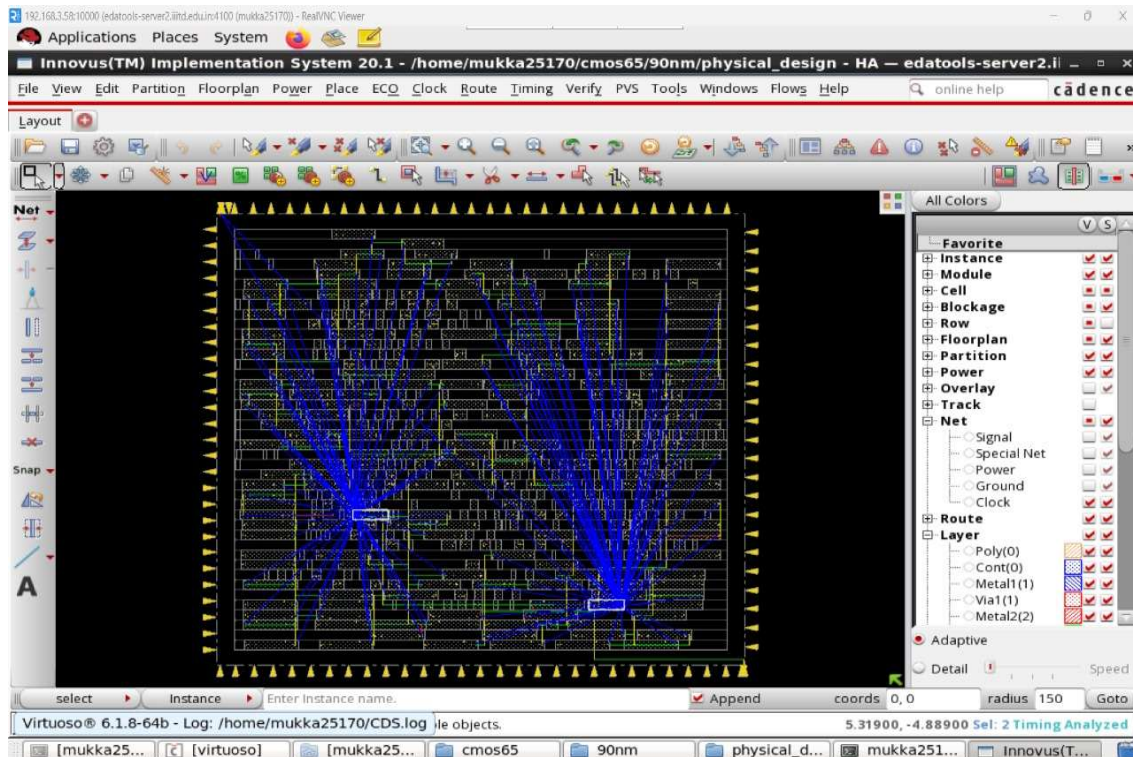
Standard Cells and cell layout:



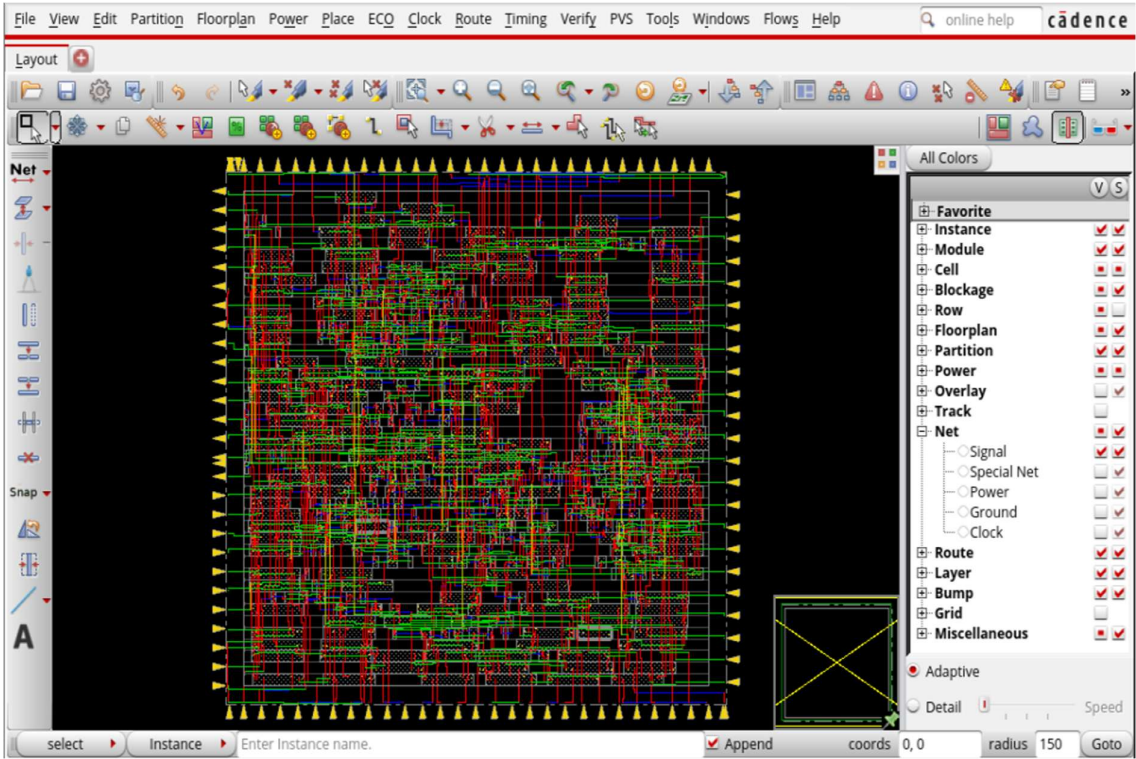
Power and grnd rails:



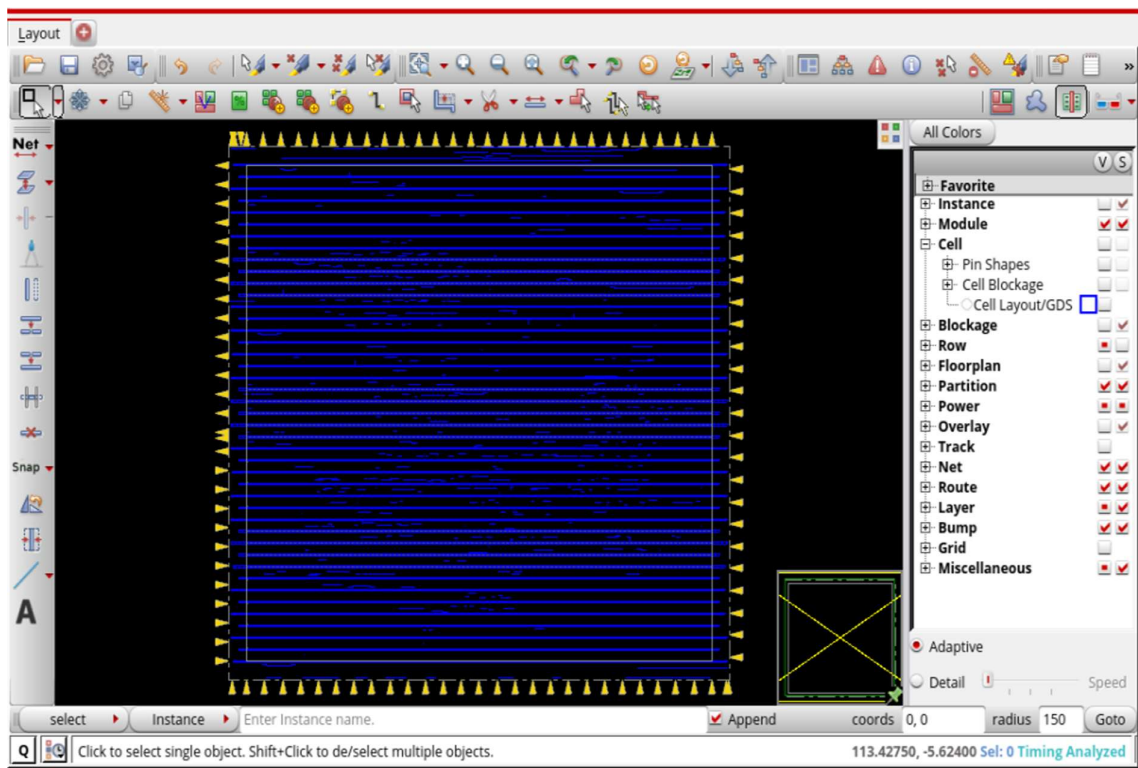
Fly-lines:



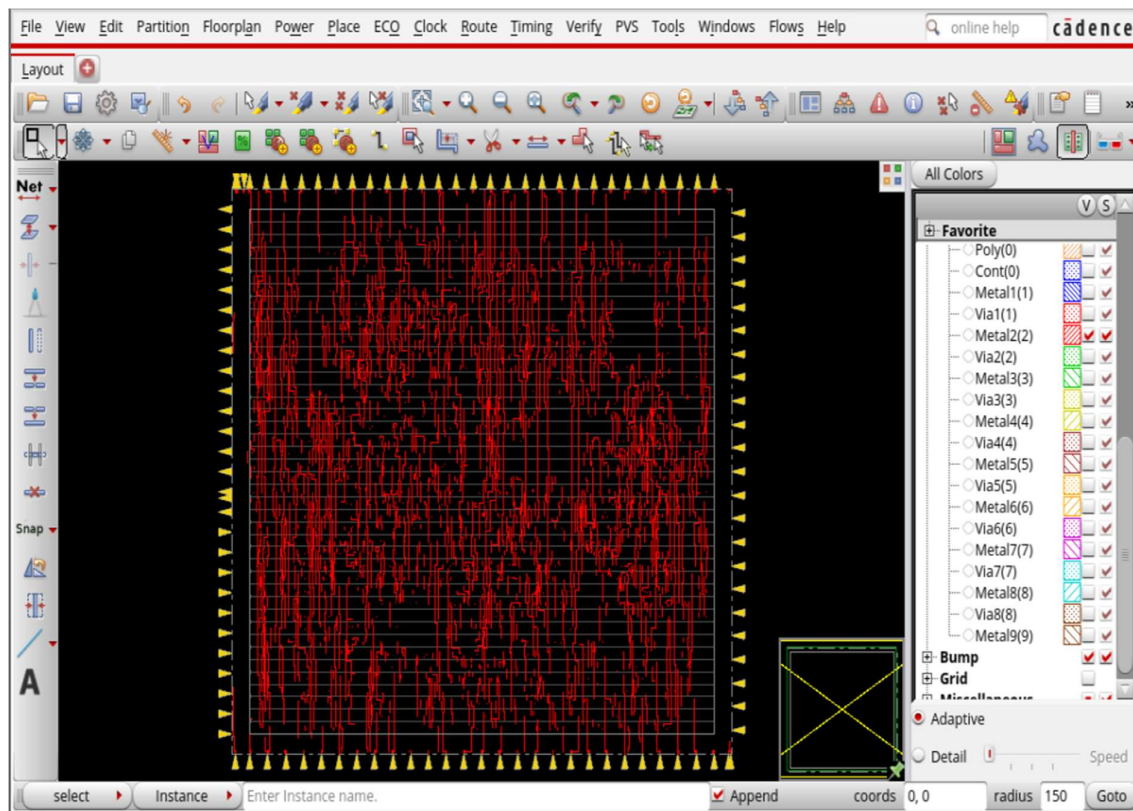
Signals from ports:



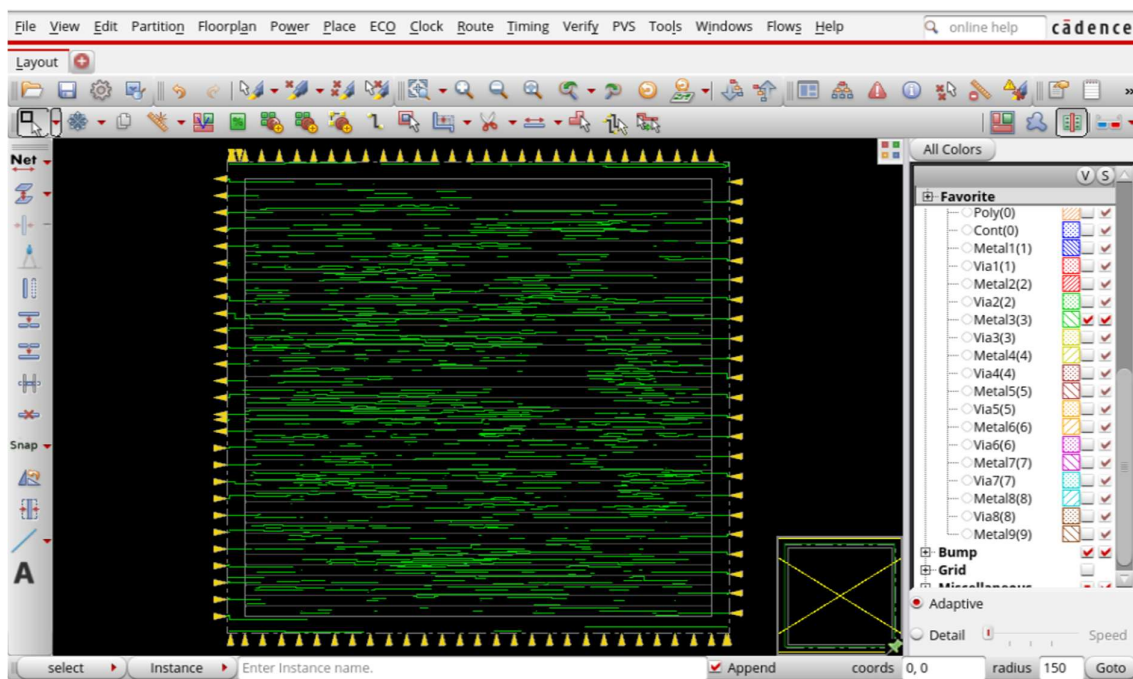
M1 routing:



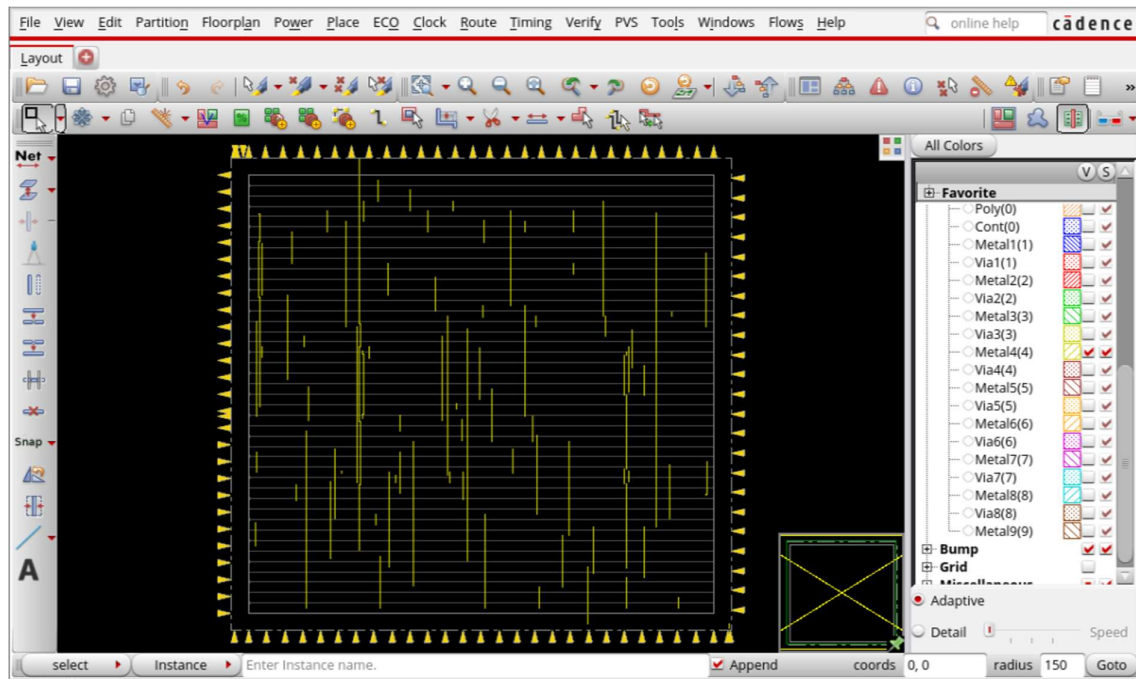
M2 routing:



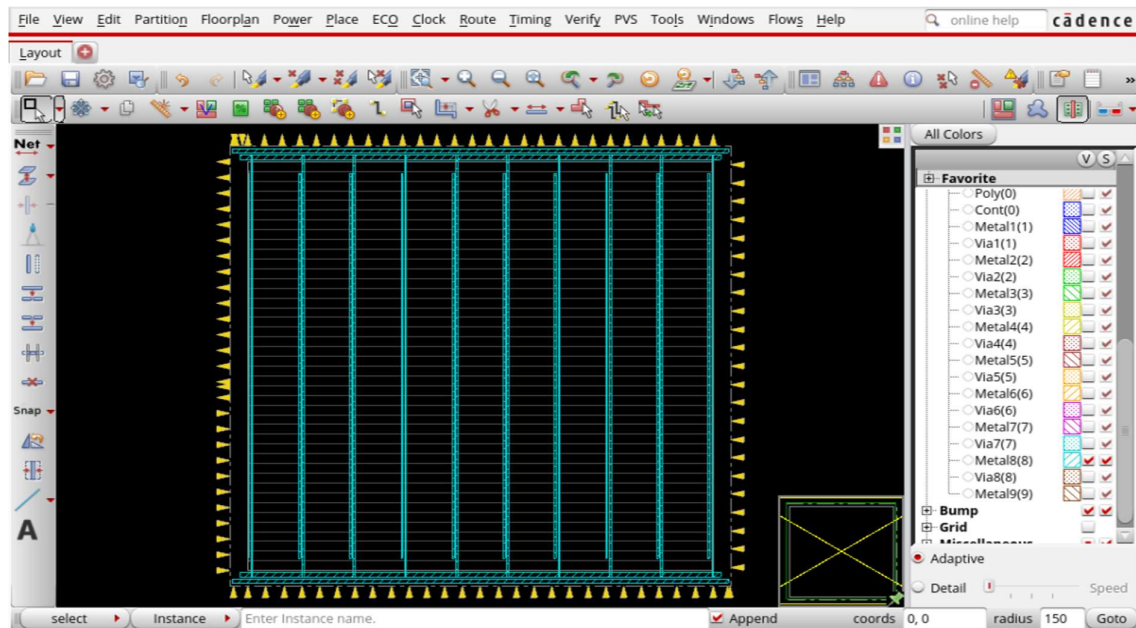
M3 routing:



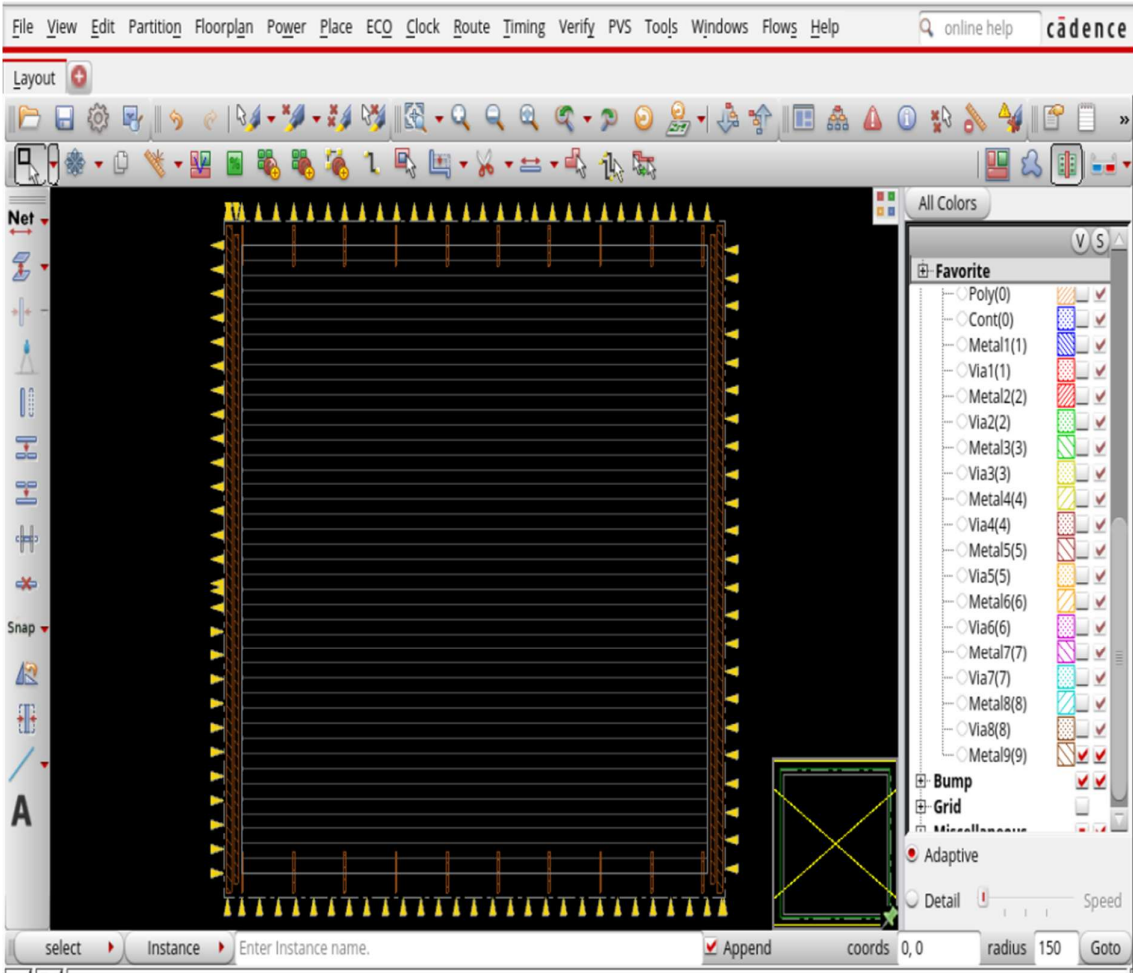
M4 routing



M8 routing:



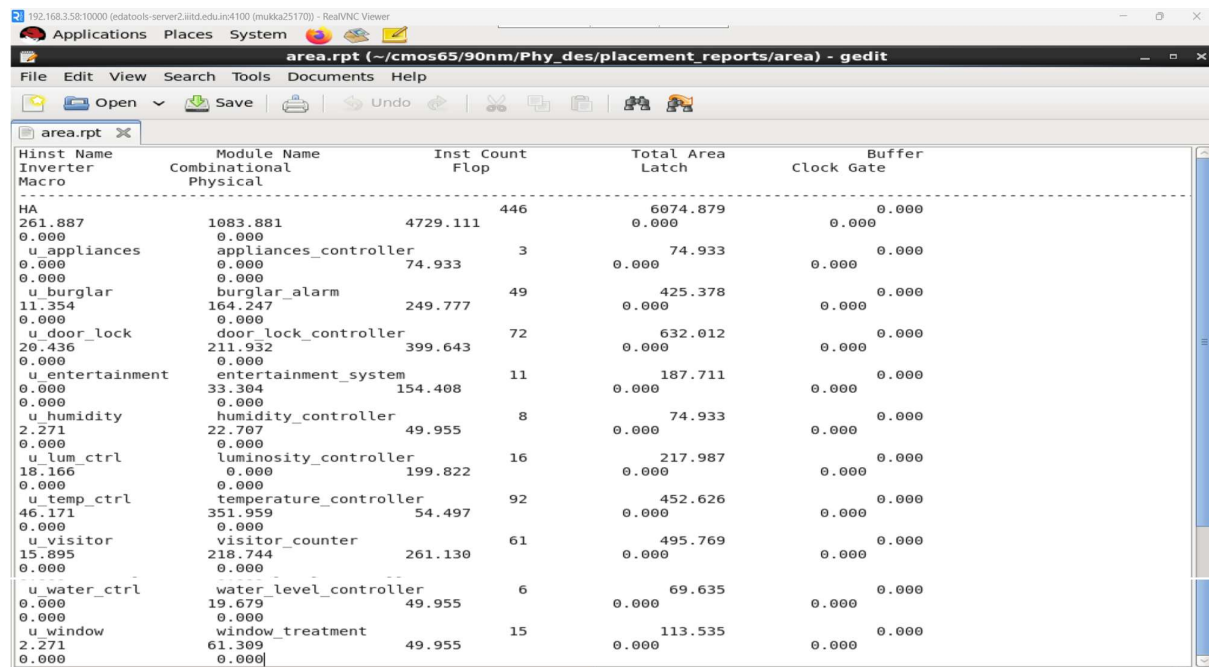
M9 routing:



Utilization Factor = 0.8

Placement reports:

1. Area:



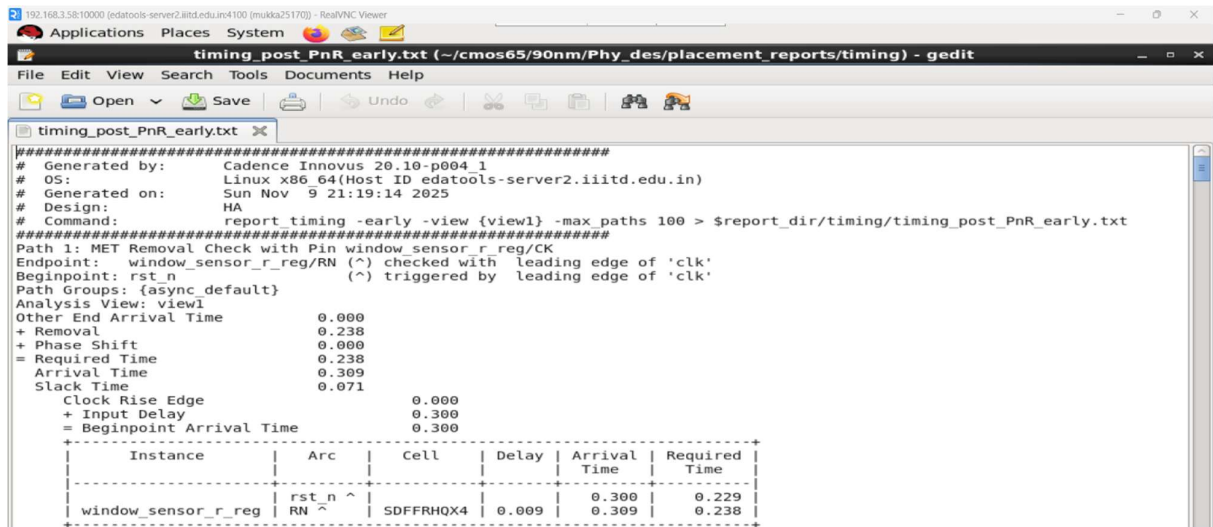
Hinst Name	Module Name	Inst Count	Total Area	Buffer
Inverter	Combinational	Flop	Latch	Clock Gate
Macro	Physical			
HA		446	6074.879	0.000
261.887	1083.881	4729.111	0.000	0.000
0.000	0.000			
u_appliances	appliances_controller	3	74.933	0.000
0.000	0.000	74.933	0.000	0.000
0.000	0.000			
u_burglar	burglar_alarm	49	425.378	0.000
11.354	164.247	249.777	0.000	0.000
0.000	0.000			
u_door_lock	door_lock_controller	72	632.012	0.000
20.436	211.932	399.643	0.000	0.000
0.000	0.000			
u_entertainment	entertainment_system	11	187.711	0.000
0.000	33.304	154.408	0.000	0.000
0.000	0.000			
u_humidity	humidity_controller	8	74.933	0.000
2.271	22.707	49.955	0.000	0.000
0.000	0.000			
u_lum_ctrl	luminosity_controller	16	217.987	0.000
18.166	0.000	199.822	0.000	0.000
0.000	0.000			
u_temp_ctrl	temperature_controller	92	452.626	0.000
46.171	351.959	54.497	0.000	0.000
0.000	0.000			
u_visitor	visitor_counter	61	495.769	0.000
15.895	218.744	261.130	0.000	0.000
0.000	0.000			
u_water_ctrl	water_level_controller	6	69.635	0.000
0.000	19.679	49.955	0.000	0.000
0.000	0.000			
u_window	window_treatment	15	113.535	0.000
2.271	61.309	49.955	0.000	0.000
0.000	0.000			

Area of Standard Cells

- Inverters: The report shows a total inverter area of 261.887. This value includes all placed inverter instances contributing to signal inversion throughout the design.
- Buffers: The dedicated buffer area is shown as 0.000. This means no isolated buffer cells (for pure drive strength) are present, or they are categorized differently in your report.
- Flip-Flops: The area taken by flip-flops (labeled as "Flop") sums to 4729.111. Flip-flops are used for state retention and sequential logic, and occupy the largest portion among standard cell types.
- Combinational Cells (Other): Cells such as multiplexers, gates, arithmetic units, etc. are grouped as "Combinational" and total 1083.881 in area. These implement various logic functions apart from inverters and buffers.
- Latches: The area for latches is listed as 446. Latches are commonly used for level-sensitive storage in timing-critical paths.
- Macros: "Macro" area is 0.000, suggesting no non-standard macro cells used.
- Total Standard Cell Area: The total area sums up as 6074.879, which is the sum of all above cell types deployed on silicon

2. Timing Reports

Hold slack:



```
#####
# Generated by: Cadence Innovus 20.10-p004.1
# OS: Linux x86_64 (Host ID: edatools-server2.iitd.edu.in)
# Generated on: Sun Nov 9 21:19:14 2025
# Design: HA
# Command: report timing -early -view {view1} -max_paths 100 > $report_dir/timing/timing_post_PnR_early.txt
#####
Path 1: MET Removal Check with Pin window_sensor_r_reg/CK
Endpoint: window_sensor_r_reg/RN (^) checked with leading edge of 'clk'
Beginpoint: rst_n (^) triggered by leading edge of 'clk'
Path Groups: {async default}
Analysis View: view1
Other End Arrival Time 0.000
+ Removal 0.238
+ Phase Shift 0.000
= Required Time 0.238
Arrival Time 0.309
Slack Time 0.071
Clock Rise Edge 0.000
+ Input Delay 0.300
= Beginpoint Arrival Time 0.300
#####
```

Instance	Arc	Cell	Delay	Arrival Time	Required Time
window_sensor_r_reg	rst_n ^	SDFFRHQX4	0.009	0.300	0.229
	RN ^			0.309	0.238

Arrival Time: 0.309 ns (when reset reaches the flip-flop)

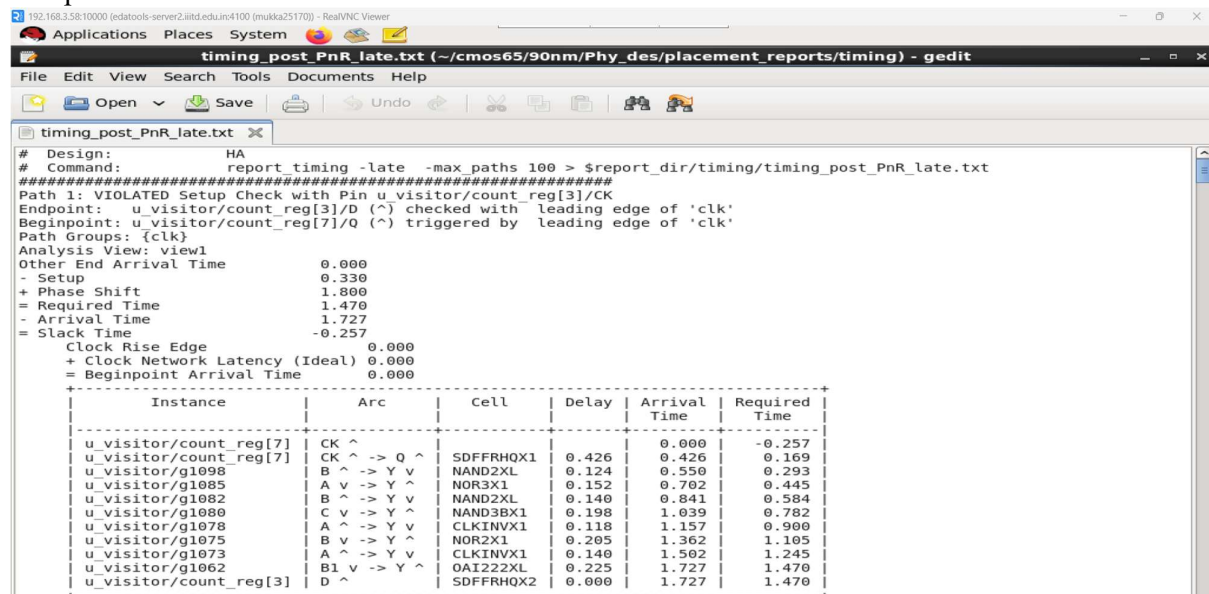
Required Time: 0.238 ns (latest safe de-assertion time)

Slack Calculation:

Slack = Arrival Time - Required Time

0.309 - 0.238 = +0.071 ns

Setup slack:



```
#####
# Design: HA
# Command: report timing -late -max_paths 100 > $report_dir/timing/timing_post_PnR_late.txt
#####
Path 1: VIOLATED Setup Check with Pin u_visitor/count_reg[3]/CK
Endpoint: u_visitor/count_reg[3]/D (^) checked with leading edge of 'clk'
Beginpoint: u_visitor/count_reg[7]/Q (^) triggered by leading edge of 'clk'
Path Groups: {clk}
Analysis View: view1
Other End Arrival Time 0.000
- Setup 0.330
+ Phase Shift 1.800
= Required Time 1.470
Arrival Time 1.727
= Slack Time -0.257
Clock Rise Edge 0.000
+ Clock Network Latency (Ideal) 0.000
= Beginpoint Arrival Time 0.000
#####
```

Instance	Arc	Cell	Delay	Arrival Time	Required Time
u_visitor/count_reg[7]	CK ^	SDFFRHQX1	0.426	0.426	0.169
u_visitor/count_reg[7]	CK ^ -> Q ^			0.550	0.293
u_visitor/g1098	B ^ -> Y v	NAND2XL	0.124	0.702	0.445
u_visitor/g1085	A v -> Y ^	NOR3X1	0.152	0.841	0.584
u_visitor/g1082	B ^ -> Y v	NAND2XL	0.140	1.039	0.782
u_visitor/g1080	C v -> Y ^	NAND3BX1	0.198	1.157	0.900
u_visitor/g1079	A ^ -> Y v	CLKINVX1	0.118	1.362	1.105
u_visitor/g1075	B v -> Y ^	NOR2X1	0.205	1.502	1.245
u_visitor/g1073	A ^ -> Y v	CLKINVX1	0.140	1.727	1.470
u_visitor/g1062	B1 v -> Y ^	OAI222XL	0.225	1.727	1.470
u_visitor/count_reg[3]	D ^	SDFFRHQX2	0.000	1.727	1.470

Required Time: 1.470 ns (latest safe arrival time)

Arrival Time: 1.727 ns (actual data arrival time)

Slack Calculation:

Slack = Required Time - Arrival Time

1.470 - 1.727 = -0.257 ns

3. Power report

192.168.3.58:10000 (edatools-server2.mind.edu.cn:4100 (mukka25170)) - RealVNC Viewer

Applications Places System

power.rpt (~/.cmos65/90nm/Phy_des/placement_reports/power) - gedit

File Edit View Search Tools Documents Help

Open Save Undo Redo Print Copy Paste

power.rpt

```
-----
Innovos 20.10-p004_1 (64bit) 05/07/2020 20:02 (Linux 2.6.32-431.11.2.el6.x86_64)
*
*
*
Date & Time: 2025-Nov-09 21:19:14 (2025-Nov-09 15:49:14 GMT)
*
*
-----
Design: HA
*
Liberty Libraries used:
view1: /home/mukka25170/cmos65/90nm/Phy_des/slow.lib
*
Power Domain used:
Rail: VDD Voltage: 0.9
*
Power View : view1
*
User-Defined Activity : N.A.
*
Activity File: N.A.
*
Hierarchical Global Activity: N.A.
*
Global Activity: N.A.
*
Sequential Element Activity: N.A.
*
Primary Input Activity: 0.200000
*
Default icg ratio: N.A.
*
Global Comb ClockGate Ratio: N.A.
*
*
Power Units = 1mW
Time Units = 1e-09 secs
Temperature = 125
*
report_power -outfile placement_reports/power/power.rpt -rail_analysis_format VS
*
-----
```

cell	Internal Power	Switching Power	Total Power	Leakage Power	Cell Name
water_valve_close_r_reg	0.04581	0.002113	0.04823	0.0003009	SDFFRHQX8
visitor_count_r_reg[3]	0.04635	0.00156	0.04821	0.0003009	SDFFRHQX8
visitor_count_r_reg[2]	0.04637	0.001402	0.04807	0.0003009	SDFFRHQX8
visitor_count_r_reg[6]	0.04641	0.001128	0.04784	0.0003009	SDFFRHQX8
visitor_count_r_reg[1]	0.04639	0.001112	0.0478	0.0003009	SDFFRHQX8
visitor_count_r_reg[7]	0.04544	0.001422	0.04717	0.0003009	SDFFRHQX8
visitor_count_r_reg[5]	0.04532	0.001404	0.04702	0.0003009	SDFFRHQX8
visitor_count_r_reg[0]	0.04494	0.001377	0.04662	0.0003009	SDFFRHQX8
volume_level_r_reg[0]	0.04378	0.001454	0.04554	0.0003009	SDFFRHQX8
visitor_count_r_reg[4]	0.0434	0.0008572	0.04455	0.0003009	SDFFRHQX8
volume_level_r_reg[1]	0.04282	0.00107	0.04419	0.0003009	SDFFRHQX8
volume_level_r_reg[2]	0.04247	0.00137	0.04414	0.0003009	SDFFRHQX8
lock_status_r_reg[1]	0.04251	0.0008939	0.0437	0.0003009	SDFFRHQX8
lock_status_r_reg[0]	0.04238	0.001044	0.04363	0.0003009	SDFFRHQX8
volume_level_r_reg[3]	0.04232	0.0009971	0.04362	0.0003009	SDFFRHQX8
led_brightness_r_reg[4]	0.04156	0.001262	0.04313	0.0003009	SDFFRHQX8
led_brightness_r_reg[3]	0.04152	0.0008897	0.04271	0.0003009	SDFFRHQX8
led_brightness_r_reg[5]	0.04087	0.0009758	0.04214	0.0003009	SDFFRHQX8
u_burglar/g717	0.0001513	7.813e-05	0.0002387	9.289e-06	NAND2XL
u_visitor/g1085	0.0001792	3.045e-05	0.00023	2.034e-05	NOR3X1
u_temp_ctrl/g1685	0.0001168	8.967e-05	0.0002164	9.93e-06	NAND3XL
u_temp_ctrl/g167	6.506e-05	0.0001196	0.0001939	9.289e-06	NAND2XL
u_door_lock/g2257	0.0001271	3.173e-05	0.0001712	1.236e-05	NOR2XL
u_temp_ctrl/g1665	7.62e-05	5.491e-05	0.0001416	1.046e-05	NAND4XL
u_visitor/g1057	3.037e-05	7.712e-05	0.0001168	9.289e-06	NAND2XL
u_burglar/g710	3.131e-05	3.051e-05	0.0001044	4.256e-05	OR2X1
u_burglar/g714	6.136e-05	2.84e-05	0.0001028	1.301e-05	INVX1
u_burglar/g715	3.971e-05	5.113e-05	0.0001013	1.046e-05	NAND4XL
u_door_lock/g2258	2.687e-05	1.805e-05	8.091e-05	3.499e-05	NAND2BX1
u_door_lock/g2245	1.328e-05	9.839e-06	5.811e-05	3.499e-05	NAND2BX1
u_burglar/g701	2.25e-05	9.145e-06	4.4e-05	1.236e-05	NOR2XL
u_door_lock/g2243	2.475e-05	6.638e-06	4.375e-05	1.236e-05	NOR2XL
u_burglar/g702	4.375e-06	4.07e-06	4.343e-05	3.499e-05	NAND2BX1
u_door_lock/g2240	2.119e-06	1.337e-06	3.844e-05	3.499e-05	NAND2BX1
u_burglar/g689	1.151e-06	6.926e-07	3.683e-05	3.499e-05	NAND2BX1
u_burglar/g692	4.64e-07	2.512e-07	3.491e-05	3.419e-05	NAND3X1
u_burglar/g705	3.53e-06	1.827e-05	3.481e-05	1.301e-05	CLKINX1
u_burglar/g713	1.213e-05	7.449e-06	3.368e-05	1.41e-05	OAI211X1
u_door_lock/g2238	1.117e-05	3.254e-06	2.678e-05	1.236e-05	NOR2XL
u_burglar/g707	4.839e-06	3.11e-06	2.031e-05	1.236e-05	NOR2XL
u_visitor/g1082	4.676e-06	5.925e-06	1.989e-05	9.289e-06	NAND2XL
u_burglar/g678	4.282e-06	1.76e-06	1.84e-05	1.236e-05	NOR2XL
u_door_lock/g2235	1.432e-06	1.056e-06	1.55e-05	1.301e-05	CLKINX1
u_burglar/g684	2.944e-07	1.151e-07	1.277e-05	1.236e-05	NOR2XL
u_burglar/g703	1.927e-07	6.28e-08	1.261e-05	1.236e-05	NOR2XL

Total (446 of 446)	4.254	0.5005	4.794	0.03948	
Total Capacitance	3.796e-12 F				
Power Density	*** No Die Area ***				

Power Distribution

Total Power: 4.794mW

Internal Power: 4.254mW (88.7%) - within cells during switching

Switching Power: 0.5005mW (10.4%) - capacitive charging

Leakage Power: 0.03948mW (0.9%) - static power

CTS Reports:

1. Area report

Hinst Name	Module Name	Inst Count	Total Area	Buffer
Inverter	Combinational	Flop	Latch	Clock Gate
Macro	Physical			
HA		465	6130.133	66.607
176.358	1145.190	4741.979	0.000	0.000
0.000	0.000			
u.appliances	appliances_controller	3	74.933	0.000
0.000	0.000	74.933	0.000	0.000
0.000	0.000			
u.burglar	burglar_alarm	49	426.892	0.000
11.354	165.761	249.777	0.000	0.000
0.000	0.000			
u.door_lock	door_lock_controller	73	638.067	0.000
20.436	217.987	399.643	0.000	0.000
0.000	0.000			
u.entertainment	entertainment_system	11	187.711	0.000
0.000	33.304	154.408	0.000	0.000
0.000	0.000			
u.humidity	humidity_controller	8	74.933	0.000
2.271	22.707	49.955	0.000	0.000
0.000	0.000			
u.lum_ctrl	luminosity_controller	16	217.987	0.000
18.166	0.000	199.822	0.000	0.000
0.000	0.000			
u.temp_ctrl	temperature_controller	100	501.825	0.000
59.038	388.290	54.497	0.000	0.000
0.000	0.000			
u.visitor	visitor_counter	65	536.642	0.000
26.492	236.153	273.998	0.000	0.000
0.000	0.000			
u.water_ctrl	water_level_controller	6	69.635	0.000
0.000	19.679	49.955	0.000	0.000
0.000	0.000			
u.window	window_treatment	15	113.535	0.000
2.271	61.309	49.955	0.000	0.000
0.000	0.000			

- Inverters: The inverter area is 176.358. These provide signal inversion across the design and are usually widely distributed.
- Buffers: Buffers are now reported with a total area of 66.607, indicating explicit buffer insertion (often post-CTS or for drive strength).
- Flip-Flops: The total area for flip-flops is 4741.979, which again represents the largest block of standard cell area, corresponding to state elements.
- Combinational Cells (Other): Cells for general logic (AND, OR, NAND, arithmetic, etc.) collectively consume 1145.190 in area, up from the previous stage, possibly reflecting optimization or mapping changes.
- Latches: The area for latches is listed as 465, showing a minor increase likely due to further register retiming or similar optimizations.
- Macros: Macro area is 0.000, so no black-box memory macros or similar large blocks are used.
- Total Standard Cell Area: The cumulative area utilized by all standard cells is 6130.133, reflecting design changes after place-and-route steps.

2. Timing Report:

Hold slack

```

192.168.3.58:10000 (edatools-server2.iiitd.edu.in:4100 (mukka25170)) - RealVNC Viewer
Applications Places System
timing_post_PnR_early.txt (~/.cmos65/90nm/Phy_des/cts_reports/timing) - gedit
File Edit View Search Tools Documents Help
Open Save Undo
timing_post_PnR_early.txt
#####
# Generated by: Cadence Innovus 20.10-p004.1
# OS: Linux x86_64 (Host ID edatools-server2.iiitd.edu.in)
# Generated on: Sun Nov 9 21:20:02 2025
# Design: HA
# Command: report timing -early -view {view1} -max_paths 100 > cts_reports/timing/timing_post_PnR_early.txt
#####
Path 1: MET Removal Check with Pin u_burglar/delay_counter_reg[2]/CK
Endpoint: u_burglar/delay_counter_reg[2]/RN (^) checked with leading edge of
'clk'
Beginpoint: rst_n (^) triggered by leading edge of
'clk'
Path Groups: {async default}
Analysis View: view1
Other End Arrival Time -0.004
+ Removal 0.207
+ Phase Shift 0.000
= Required Time 0.203
Arrival Time 0.307
Slack Time 0.103
Clock Rise Edge 0.000
+ Input Delay 0.300
= Beginpoint Arrival Time 0.300
+-----+-----+-----+-----+-----+-----+
| Instance | Arc | Cell | Delay | Arrival Time | Required Time |
+-----+-----+-----+-----+-----+-----+
| u_burglar/delay_counter_reg[2] | rst_n ^ | SDFFRHQX1 | 0.007 | 0.300 | 0.197 |
| | RN ^ | | | 0.307 | 0.203 |
+-----+-----+-----+-----+-----+-----+

```

Arrival Time: 0.307 ns (when reset reaches the flip-flop)

Required Time: 0.203 ns (latest safe de-assertion time)

Slack Calculation:

Slack = Arrival Time - Required Time

0.307 - 0.203 = +0.104 ns

Setup slack:

```

192.168.3.58:10000 (edatools-server2.iiitd.edu.in:4100 (mukka25170)) - RealVNC Viewer
#####
# Generated by: Cadence Innovus 20.10-p004.1
# OS: Linux x86_64 (Host ID edatools-server2.iiitd.edu.in)
# Generated on: Sun Nov 9 21:20:02 2025
# Design: HA
# Command: report timing -late -max_paths 100 > cts_reports/timing/timing_post_PnR_late.txt
#####
Path 1: MET Setup Check with Pin u_temp_ctrl/cooler_on reg/CK
Endpoint: u_temp_ctrl/cooler_on reg/D (^) checked with leading edge of 'clk'
Beginpoint: temp_setpoint_r_reg[4]/Q (^) triggered by leading edge of 'clk'
Path Groups: {clk}
Analysis View: view1
Other End Arrival Time 0.000
- Setup 0.216
+ Phase Shift 1.800
= Required Time 1.584
Arrival Time 1.549
= Slack Time 0.035
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) 0.000
= Beginpoint Arrival Time 0.000
+-----+-----+-----+-----+-----+-----+
| Instance | Arc | Cell | Delay | Arrival Time | Required Time |
+-----+-----+-----+-----+-----+-----+
| temp_setpoint_r_reg[4] | CK ^ | | | 0.000 | 0.036 |
| temp_setpoint_r_reg[4] | CK ^ -> Q ^ | SDFFRHQX4 | 0.437 | 0.437 | 0.473 |
| u_temp_ctrl/g1715 | B ^ -> Y v | NAND2X2 | 0.080 | 0.517 | 0.552 |
| u_temp_ctrl/g1712 | A v -> Y ^ | CLKINVX1 | 0.061 | 0.578 | 0.614 |
| u_temp_ctrl/g1685 | B ^ -> Y v | NAND3XL | 0.141 | 0.720 | 0.755 |
| u_temp_ctrl/g1664 | A1 v -> Y ^ | AOI21X1 | 0.195 | 0.914 | 0.950 |
| u_temp_ctrl/g1648 | A1 ^ -> Y v | AOI21X1 | 0.099 | 1.013 | 1.049 |
| u_temp_ctrl/FE RC 10 0 | A v -> Y ^ | CLKINVX1 | 0.056 | 1.069 | 1.105 |
| u_temp_ctrl/FE RC 8 0 | A1N ^ -> Y ^ | AOI2BB2X1 | 0.189 | 1.258 | 1.294 |
| u_temp_ctrl/g1636 | A ^ -> Y v | NAND2X2 | 0.107 | 1.365 | 1.400 |
| u_temp_ctrl/g1635 | A v -> Y ^ | INVX1 | 0.059 | 1.424 | 1.459 |
| u_temp_ctrl/g1632 | A ^ -> Y v | NAND2X1 | 0.067 | 1.491 | 1.527 |
| u_temp_ctrl/g1631 | B0 v -> Y ^ | AOI21X1 | 0.057 | 1.549 | 1.584 |
| u_temp_ctrl/cooler_on reg | D ^ | SDFFRXL | 0.000 | 1.549 | 1.584 |
+-----+-----+-----+-----+-----+-----+

```

Arrival Time: 0.307 ns (when reset reaches the flip-flop)

Required Time: 0.203 ns (latest safe de-assertion time)

Slack Calculation:

Slack = Arrival Time - Required Time

0.307 - 0.203 = +0.104 ns

3. Power Report

192.168.3.58:10000 (edatools-server2.lit.edu.in:4100 (mukka25170)) - RealVNC Viewer

Applications Places System

ctsafteranalysis.rpt (~/.cmos65/90nm/Phy_des/cts_reports) - gedit

File Edit View Search Tools Documents Help

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ctsafteranalysis.rpt

```
-----
Innovus 20.10-p004_1 (64bit) 05/07/2020 20:02 (Linux 2.6.32-431.11.2.el6.x86_64)
*
*
* Date & Time: 2025-Nov-09 21:20:02 (2025-Nov-09 15:50:02 GMT)
*
*
* -----
*
* Design: HA
*
* Liberty Libraries used:
*   view1: /home/mukka25170/cmos65/90nm/Phy_des/slow.lib
*
* Power Domain used:
*   Rail: VDD Voltage: 0.9
*
* Power View : view1
*
* User-Defined Activity : N.A.
*
* Activity File: N.A.
*
* Hierarchical Global Activity: N.A.
*
* Global Activity: N.A.
*
* Sequential Element Activity: 0.200000
*
* Primary Input Activity: 0.200000
*
* Default icg ratio: N.A.
*
* Global Comb ClockGate Ratio: N.A.
*
* Power Units = 1mW
*
* Time Units = 1e-09 secs
*
* Temperature = 125
*
* report_power -outfile cts_reports/ctsafteranalysis.rpt -rail_analysis_format VS
*
* -----
```

Cell	Internal Power	Switching Power	Total Power	Leakage Power	Cell Name
CTS_ccl_a_buf_00006	0.07976	0.08821	0.1682	0.0001864	CLKBUF16
CTS_ccl_a_buf_00004	0.07975	0.08759	0.1675	0.0001864	CLKBUF16
FE_OF04_rst_n	0.001582	0.02049	0.02215	7.504e-05	CLKBUF4
led_brightness_r_reg[0]	0.01976	0.0002886	0.02035	0.0003009	SDFFRHQX8
window_state_r_reg[0]	0.01938	0.0005358	0.02021	0.0003009	SDFFRHQX8
window_state_r_reg[1]	0.01939	0.0004812	0.02017	0.0003009	SDFFRHQX8
water_valve_close_r_reg	0.0194	0.0004138	0.02011	0.0003009	SDFFRHQX8
visitor_count_r_reg[3]	0.01949	0.0003046	0.0201	0.0003009	SDFFRHQX8
dehumidifier_on_r_reg	0.01941	0.000366	0.02008	0.0003009	SDFFRHQX8
humidifier_on_r_reg	0.01943	0.0003519	0.02008	0.0003009	SDFFRHQX8
music_on_r_reg	0.01941	0.0003616	0.02007	0.0003009	SDFFRHQX8
FE_OF05_scan_en	0.001583	0.01841	0.02006	7.504e-05	CLKBUF4
led_brightness_r_reg[4]	0.01948	0.0002804	0.02006	0.0003009	SDFFRHQX8
led_brightness_r_reg[2]	0.01948	0.0002788	0.02006	0.0003009	SDFFRHQX8
visitor_count_r_reg[2]	0.01948	0.0002751	0.02006	0.0003009	SDFFRHQX8
visitor_count_r_reg[0]	0.01948	0.0002735	0.02005	0.0003009	SDFFRHQX8
tv_on_r_reg	0.01942	0.0003336	0.02005	0.0003009	SDFFRHQX8
led_brightness_r_reg[6]	0.01948	0.0002617	0.02005	0.0003009	SDFFRHQX8
led_brightness_r_reg[7]	0.01949	0.0002469	0.02003	0.0003009	SDFFRHQX8
volume_level_r_reg[0]	0.01943	0.0003049	0.02003	0.0003009	SDFFRHQX8

192.168.3.58:10000 (edatools-server2.lit.edu.in:4100 (mukka25170)) - RealVNC Viewer

Applications Places System

ctsafteranalysis.rpt (~/.cmos65/90nm/Phy_des/cts_reports) - gedit

File Edit View Search Tools Documents Help

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ctsafteranalysis.rpt

```
u_burglar/FE_OF00_n_6 0.0001392 9.197e-05 0.0002442 1.301e-05 INVX1
u_door_lock/g2257 0.0001847 4.576e-05 0.0002428 1.236e-05 NOR2XL
u_temp_ctrl/g1660 0.000137 8.055e-05 0.0002269 9.289e-06 NAND2XL
u_visitor/g1077 0.0001092 8.87e-05 0.000216 1.816e-05 NOR2X1
u_temp_ctrl/g1700 0.0001297 7.266e-05 0.0002117 9.289e-06 NAND2XL
u_visitor/g1086 5.621e-05 0.0001248 0.0001903 9.289e-06 NAND2XL
u_temp_ctrl/g1694 8.81e-05 8.701e-05 0.0001844 9.289e-06 NAND2XL
u_burglar/g701 0.0001156 4.666e-05 0.0001746 1.236e-05 NOR2XL
u_burglar/g713 9.36e-05 6.382e-05 0.0001715 1.41e-05 OAI211X1
u_temp_ctrl/FE_RC_13_0 8.988e-05 6.802e-05 0.0001672 9.289e-06 NAND2XL
u_door_lock/g2243 0.0001154 3.316e-05 0.0001609 1.236e-05 NOR2XL
u_burglar/g689 7.749e-05 4.703e-05 0.0001595 3.499e-05 NAND2BX1
u_door_lock/g2240 7.72e-05 4.553e-05 0.0001577 3.499e-05 NAND2BX1
u_visitor/g1057 5.344e-05 8.648e-05 0.000154 1.403e-05 NAND2X1
u_temp_ctrl/g1697 5.112e-05 5.727e-05 0.0001224 1.403e-05 NAND2X1
u_temp_ctrl/g1685 5.835e-05 4.701e-05 0.0001153 9.93e-06 NAND3XL
u_burglar/g692 4.789e-05 2.8e-05 0.0001101 3.419e-05 NAND3BX1
u_burglar/g707 5.438e-05 4.308e-05 0.0001098 1.236e-05 NOR2XL
u_door_lock/g2238 6.914e-05 1.79e-05 9.939e-05 1.236e-05 NOR2XL
u_door_lock/g2235 4.932e-05 3.629e-05 9.863e-05 1.301e-05 CLKINVX1
u_temp_ctrl/FE_RC_12_0 6.405e-05 1.559e-05 9.78e-05 1.816e-05 NOR2X1
u_temp_ctrl/g1665 3.794e-05 2.612e-05 7.451e-05 1.046e-05 NAND4XL
u_burglar/g678 4.022e-05 1.63e-05 6.888e-05 1.236e-05 NOR2XL
u_visitor/g1082 1.742e-05 2.121e-05 5.266e-05 1.403e-05 NAND2X1
u_visitor/g1064 2.026e-05 1.642e-05 4.597e-05 9.289e-06 NAND2XL
u_burglar/g684 2.307e-05 8.959e-06 4.439e-05 1.236e-05 NOR2XL
u_burglar/g703 2.364e-05 7.728e-06 4.372e-05 1.236e-05 NOR2XL
```

Total (465 of 465) 2.781 0.3986 3.219 0.0394

Total Capacitance 5.284e-12 F

Power Density *** No Die Area ***

Plain Text v Phy_des b Ln 1, Col 1 INS

[mukka... [virtuo... mukka... [cmos... 90nm [mukka... [Innov... Phy_des cts_re... ctsafte...

Power Distribution

Total Power: 3.219mW

Internal Power: 2.781mW (86.4%)

Switching Power: 0.3986mW (12.4%)

Leakage Power: 0.0394mW (1.2%)

Optimized CTS reports:

1. Area Report

Hinst Name	Module Name	Inst Count	Total Area	Latch	Buffer
HA	Combinational	465	5797.854		64.337
176.358	Physical		0.000		0.000
0.000					
u_appliances	appliances_controller	3	74.933	0.000	0.000
0.000					
0.000					
u_burglar	burglar_alarm	49	426.892	0.000	0.000
11.354					
0.000					
u_door_lock	door_lock_controller	73	638.067	0.000	0.000
20.436					
0.000					
u_entertainment	entertainment_system	11	187.711	0.000	0.000
0.000					
0.000					
u_humidity	humidity_controller	8	74.933	0.000	0.000
2.271					
0.000					
u_lum_ctrl	luminescence_controller	16	217.987	0.000	0.000
18.166					
0.000					
u_temp_ctrl	temperature_controller	100	501.825	0.000	0.000
59.038					
0.000					
u_visitor	visitor_counter	65	535.885	0.000	0.000
26.492					
0.000					
u_water_ctrl	water_level_controller	6	69.635	0.000	0.000

- Inverters: Inverters take up an area of 176.358, used for signal inversion tasks throughout the design.
- Buffers: Buffers have a total area of 64.337, implemented for signal integrity and drive-strength reasons.
- Flip-Flops: The total flip-flop area is 4411.979, representing the primary storage (state-keeping) elements used in sequential design.
- Combinational Cells (Other): General logic gates and arithmetic/other complex combinational cells collectively take 1145.190 area, responsible for the main logic implementation.
- Latches: Latch area is 465, for level-sensitive storage and timing adjustments.
- Macros: Macro area remains 0.000, meaning there are no embedded memories or custom macros.
- Total Standard Cell Area: The sum of all standard cell areas amounts to 5797.854, showing overall silicon occupied by these cells.

2. Timing Report

Setup slack:

```

192.168.3.58:10000 (edatools-server2.iitd.edu.in:4100 (mukka25170)) - RealVNC Viewer
#####
# Generated by: Cadence Innovus 20.10-p004_1
# OS: Linux x86_64 (Host ID: edatools-server2.iitd.edu.in)
# Generated on: Sun Nov 9 21:20:56 2025
# Design: HA
# Command: report_timing -late -max_paths 100 > optimised_cts_reports/timing/timing_post_PnR_late.txt
#####
Path 1: MET Setup Check with Pin u_temp_ctrl/cooler_on reg/CK
Endpoint: u_temp_ctrl/cooler_on reg/D (^) checked with leading edge of 'clk'
Beginpoint: temp_setpoint_r_reg[4]/Q (^) triggered by leading edge of 'clk'
Path Groups: {clk}
Analysis View: view1
Other End Arrival Time 0.000
- Setup 0.216
+ Phase Shift 1.800
= Required Time 1.584
- Arrival Time 1.548
= Slack Time 0.036
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) 0.000
= Beginpoint Arrival Time 0.000
+-----+-----+-----+-----+-----+-----+
| Instance | Arc | Cell | Delay | Arrival Time | Required Time |
+-----+-----+-----+-----+-----+-----+
| temp_setpoint_r_reg[4] | CK ^ -> Q ^ | SDFFRH0X4 | 0.437 | 0.437 | 0.473 |
| temp_setpoint_r_reg[4] | B ^ -> Y v | NAND2X2 | 0.080 | 0.517 | 0.553 |
| u_temp_ctrl/g1715 | A v -> Y ^ | CLKINVX1 | 0.061 | 0.578 | 0.614 |
| u_temp_ctrl/g1712 | B ^ -> Y v | NAND3XL | 0.141 | 0.719 | 0.755 |
| u_temp_ctrl/g1685 | A1 v -> Y v | AOI21X1 | 0.195 | 0.914 | 0.950 |
| u_temp_ctrl/g1664 | A1 ^ -> Y v | OAI21X1 | 0.099 | 1.013 | 1.049 |
| u_temp_ctrl/g1648 | A v -> Y ^ | CLKINVX1 | 0.056 | 1.069 | 1.105 |
| u_temp_ctrl/FE_RC_10_0 | A1N ^ -> Y ^ | AOI2BB2X1 | 0.189 | 1.258 | 1.294 |
| u_temp_ctrl/g1636 | A ^ -> Y v | NAND2X2 | 0.107 | 1.364 | 1.400 |
| u_temp_ctrl/g1635 | A v -> Y ^ | INVX1 | 0.059 | 1.423 | 1.459 |
| u_temp_ctrl/g1632 | A ^ -> Y v | NAND2X1 | 0.068 | 1.491 | 1.527 |
| u_temp_ctrl/g1631 | B0 v -> Y ^ | OAI21X1 | 0.057 | 1.548 | 1.584 |
| u_temp_ctrl/cooler_on reg | D ^ | SDFFRXL | 0.000 | 1.548 | 1.584 |
+-----+-----+-----+-----+-----+-----+
Path 2: MET Setup Check with Pin u_temp_ctrl/heater_on reg/CK
Endpoint: u_temp_ctrl/heater_on reg/D (^) checked with leading edge of 'clk'
Beginpoint: temp_setpoint_r_reg[3]/Q (v) triggered by leading edge of 'clk'
Path Groups: {clk}
Analysis View: view1

```

Required Time: 1.584 ns (latest safe arrival time)

Arrival Time: 1.548 ns (actual data arrival time)

Slack Calculation:

Slack = Required Time - Arrival Time

1.584 - 1.548 = +0.036 ns

Hold slack:

```

192.168.3.58:10000 (edatools-server2.iitd.edu.in:4100 (mukka25170)) - RealVNC Viewer
Applications Places System
timing_post_PnR_early.txt (~:/cmos65/90nm/Phy_des/optimised_cts_reports/timing) - gedit
File Edit View Search Tools Documents Help
Open Save Undo Redo
timing_post_PnR_early.txt
#####
# Generated by: Cadence Innovus 20.10-p004_1
# OS: Linux x86_64 (Host ID: edatools-server2.iitd.edu.in)
# Generated on: Sun Nov 9 21:20:56 2025
# Design: HA
# Command: report_timing -early -view {view1} -max_paths 100 > optimised_cts_reports/timing/timing_post_PnR_early.txt
#####
Path 1: MET Removal Check with Pin u_window/window_state_reg[1]/CK
Endpoint: u_window/window_state_reg[1]/RN (^) checked with leading edge of 'clk'
Beginpoint: rst_n (^) triggered by leading edge of 'clk'
Path Groups: {async default}
Analysis View: view1
Other End Arrival Time -0.004
+ Removal 0.207
+ Phase Shift 0.000
= Required Time 0.203
- Arrival Time 0.307
= Slack Time 0.104
Clock Rise Edge 0.000
+ Input Delay 0.300
= Beginpoint Arrival Time 0.300
+-----+-----+-----+-----+-----+-----+
| Instance | Arc | Cell | Delay | Arrival Time | Required Time |
+-----+-----+-----+-----+-----+-----+
| u_window/window_state_reg[1] | rst_n ^ | SDFFRH0X1 | 0.007 | 0.300 | 0.196 |
| | RN ^ | | | 0.307 | 0.203 |
+-----+-----+-----+-----+-----+-----+
Path 2: MET Removal Check with Pin u_burglar/delay_counter_reg[3]/CK
Endpoint: u_burglar/delay_counter_reg[3]/RN (^) checked with leading edge of

```

Arrival Time: 0.307 ns (when reset reaches the flip-flop)

Required Time: 0.203 ns (latest safe de-assertion time)

Slack Calculation:

Slack = Arrival Time - Required Time

0.307 - 0.203 = +0.104 ns

3. Power Report

Innovus 20.10-p004_1 (64bit) 05/07/2020 20:02 (Linux 2.6.32-431.11.2.el6.x86_64)					
Date & Time: 2025-Nov-09 21:20:57 (2025-Nov-09 15:50:57 GMT)					
Design: HA					
Liberty Libraries used:					
view1: /home/mukka25170/cmos65/90nm/Phy_des/slow.lib					
Power Domain used:					
Rail: VDD Voltage: 0.9					
Power View : view1					
User-Defined Activity : N.A.					
Activity File: N.A.					
Hierarchical Global Activity: N.A.					
Global Activity: N.A.					
Sequential Element Activity: 0.200000					
Primary Input Activity: 0.200000					
Default icg ratio: N.A.					
Global Comb ClockGate Ratio: N.A.					
Power Units = 1mW					
Time Units = 1e-09 secs					
Temperature = 125					
report_power -outfile optimised_cts_reports/ctsafteranalysispostoptimization.rpt -rail_analysis_format VS					
Cell	Internal Power	Switching Power	Total Power	Leakage Power	Cell Name
CTS_ccl_a_buf_00006	0.07976	0.08834	0.1683	0.0001864	CLKBUF16
CTS_ccl_a_buf_00004	0.07976	0.08781	0.1677	0.0001864	CLKBUF16
FE_OFCS_scan_en	0.001583	0.01838	0.02004	7.504e-05	CLKBUF4
u_visitor/count_reg[5]	0.01685	0.001388	0.01846	0.0002171	SDFFRHQX4
u_visitor/count_reg[0]	0.01693	0.001132	0.01828	0.0002171	SDFFRHQX4
u_visitor/count_reg[1]	0.01698	0.0009845	0.01818	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[4]	0.01632	0.001583	0.01812	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[1]	0.01632	0.001319	0.01786	0.0002171	SDFFRHQX4
temp_sensor_r_reg[5]	0.01664	0.0009644	0.01782	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[5]	0.01632	0.001276	0.01782	0.0002171	SDFFRHQX4
window_auto_mode_r_reg	0.01664	0.0009138	0.01777	0.0002171	SDFFRHQX4
humidity_sensor_r_reg[1]	0.01664	0.0009027	0.01776	0.0002171	SDFFRHQX4
temp_sensor_r_reg[2]	0.01664	0.0008997	0.01776	0.0002171	SDFFRHQX4
temp_sensor_r_reg[3]	0.01664	0.0008551	0.01771	0.0002171	SDFFRHQX4
volume_cmd_r_reg[2]	0.01664	0.0008422	0.0177	0.0002171	SDFFRHQX4
volume_cmd_r_reg[3]	0.01664	0.0008134	0.01767	0.0002171	SDFFRHQX4
entry_sensor_r_reg	0.01664	0.0007923	0.01765	0.0002171	SDFFRHQX4
volume_cmd_r_reg[1]	0.01665	0.000757	0.01762	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[3]	0.01633	0.001066	0.01761	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[2]	0.01633	0.001055	0.0176	0.0002171	SDFFRHQX4
volume_cmd_r_reg[0]	0.01665	0.0007389	0.0176	0.0002171	SDFFRHQX4
light_sensor_r_reg[6]	0.01665	0.0007303	0.0176	0.0002171	SDFFRHQX4
light_sensor_r_reg[2]	0.01668	0.0006787	0.01758	0.0002171	SDFFRHQX4
light_sensor_r_reg[1]	0.01668	0.0006745	0.01757	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[0]	0.01668	0.0006723	0.01757	0.0002171	SDFFRHQX4
light_sensor_r_reg[5]	0.01665	0.0006807	0.01755	0.0002171	SDFFRHQX4
exit_sensor_r_reg	0.01665	0.0006754	0.01754	0.0002171	SDFFRHQX4
light_sensor_r_reg[3]	0.01668	0.0006386	0.01754	0.0002171	SDFFRHQX4
light_sensor_r_reg[7]	0.01665	0.0006691	0.01754	0.0002171	SDFFRHQX4
light_sensor_r_reg[4]	0.01665	0.0006557	0.01753	0.0002171	SDFFRHQX4
temp_sensor_r_reg[4]	0.01665	0.0006505	0.01752	0.0002171	SDFFRHQX4
humidity_sensor_r_reg[7]	0.01665	0.0006414	0.01751	0.0002171	SDFFRHQX4
humidity_sensor_r_reg[3]	0.01666	0.0005354	0.01741	0.0002171	SDFFRHQX4
tv_on_cmd_r_reg	0.01666	0.0005238	0.0174	0.0002171	SDFFRHQX4
music_on_cmd_r_reg	0.01666	0.0005047	0.01738	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[6]	0.01634	0.0008291	0.01738	0.0002171	SDFFRHQX4
light_sensor_r_reg[0]	0.0167	0.0004694	0.01738	0.0002171	SDFFRHQX4
temp_sensor_r_reg[6]	0.01634	0.0008149	0.01737	0.0002171	SDFFRHQX4
temp_sensor_r_reg[0]	0.01666	0.0004879	0.01737	0.0002171	SDFFRHQX4
humidity_sensor_r_reg[5]	0.01666	0.0004806	0.01736	0.0002171	SDFFRHQX4
door_lock_cmd_r_reg	0.01666	0.0004632	0.01734	0.0002171	SDFFRHQX4
window_manual_open_r_reg	0.01666	0.0004584	0.01734	0.0002171	SDFFRHQX4

u_temp_ctrl/g1667	0.0002545	2.831e-05	0.0003037	2.093e-05	NOR4X1
u_door_lock/g2287	0.0001749	0.0001163	0.0003006	9.289e-06	NAND2XL
u_burglar/g705	0.0001629	0.0001111	0.0002964	1.612e-05	NAND4X1
u_window/g369	0.0001629	0.0001043	0.0002959	2.877e-05	NAND2BXL
u_temp_ctrl/g1706	0.0001785	7.18e-05	0.0002853	3.499e-05	NAND2BXL
u_burglar/g702	0.0001276	0.0001166	0.0002792	3.499e-05	NAND2BXL
u_burglar/g717	0.0001742	9.192e-05	0.0002754	9.289e-06	NAND2XL
u_temp_ctrl/g1729	0.0001681	9.135e-05	0.0002682	8.757e-06	INVXL
u_humidity/g204	0.0001672	9.226e-05	0.0002682	8.757e-06	INVXL
u_burglar/g705	3.99e-05	0.0002057	0.0002586	1.301e-05	CLKINVX1
u_door_lock/g2245	0.0001271	8.952e-05	0.0002516	3.499e-05	NAND2BXL
u_visitor/g1091	0.0001312	0.0001078	0.0002482	9.289e-06	NAND2XL
u_burglar/FE_0FC0_n_6	0.0001392	9.192e-05	0.0002442	1.301e-05	INVX1
u_door_lock/g2257	0.0001847	4.577e-05	0.0002429	1.236e-05	NOR2XL
u_temp_ctrl/g1660	0.000137	8.054e-05	0.0002269	9.289e-06	NAND2XL
u_visitor/g1077	0.0001092	8.859e-05	0.0002159	1.816e-05	NOR2X1
u_temp_ctrl/g1700	0.0001297	7.269e-05	0.0002117	9.289e-06	NAND2XL
u_visitor/g1006	5.62e-05	0.0001246	0.0001901	9.289e-06	NAND2XL
u_temp_ctrl/g1694	8.81e-05	8.704e-05	0.0001844	9.289e-06	NAND2XL
u_burglar/g701	0.0001156	4.669e-05	0.0001746	1.236e-05	NOR2XL
u_burglar/g713	9.36e-05	6.372e-05	0.0001714	1.41e-05	OAI211X1
u_temp_ctrl/FE_RC_13_0	8.988e-05	6.804e-05	0.0001672	9.289e-06	NAND2XL
u_door_lock/g2243	0.0001154	3.31e-05	0.0001608	1.236e-05	NOR2XL
u_burglar/g680	7.749e-05	4.702e-05	0.0001595	3.499e-05	NAND2BXL
u_door_lock/g2240	7.72e-05	0.0001595	0.0001577	3.499e-05	NAND2BXL
u_visitor/g1057	5.344e-05	8.602e-05	0.0001535	1.403e-05	NAND2X1
u_temp_ctrl/g1697	5.112e-05	5.73e-05	0.0001224	1.403e-05	NAND2X1
u_temp_ctrl/g1685	5.835e-05	4.697e-05	0.0001152	9.93e-06	NAND3XL
u_burglar/g692	4.789e-05	2.8e-05	0.0001101	3.419e-05	NAND3BXL
u_burglar/g707	5.437e-05	4.299e-05	0.0001097	1.236e-05	NOR2XL
u_door_lock/g2238	6.913e-05	1.945e-05	0.0001009	1.236e-05	NOR2XL
u_door_lock/g2235	4.932e-05	3.704e-05	9.938e-05	1.301e-05	CLKINVX1
u_temp_ctrl/FE_RC_12_0	6.405e-05	1.56e-05	9.78e-05	1.816e-05	NOR2X1
u_temp_ctrl/g1665	3.794e-05	2.609e-05	7.449e-05	1.046e-05	NAND4XL
u_burglar/g678	4.023e-05	1.63e-05	6.888e-05	1.236e-05	NOR2XL
u_visitor/g1082	1.742e-05	2.094e-05	5.238e-05	1.403e-05	NAND2X1
u_visitor/g1064	2.026e-05	1.64e-05	4.595e-05	9.289e-06	NAND2XL
u_burglar/g684	2.307e-05	8.963e-06	4.439e-05	1.236e-05	NOR2XL
u_burglar/g703	2.364e-05	7.732e-06	4.373e-05	1.236e-05	NOR2XL

Total (465 of 465)	2.592	0.391	3.018	0.03504	
Total Capacitance	5.109e-12 F				
Power Density	*** No Die Area ***				

Power Distribution

Total Power: 3.018mW (slightly reduced from previous CTS)

Internal Power: 2.592mW (85.9%)

Switching Power: 0.391mW (13.0%)

Leakage Power: 0.03504mW (1.1%)

Post route reports:

1. Area report:

Hinst Name	Module Name	Inst Count	Total Area	Buffer Macro	Inverter Physical
Combinational	Flop	Latch	Clock Gate		
HA		465	5797.854	64.337	
176.358	1145.190	4411.970	0.000	0.000	
0.000	0.000				
u_appliances	appliances_controller	3	74.933	0.000	0.000
0.000	0.000	74.933	0.000	0.000	0.000
0.000					
u_burglar	burglar_alarm	49	426.892	0.000	0.000
11.354	165.761	249.777	0.000	0.000	0.000
0.000					
u_door_lock	door_lock_controller	73	638.067	0.000	0.000
20.436	217.987	399.643	0.000	0.000	0.000
0.000					
u_entertainment	entertainment_system	11	187.711	0.000	0.000
0.000	33.304	154.408	0.000	0.000	0.000
0.000					
u_humidity	humidity_controller	8	74.933	0.000	0.000
2.271	22.707	49.955	0.000	0.000	0.000
0.000					
u_lum_ctrl	luminosity_controller	16	217.987	0.000	0.000
18.166	0.000	199.822	0.000	0.000	0.000
0.000					
u_temp_ctrl	temperature_controller	100	501.825	0.000	0.000
59.038	388.290	54.497	0.000	0.000	0.000
0.000					
u_visitor	visitor_counter	65	535.885	0.000	0.000
26.492	236.153	273.241	0.000	0.000	0.000
0.000					
u_water_ctrl	water_level_controller	6	69.635	0.000	0.000
0.000	19.679	49.955	0.000	0.000	0.000
0.000					
u_window	window_treatment	15	113.535	0.000	0.000
2.271	61.309	49.955	0.000	0.000	0.000
0.000					

- Inverters: The inverters occupy an area of 176.358, which are used for signal inversion throughout the circuit.
- Buffers: Total buffer area is 64.337, showing buffers inserted for drive strength or signal integrity improvement.
- Flip-Flops: The flip-flop area is 4411.970, which forms the largest group, mainly for sequential logic and state retention.

- Combinational Cells (Other): The combinational (other logic) cells take up 1145.190 area, crucial for logic implementation across all design modules.
- Latches: Area for latches is 465, supporting timing closure and level-sensitive data storage.
- Macros: Macro area is 0.000, meaning no hard macros or memory blocks.
- Total Standard Cell Area: The total area used by all standard cells is 5797.854, representing cumulative silicon utilization for all placed standard logic.

2. Timing Report:

Setup slack:

```
192.168.3.58:10000 (edatools-server2.iitd.edu.in:4100 (mukka25170)) - RealVNC Viewer
#####
# Generated by: Cadence Innovus 20.10-p004.1
# OS: Linux x86_64 (Host ID edatools-server2.iitd.edu.in)
# Generated on: Sun Nov 9 21:21:26 2025
# Design: HA
# Command: report_timing -late -max_paths 100 > post_route_reports/timing/timing_post_PnR_late.txt
#####
Path 1: MET Setup Check with Pin u_temp_ctrl/cooler_on_reg/CK
Endpoint: u_temp_ctrl/cooler_on_reg/D (^) checked with leading edge of 'clk'
Beginpoint: temp_setpoint_r_reg[4]/Q (^) triggered by leading edge of 'clk'
Path Groups: {clk}
Analysis View: view1
Other End Arrival Time -0.000
- Setup 0.216
+ Phase Shift 1.800
= Required Time 1.584
- Arrival Time 1.541
= Slack Time 0.042
Clock Rise Edge 0.000
+ Clock Network Latency (Prop) -0.001
= Beginpoint Arrival Time -0.001
+-----+-----+-----+-----+-----+-----+
| Instance | Arc | Cell | Delay | Arrival Time | Required Time |
+-----+-----+-----+-----+-----+-----+
| temp_setpoint_r_reg[4] | CK ^ | SDFFRH0X4 | 0.436 | 0.435 | 0.477 |
| temp_setpoint_r_reg[4] | CK ^ -> Q ^ | SDFFRH0X4 | 0.436 | 0.435 | 0.477 |
| u_temp_ctrl/g1715 | B ^ -> Y v | NAND2X2 | 0.079 | 0.514 | 0.556 |
| u_temp_ctrl/g1712 | A v -> Y ^ | CLKINXV1 | 0.061 | 0.575 | 0.617 |
| u_temp_ctrl/g1685 | B ^ -> Y v | NAND3XL | 0.142 | 0.717 | 0.759 |
| u_temp_ctrl/g1664 | A1 v -> Y ^ | AOI21X1 | 0.197 | 0.914 | 0.957 |
| u_temp_ctrl/g1648 | A1 ^ -> Y v | AOI21X1 | 0.098 | 1.012 | 1.054 |
| u_temp_ctrl/FE_RC_10_0 | A v -> Y ^ | CLKINXV1 | 0.055 | 1.067 | 1.109 |
| u_temp_ctrl/FE_RC_8_0 | A1N ^ -> Y ^ | AOI2BB2X1 | 0.188 | 1.255 | 1.297 |
| u_temp_ctrl/g1636 | A ^ -> Y v | NAND2X2 | 0.106 | 1.360 | 1.403 |
| u_temp_ctrl/g1635 | A v -> Y ^ | INVX1 | 0.058 | 1.419 | 1.461 |
| u_temp_ctrl/g1632 | A ^ -> Y v | NAND2X1 | 0.066 | 1.485 | 1.527 |
| u_temp_ctrl/g1631 | B0 v -> Y ^ | AOI21X1 | 0.057 | 1.541 | 1.584 |
| u_temp_ctrl/cooler_on_reg | D ^ | SDFFRXL | 0.000 | 1.541 | 1.584 |
+-----+-----+-----+-----+-----+-----+

```

Required Time: 1.584 ns (latest safe arrival time)

Arrival Time: 1.541 ns (actual data arrival time)

Slack Calculation:

Slack = Required Time - Arrival Time

1.584 - 1.541 = +0.043 ns

Hold slack:

```
192.168.3.58:10000 (edatools-server2.iitd.edu.in:4100 (mukka25170)) - RealVNC Viewer
Applications Places System
#####
File Edit View Search Tools Documents Help
#####
timing_post_PnR_early.txt (~:/cmos65/90nm/Phy_des/post_route_reports/timing) - gedit
#####
# Generated by: Cadence Innovus 20.10-p004.1
# OS: Linux x86_64 (Host ID edatools-server2.iitd.edu.in)
# Generated on: Sun Nov 9 21:21:21 2025
# Design: HA
# Command: report_timing -early -view {view1} -max_paths 100 > post_route_reports/timing/timing_post_PnR_early.txt
#####
Path 1: MET Removal Check with Pin humidity_sensor_r_reg[7]/CK
Endpoint: humidity_sensor_r_reg[7]/RN (^) checked with leading edge of 'clk'
Beginpoint: rst_n (^) triggered by leading edge of 'clk'
Path Groups: {async default}
Analysis View: view1
Other End Arrival Time -0.002
+ Removal 0.202
+ Phase Shift 0.000
= Required Time 0.200
- Arrival Time 0.306
= Slack Time 0.106
Clock Rise Edge 0.000
+ Input Delay 0.300
= Beginpoint Arrival Time 0.300
+-----+-----+-----+-----+-----+-----+
| Instance | Arc | Cell | Delay | Arrival Time | Required Time |
+-----+-----+-----+-----+-----+-----+
| humidity_sensor_r_reg[7] | RN ^ | SDFFRH0X4 | 0.006 | 0.300 | 0.194 |
+-----+-----+-----+-----+-----+-----+

```

Arrival Time: 0.306 ns (when reset reaches the flip-flop)

Required Time: 0.200 ns (latest safe de-assertion time)

Slack Calculation:

Slack = Arrival Time - Required Time

0.306 - 0.200 = +0.106 ns

3. Power Report

```
192.168.3.58:10000 (edatools-server2.intl.edun4100 (mukka25170)) - RealVNC Viewer
-----
Innovus 20.10-p004_1 (64bit) 05/07/2020 20:02 (Linux 2.6.32-431.11.2.el6.x86_64)
-----
Date & Time: 2025-Nov-09 21:21:28 (2025-Nov-09 15:51:28 GMT)
-----
Design: HA
Liberty Libraries used:
view1: /home/mukka25170/cmos65/90nm/Phy_des/slow.lib
Power Domain used:
Rail: VDD Voltage: 0.9
Power View : view1
User-Defined Activity : N.A.
Activity File: N.A.
Hierarchical Global Activity: N.A.
Global Activity: N.A.
Sequential Element Activity: 0.200000
Primary Input Activity: 0.200000
Default icg ratio: N.A.
Global Comb ClockGate Ratio: N.A.
Power Units = 1mW
Time Units = 1e-09 secs
Temperature = 125
report_power -outfile post_route_reports/routeafteranalysis.rpt -rail_analysis_format VS
```

Cell	Internal Power	Switching Power	Total Power	Leakage Power	Cell Name
CTS_ccl_a_buf_00004	0.07978	0.09141	0.1714	0.0001864	CLKBUF16
CTS_ccl_a_buf_00006	0.07976	0.08822	0.1682	0.0001864	CLKBUF16
FE_OFCS_scan_en	0.001583	0.01856	0.02022	7.504e-05	CLKBUF4
u_visitor/count_reg[5]	0.01685	0.001411	0.01848	0.0002171	SDFFRHQX4
u_visitor/count_reg[0]	0.01693	0.001131	0.01828	0.0002171	SDFFRHQX4
u_visitor/count_reg[1]	0.01698	0.0009866	0.01819	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[4]	0.01632	0.001569	0.0181	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[1]	0.01632	0.001309	0.01785	0.0002171	SDFFRHQX4
window_auto_mode_r_reg	0.01664	0.0009805	0.01784	0.0002171	SDFFRHQX4
temp_sensor_r_reg[5]	0.01664	0.0009559	0.01782	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[5]	0.01632	0.001272	0.01781	0.0002171	SDFFRHQX4
humidity_sensor_r_reg[1]	0.01665	0.0009454	0.01781	0.0002171	SDFFRHQX4
temp_sensor_r_reg[2]	0.01664	0.0009083	0.01777	0.0002171	SDFFRHQX4
volume_cmd_r_reg[2]	0.01664	0.0008746	0.01774	0.0002171	SDFFRHQX4
temp_sensor_r_reg[3]	0.01665	0.000837	0.0177	0.0002171	SDFFRHQX4
volume_cmd_r_reg[3]	0.01665	0.0008246	0.01769	0.0002171	SDFFRHQX4
entry_sensor_r_reg	0.01665	0.0007996	0.01767	0.0002171	SDFFRHQX4
volume_cmd_r_reg[1]	0.01665	0.0007909	0.01766	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[3]	0.01633	0.001074	0.01762	0.0002171	SDFFRHQX4
volume_cmd_r_reg[0]	0.01665	0.0007477	0.01762	0.0002171	SDFFRHQX4
light_sensor_r_reg[6]	0.01666	0.0007341	0.01761	0.0002171	SDFFRHQX4
light_sensor_r_reg[2]	0.01669	0.0006988	0.0176	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[2]	0.01633	0.001052	0.0176	0.0002171	SDFFRHQX4
light_sensor_r_reg[1]	0.01669	0.0006727	0.01758	0.0002171	SDFFRHQX4
light_sensor_r_reg[3]	0.01669	0.0006511	0.01756	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[0]	0.01669	0.000651	0.01755	0.0002171	SDFFRHQX4
exit_sensor_r_reg	0.01666	0.0006725	0.01755	0.0002171	SDFFRHQX4
light_sensor_r_reg[5]	0.01666	0.0006724	0.01755	0.0002171	SDFFRHQX4
light_sensor_r_reg[7]	0.01666	0.0006712	0.01755	0.0002171	SDFFRHQX4
light_sensor_r_reg[4]	0.01666	0.0006521	0.01753	0.0002171	SDFFRHQX4
humidity_sensor_r_reg[7]	0.01666	0.0006507	0.01753	0.0002171	SDFFRHQX4
temp_sensor_r_reg[4]	0.01666	0.0006492	0.01752	0.0002171	SDFFRHQX4
light_sensor_r_reg[0]	0.0167	0.0005019	0.01742	0.0002171	SDFFRHQX4
tv_on_cmd_r_reg	0.01666	0.000537	0.01742	0.0002171	SDFFRHQX4
humidity_sensor_r_reg[3]	0.01667	0.00052	0.01741	0.0002171	SDFFRHQX4
temp_setpoint_r_reg[6]	0.01633	0.0008444	0.0174	0.0002171	SDFFRHQX4
music_on_cmd_r_reg	0.01667	0.0005128	0.0174	0.0002171	SDFFRHQX4
temp_sensor_r_reg[0]	0.01667	0.0004971	0.01738	0.0002171	SDFFRHQX4
humidity_sensor_r_reg[5]	0.01667	0.0004865	0.01737	0.0002171	SDFFRHQX4
window_manual_open_r_reg	0.01667	0.0004693	0.01735	0.0002171	SDFFRHQX4

192.168.3.58:10000 (edatools-server2.iitd.edu.in:4100 (mukka25170)) - RealVNC Viewer					
u_visitor/g1068	0.0001797	0.0001051	0.0003029	1.816e-05	NOR2X1
u_temp_ctrl/g1667	0.0002544	2.601e-05	0.0003014	2.093e-05	NOR4X1
u_burglar/g715	0.0001692	9.000115	0.0003993	1.612e-05	NAND4X1
u_window/g369	0.0001628	0.0001056	0.0002971	2.877e-05	NAND2BX1
u_temp_ctrl/g1706	0.0001785	7.244e-05	0.000286	3.499e-05	NAND2BX1
u_burglar/g702	0.0001276	0.0001145	0.0002771	3.499e-05	NAND2BX1
u_burglar/g717	0.0001741	9.181e-05	0.0002752	9.289e-06	NAND2XL
u_burglar/g705	3.992e-05	0.0002196	0.0002725	1.301e-05	CLKINVX1
u_humidity/g204	0.0001674	9.371e-05	0.0002698	8.757e-06	INVXL
u_temp_ctrl/g1729	0.000168	9.059e-05	0.0002674	8.757e-06	INVXL
u_door_lock/g2245	0.0001271	8.868e-05	0.0002507	3.499e-05	NAND2BX1
u_door_lock/g2257	0.0001847	4.821e-05	0.0002453	1.236e-05	NOR2XL
u_burglar/FE_OF0_n_6	0.0001396	9.238e-05	0.000245	1.301e-05	INVX1
u_visitor/g1091	0.000131	0.0001017	0.000242	9.289e-06	NAND2XL
u_temp_ctrl/g1660	0.000137	7.702e-05	0.000233	9.289e-06	NAND2XL
u_visitor/g1077	0.0001092	9.36e-05	0.000221	1.816e-05	NOR2X1
u_temp_ctrl/g1700	0.0001297	7.246e-05	0.0002115	9.289e-06	NAND2XL
u_visitor/g1006	5.626e-05	0.0001272	0.0001927	9.289e-06	NAND2XL
u_temp_ctrl/g1694	8.796e-05	7.921e-05	0.0001765	9.289e-06	NAND2XL
u_burglar/g701	0.0001156	4.741e-05	0.0001754	1.236e-05	NOR2XL
u_burglar/g713	9.364e-05	6.677e-05	0.0001745	1.41e-05	OAI211X1
u_temp_ctrl/FE_RC_13_0	9.01e-05	6.774e-05	0.0001671	9.289e-06	NAND2XL
u_door_lock/g2243	0.0001154	3.319e-05	0.0001609	1.236e-05	NOR2XL
u_burglar/g689	7.744e-05	4.579e-05	0.0001582	3.499e-05	NAND2BX1
u_door_lock/g2240	7.717e-05	4.616e-05	0.0001572	3.499e-05	NAND2BX1
u_visitor/g1057	5.339e-05	8.58e-05	0.0001532	1.403e-05	NAND2X1
u_temp_ctrl/g1697	5.118e-05	5.589e-05	0.0001211	1.403e-05	NAND2X1
u_temp_ctrl/g1685	5.835e-05	4.749e-05	0.0001158	9.93e-06	NAND3XL
u_burglar/g707	5.442e-05	4.41e-05	0.0001109	1.236e-05	NOR2XL
u_burglar/g692	4.787e-05	2.702e-05	0.0001091	3.419e-05	NAND3BX1
u_door_lock/g2235	4.928e-05	3.764e-05	9.994e-05	1.301e-05	CLKINVX1
u_door_lock/g2238	6.911e-05	1.737e-05	9.804e-05	1.236e-05	NOR2XL
u_temp_ctrl/FE_RC_12_0	6.405e-05	1.56e-05	9.781e-05	1.816e-05	NOR2X1
u_temp_ctrl/g1665	3.794e-05	2.684e-05	7.524e-05	1.046e-05	NAND4XL
u_burglar/g678	4.022e-05	1.624e-05	6.881e-05	1.236e-05	NOR2XL
u_visitor/g1082	1.739e-05	2.143e-05	5.284e-05	1.403e-05	NAND2X1
u_visitor/g1064	2.034e-05	1.653e-05	4.617e-05	9.289e-06	NAND2XL
u_burglar/g684	2.306e-05	9.134e-06	4.455e-05	1.236e-05	NOR2XL
u_burglar/g703	2.365e-05	7.67e-06	4.368e-05	1.236e-05	NOR2XL

Total (465 of 465)	2.593	0.3944	3.022	0.03504	
Total Capacitance	5.121e-12 F				
Power Density	*** No Die Area ***				

Power Distribution

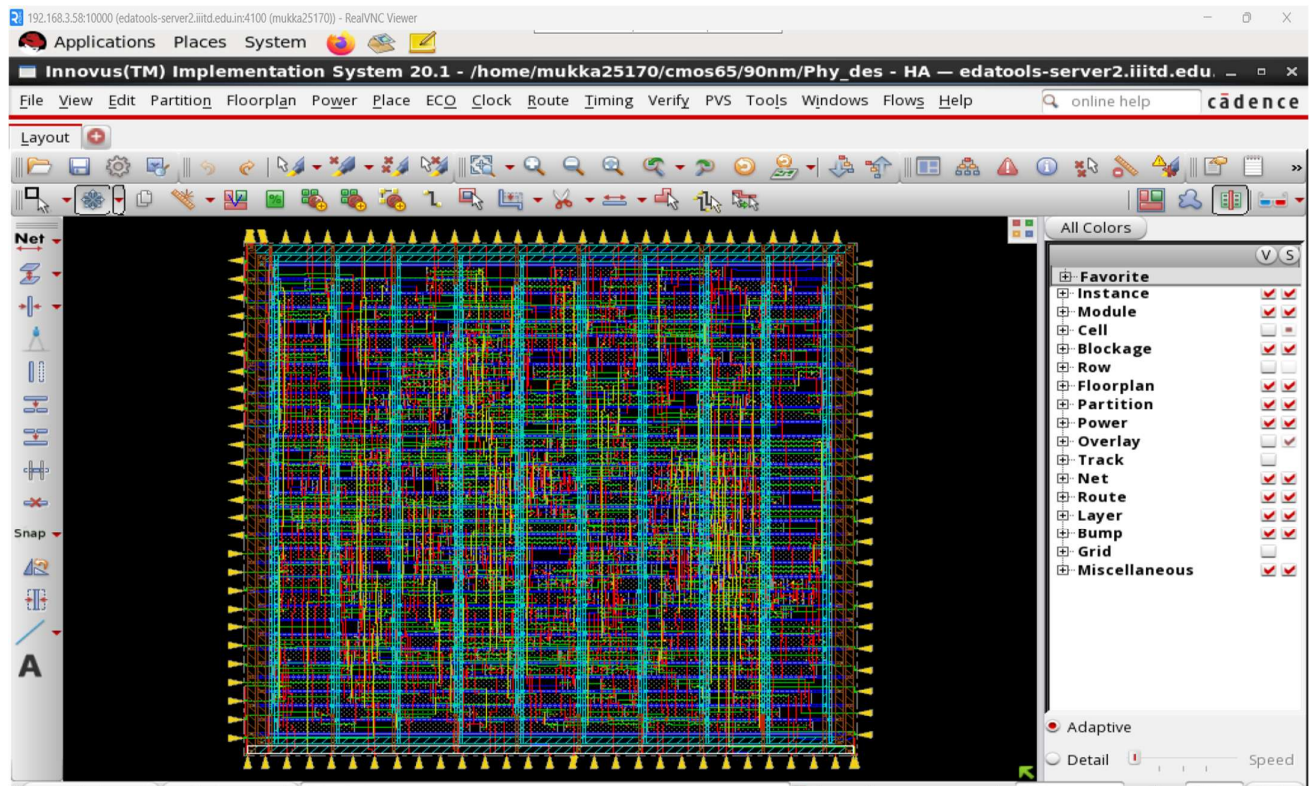
Total Power: 3.022mW (reduced from 4.794mW)

Internal Power: 2.593mW (85.8%)

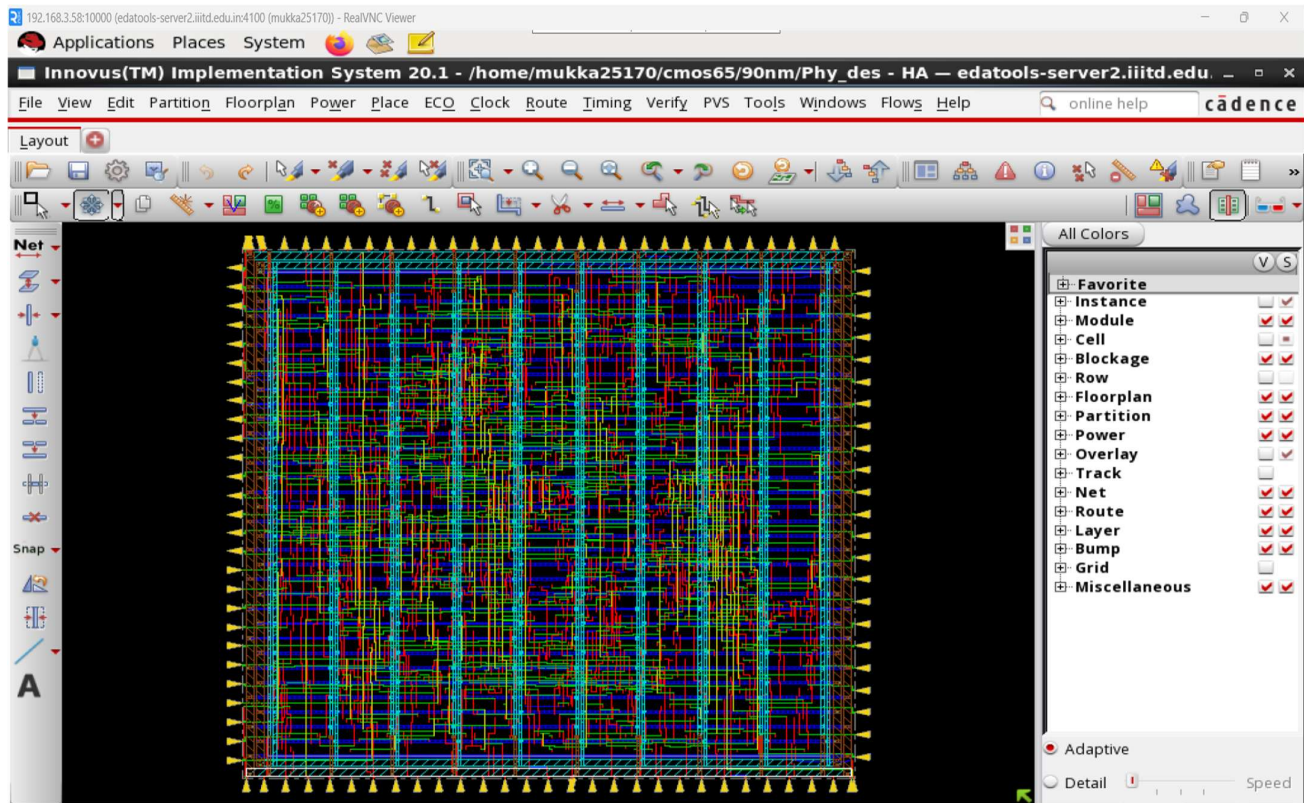
Switching Power: 0.3944mW (13.0%)

Leakage Power: 0.03504mW (1.2%)

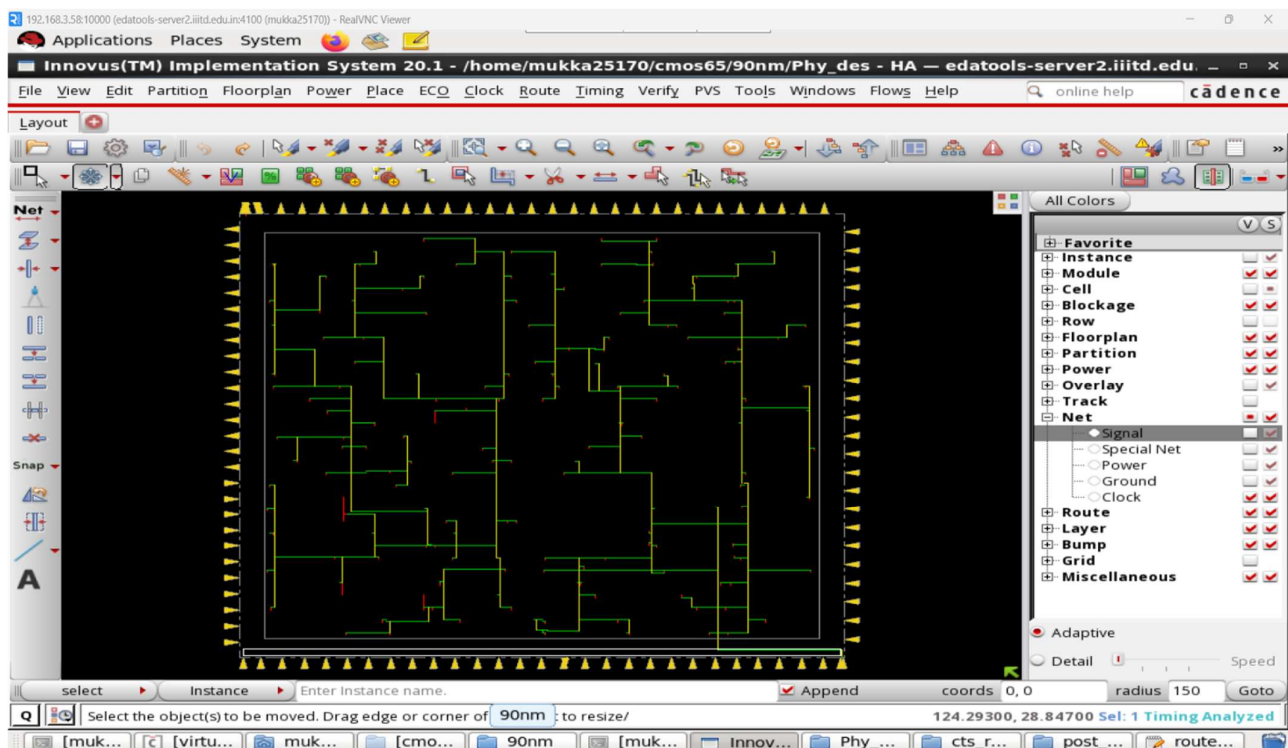
Core:



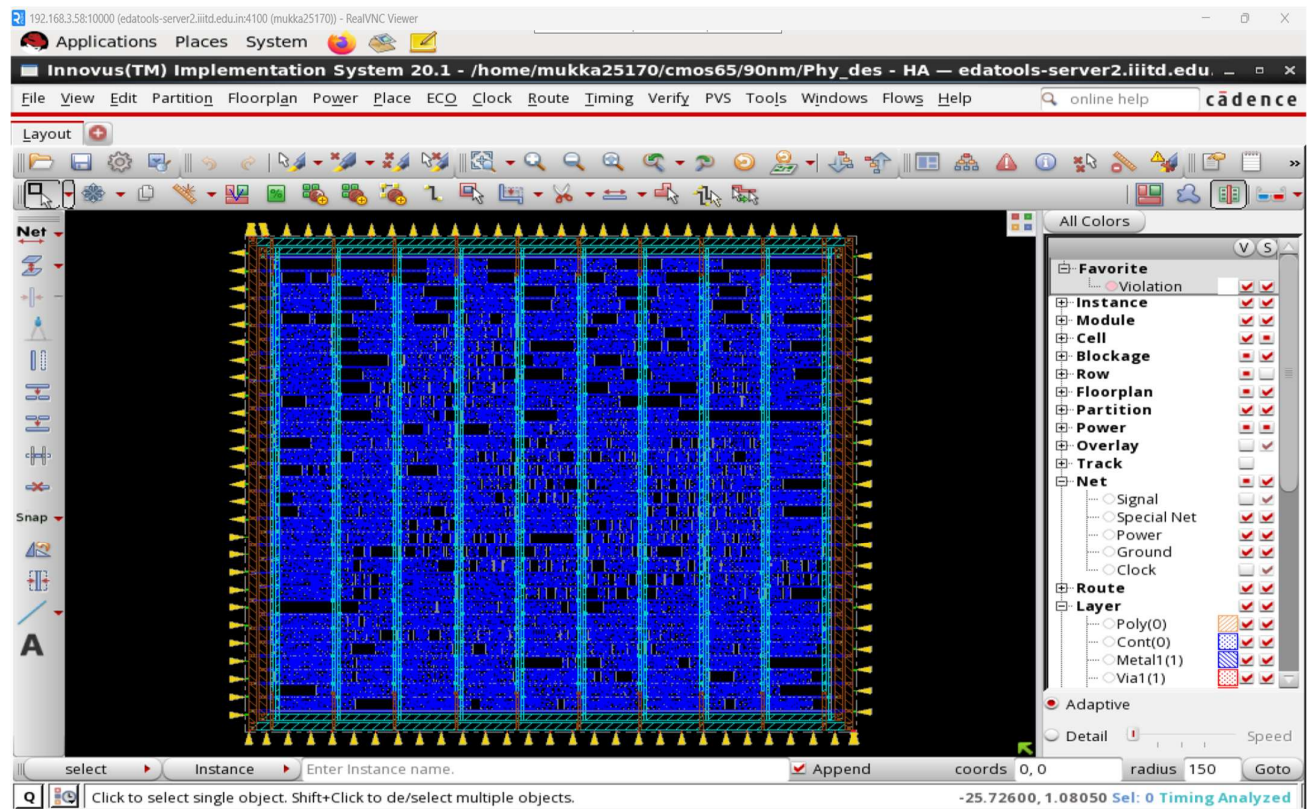
Routing of Wires:



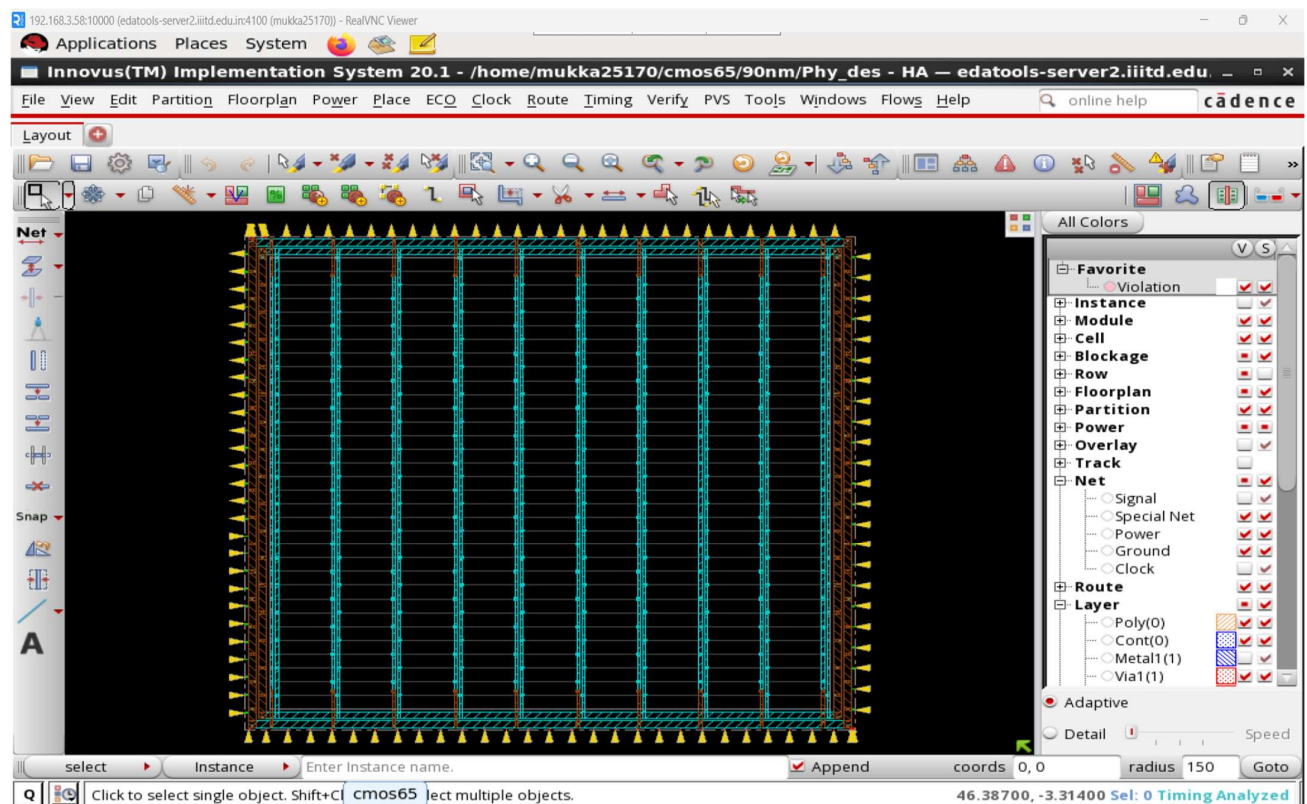
After CTS:



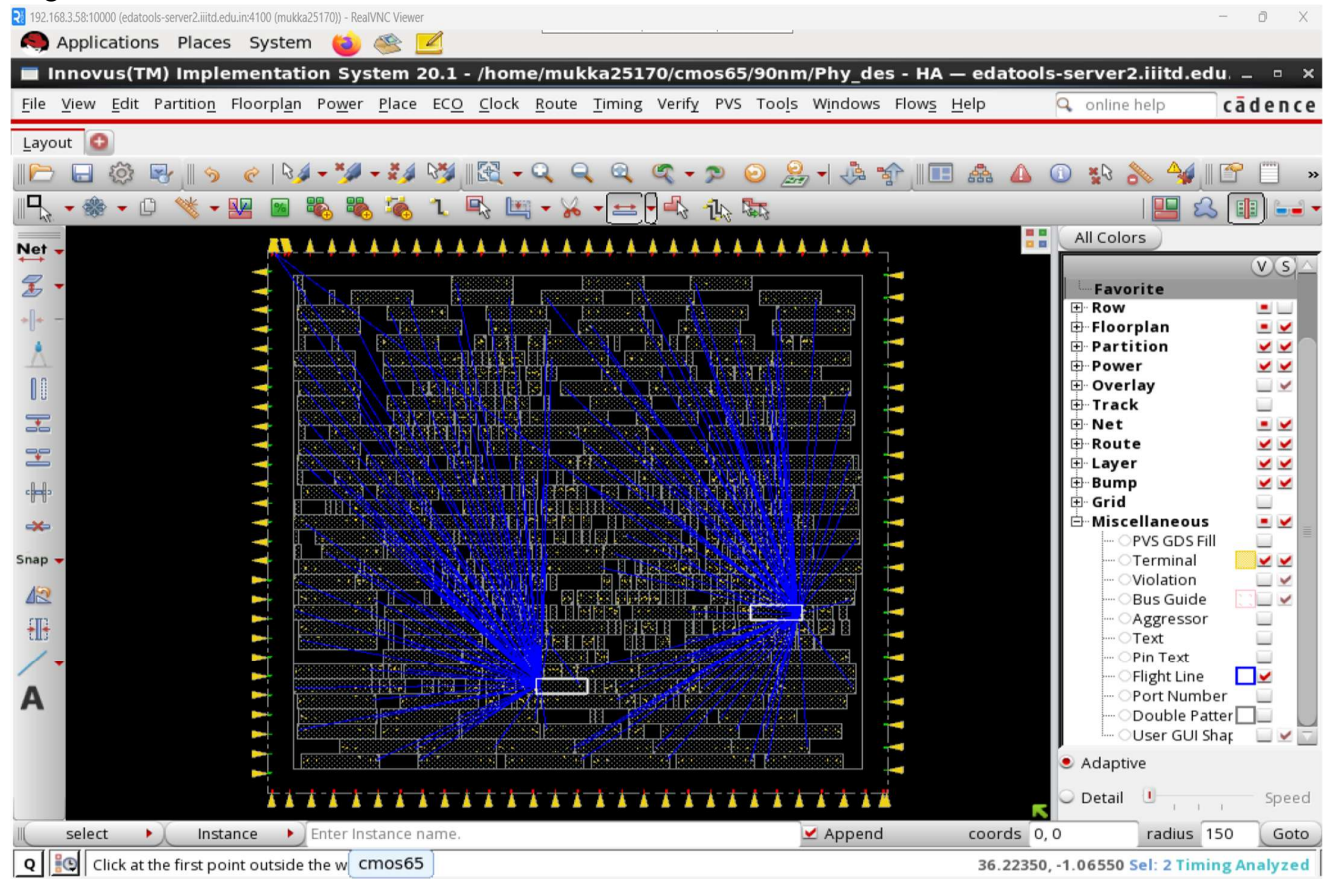
Standard Cells and Cell Layout:



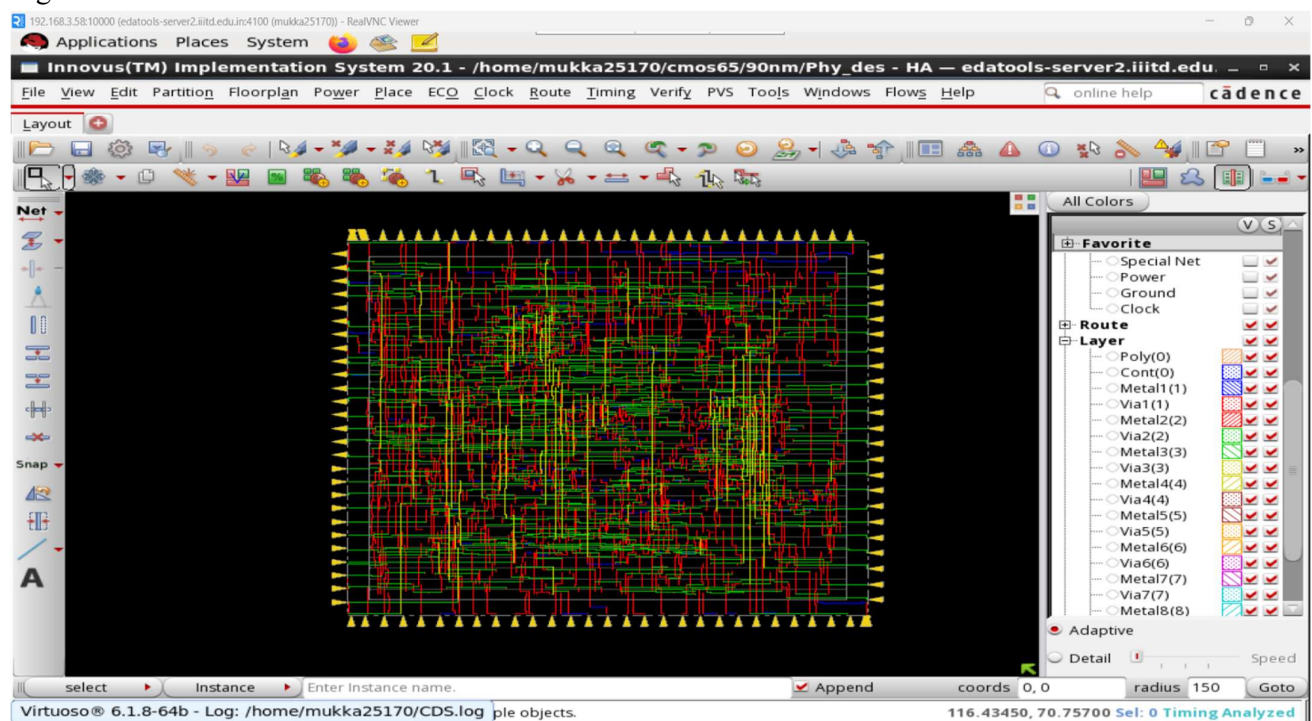
Power And Ground Rails:



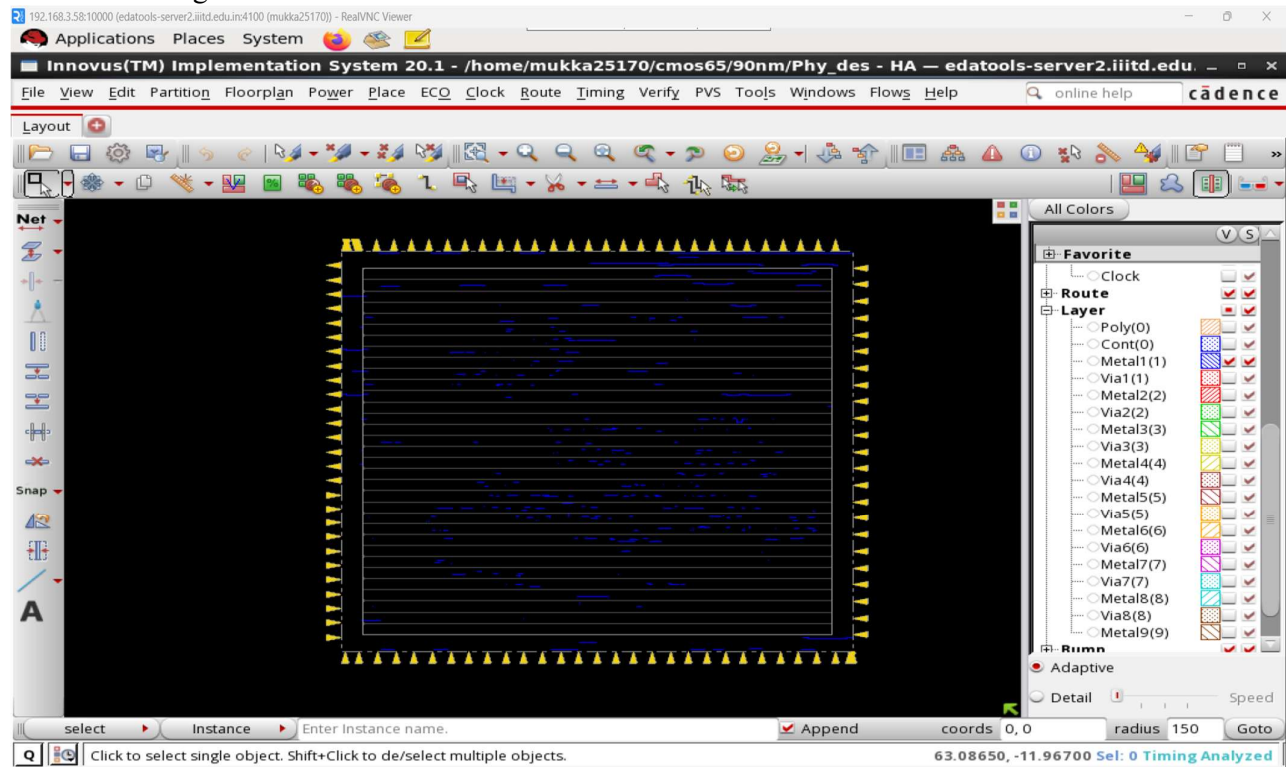
Flight Lines:



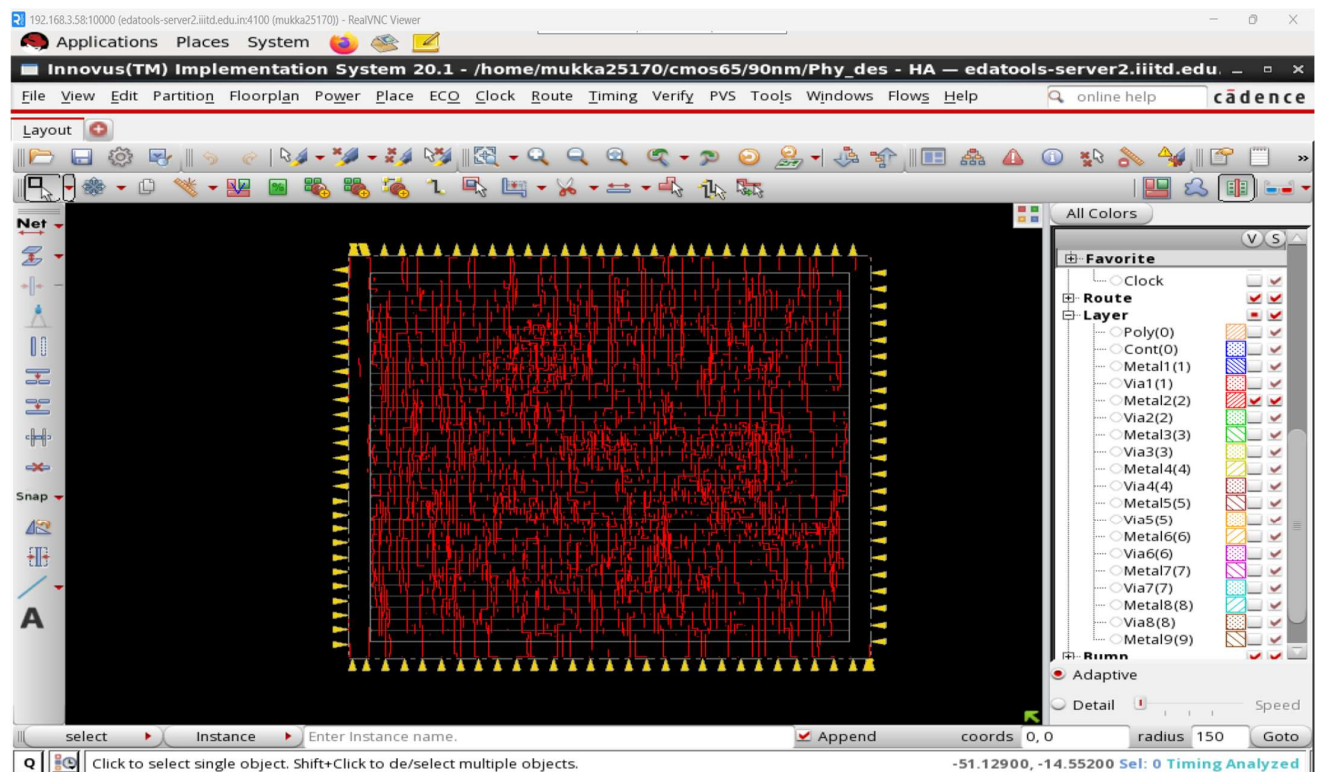
Signals From Ports:



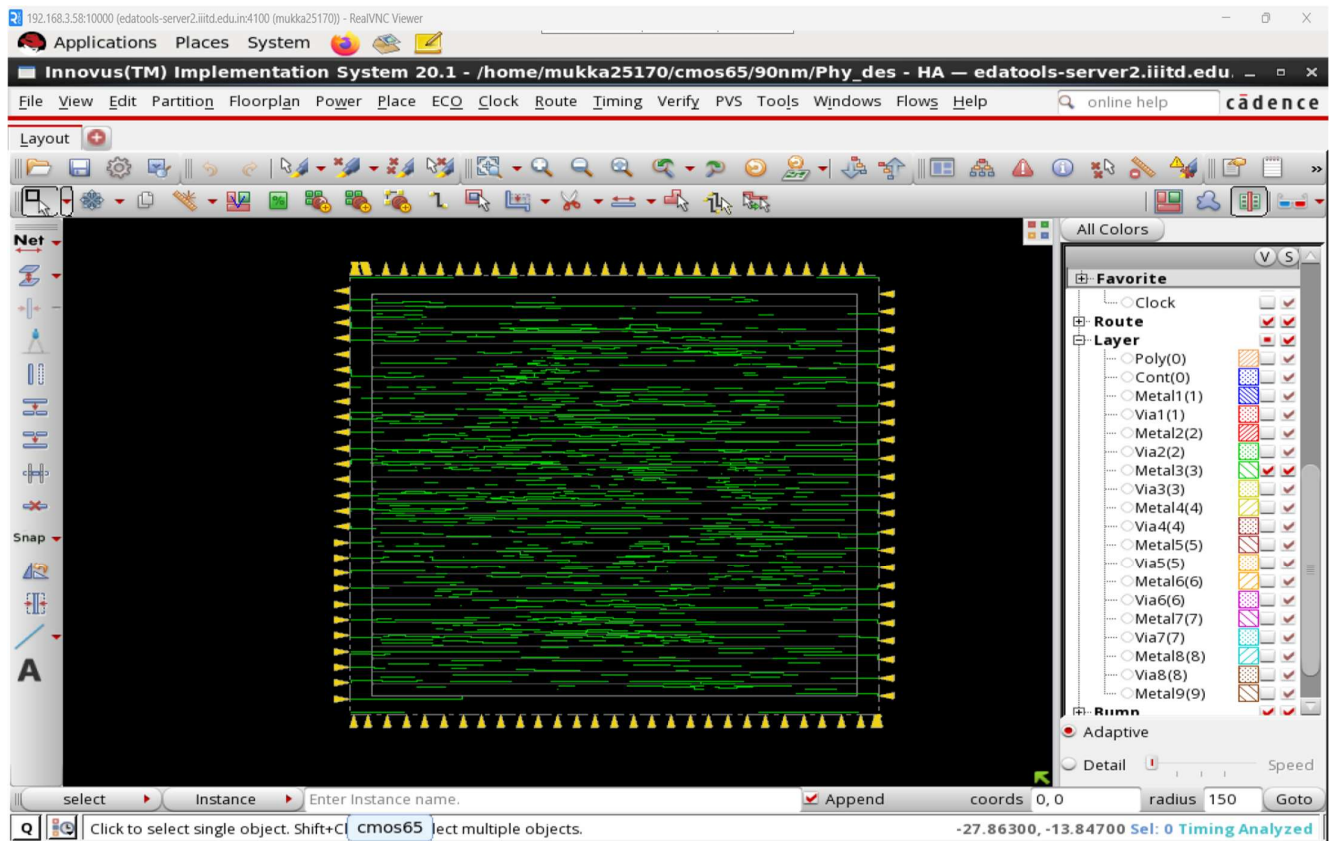
Metal 1 Routing:



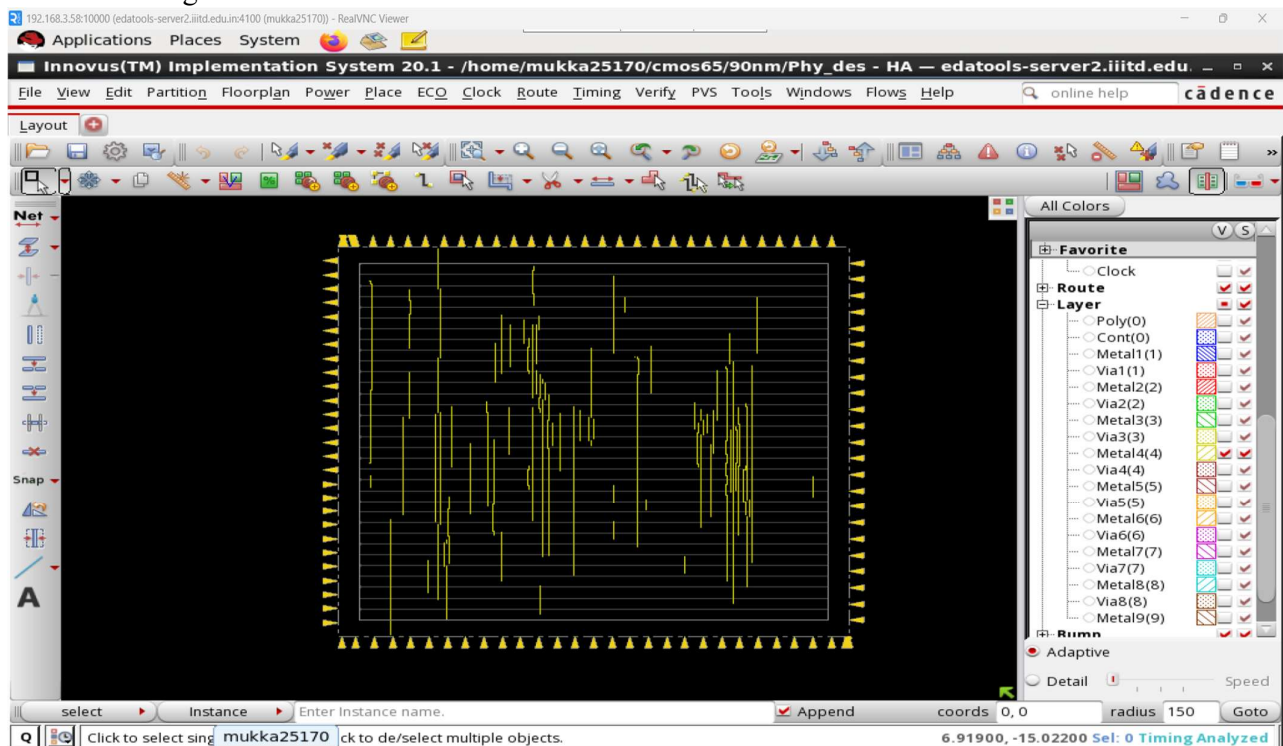
Metal 2 Routing



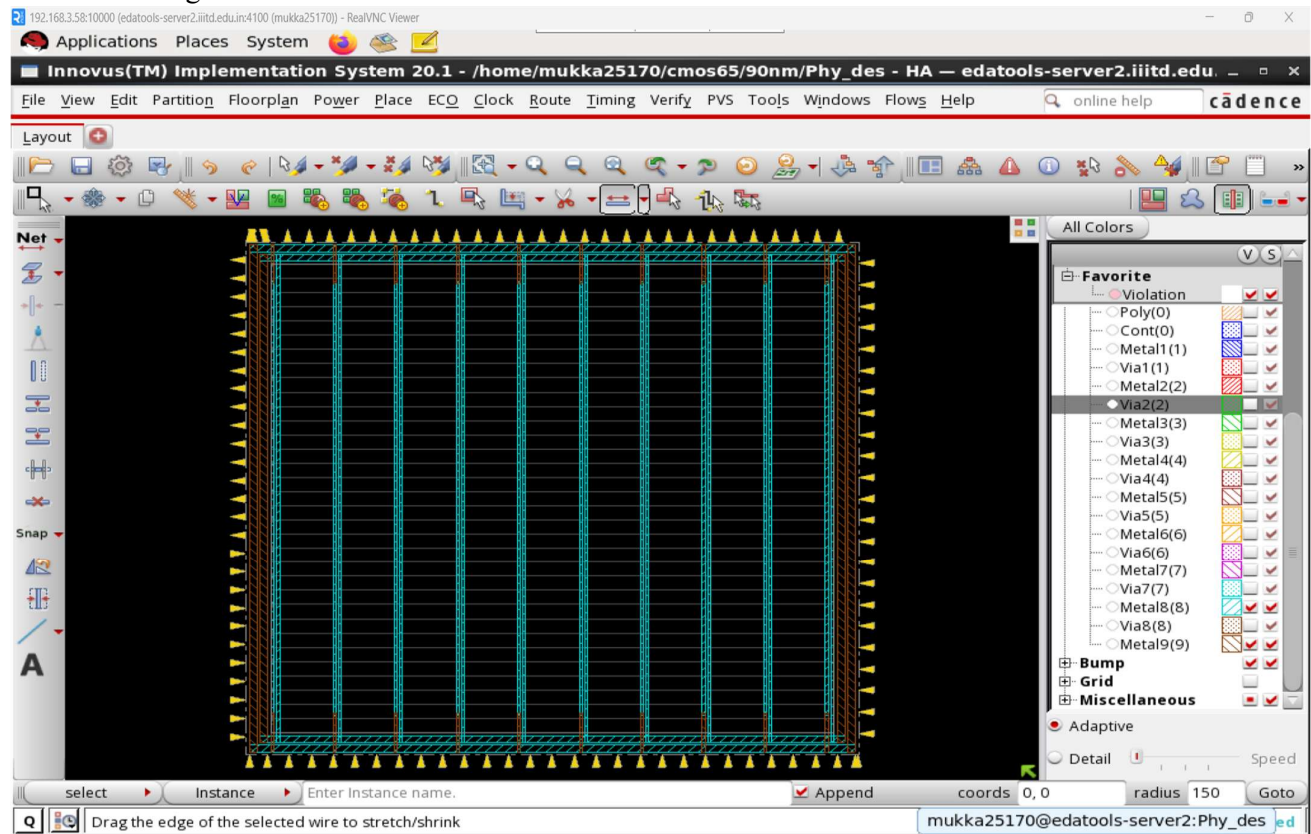
Metal3 Routing:



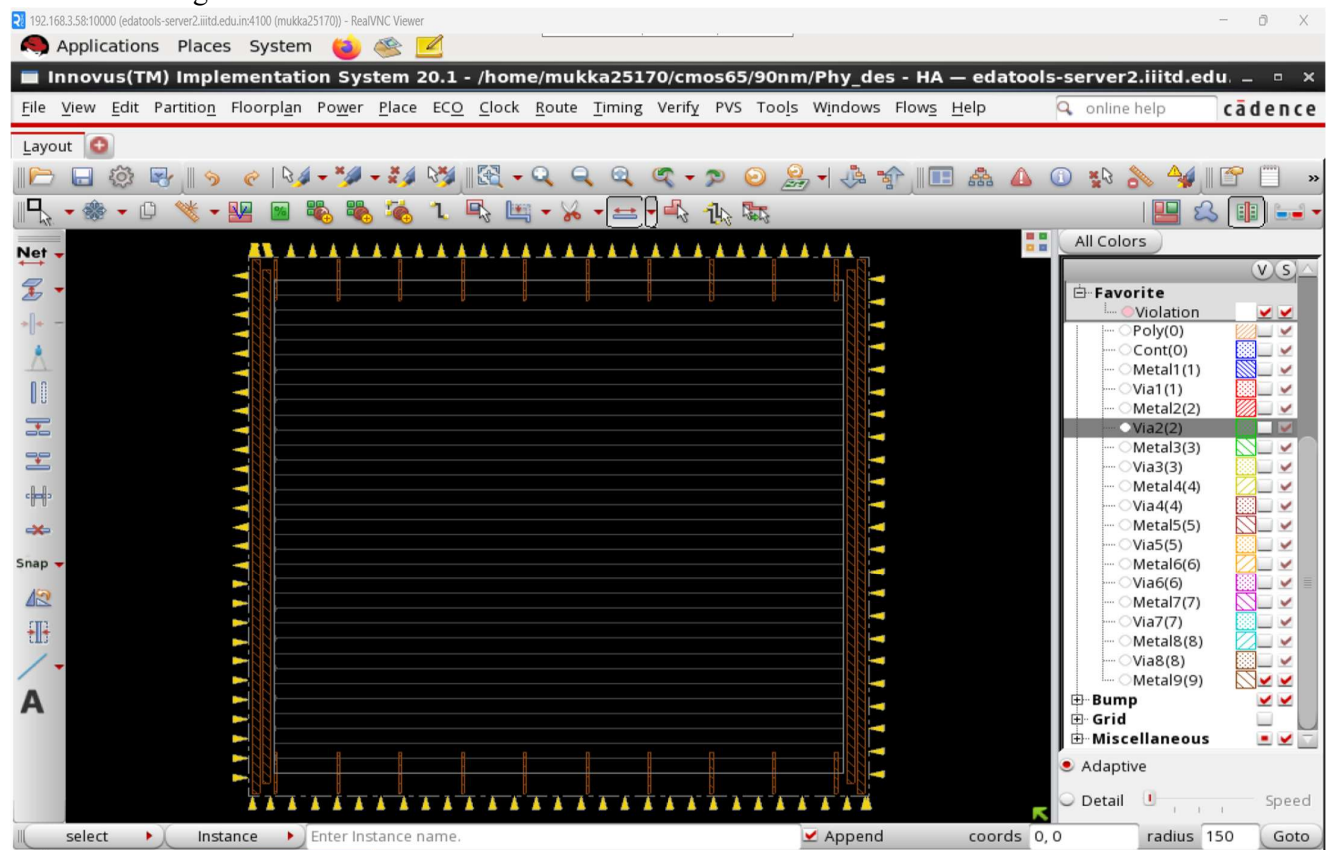
Metal4 Routing:



Metal8 Routing:

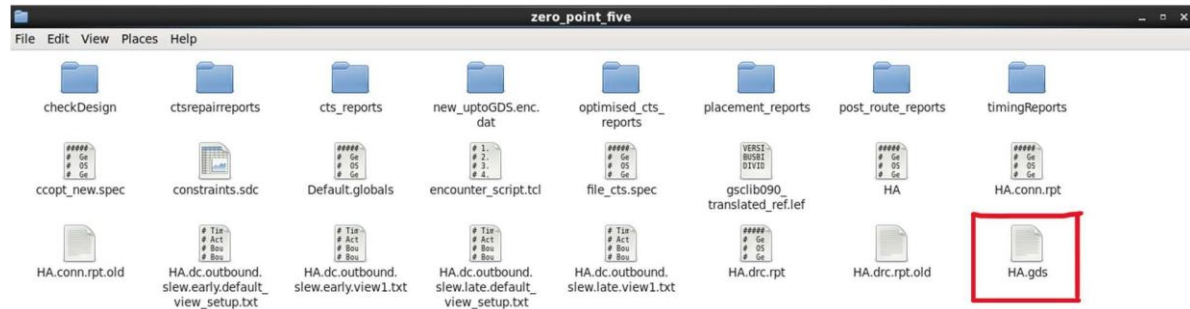


Metal 9 Routing:

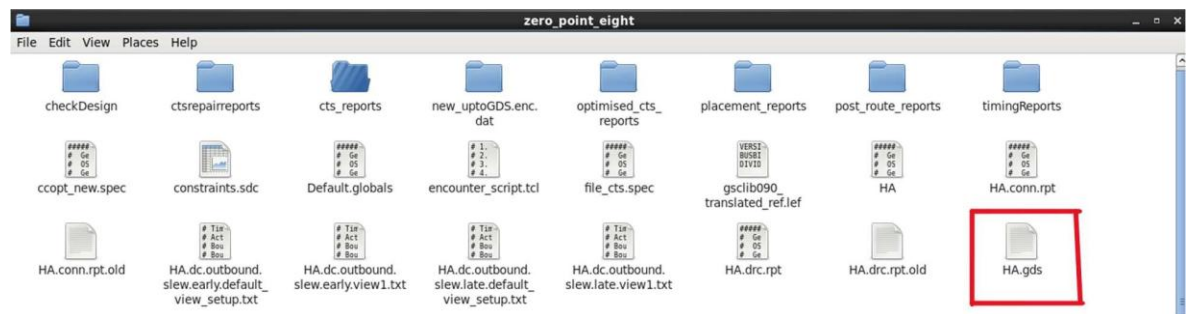


GDS files Generated:

For 0.5 utilization:



For 0.8 utilization:



Interpretation of Results Timing

Timing Utilization 0.5: Most paths meet timing with positive slack following CTS and post-routing improvements, although the timing reports indicate a setup violation (negative slack) in the placement step.

Utilization 0.8: The design satisfies timing with positive slack following CTS and postrouting, despite the timing report indicating a setup violation (negative slack) during the placement step. Higher usage results in more stringent time restrictions, necessitating more aggressive optimization to closure timing.

Area

Utilization 0.5: The total standard cell area is 6074.879 (placement), 6264.104 (CTS), and 5925.013 (post-routing). The area is larger due to more whitespace, which helps with routing and congestion avoidance.

Utilization 0.8: The total standard cell area is 6074.879 (placement), 6130.133 (CTS), and 5797.854 (post-routing). The area is smaller, reflecting denser packing, but this can lead to routing congestion.

Power

Utilization 0.5: Total power is 4.756mW (placement), 3.274mW (CTS), and 3.076mW (post-routing). The higher area and longer wire lengths contribute to higher dynamic power, but leakage is low.

Utilization 0.8: Total power is 4.794mW (placement), 3.219mW (CTS), and 3.022mW (post-routing). The power is slightly lower due to shorter wire lengths, but leakage is marginally higher due to denser packing

Routability

Utilization 0.5: The design has more whitespace, which improves routability and reduces congestion. Buffer insertion is minimal, and routing is generally easier.

Utilization 0.8: The design is more congested, requiring more buffer insertion and careful routing. The higher utilization increases the risk of unroutable nets and requires more routing iterations

CTS Shape Interpretation

- Utilization 0.5: The CTS shape is umbrella or scattered. The extra whitespace allows for more flexible routing, and the clock tree can spread out easily to reach all sequential elements without significant congestion. This results in a more distributed and less structured tree, avoiding H-type or X-type shapes due to the available area.
- Utilization 0.8: The CTS shape is also umbrella or scattered. Despite the higher density, the clock tree still spreads out from the source, but with more constraints due to limited routing resources. The tree may have more buffering and shorter branches, but it does not form a structured H-type or X-type shape, which is typical for larger, more regular designs.

For both the utilization the clock tree appears to follow an umbrella or scattered shape, which is typical for small designs where the clock source is centrally located and the tree branches out to reach all flip-flops and latches.

Analysis of each component and how it is computed in CTS and the effect of the timing of the clock-path on overall slack.

The timing report for the worst path highlights the most critical delay in the design, which is essential for ensuring the system meets its clock period constraints. Each component in the timing path contributes to the total delay, impacting whether the path passes or fails setup timing.

Components of the Worst Path

- **Setup Time:** The minimum time required before the active clock edge for data to be stable at a register input.
- **Phase Shift:** Any adjustment in the clock phase, often due to clock tree implementation.
- **Propagation Delay (Cell and Net Delay):** Sum of delays through logic gates (cells) and the interconnections (nets) between those gates.
- **Clock Network Latency/Skew:** Difference in clock arrival times between the launch and capture registers, caused by clock tree and routing variations.
- **Arrival Time:** The calculated time at which data arrives at the endpoint register input.
- **Required Time:** The latest possible time by which data must arrive, set by the clock period and register setup time.
- **Slack:** The difference between required time and arrival time; positive slack means the path meets timing, negative slack indicates a violation.

How Each Component is Computed

- **Cell Delay:** Calculated from standard cell library data under specific loading and input slope conditions.
- **Net Delay:** Comes from wire length, routing topology, resistance, and capacitance after layout extraction.
- **Clock Latency:** The actual time for the clock edge to propagate from clock origin to each register, measured after clock tree synthesis and routing.
- **Arrival Time:** Computed as the sum of clock-to-Q delay (for source FF), logic delays, and net delays up to the destination FF.
- **Required Time:** Based on the next clock edge minus setup-time and taking phase shift and clock skew into account.

Effect of Clock-Path Timing on Slack

- Clock skew and latency can significantly impact the overall slack. If the clock arrives late at the destination register or early at the source register, slack can be reduced or even become negative, causing a timing violation.
- Optimized clock tree design aims to minimize skew and balance latency, helping maximize positive slack and overall design robustness

Comparison table for 0.5 and 0.8 utilization

Criteria	Utilization 0.5 (Large Die)	Utilization 0.8 (Small Die)
Timing Closure	Easier, some initial violations	Harder, critical violations, needs aggressive optimization
Area	Larger, more whitespace	Smaller, denser packing
Power	Higher dynamic, lower leakage	Lower dynamic, slightly higher leakage
Routability	Better, less congestion	Poorer, more congestion, more buffers
Routing Iterations	Fewer	More